

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12415993
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Halperin; Schaked Omer

RNA-guided endonuclease fusion polypeptides and methods of use thereof

Abstract

The present disclosure provides a fusion polypeptide comprising: a) an enzymatically active RNA-guided endonuclease that introduces a single-stranded break in a target DNA; and b) an error-prone DNA polymerase. The present disclosure provides a system comprising: a) a fusion polypeptide of the present disclosure; and b) a guide RNA. The present disclosure provides a cell comprising a fusion polypeptide of the present disclosure, or a system of the present disclosure. The present disclosure provides a method of mutagenizing a target polynucleotide.

Inventors:	Halperin; Schaked Omer (Berkeley, CA)
Applicant:	The Regents of the University of California (Oakland, CA)
Family ID:	1000008822010
Assignee:	The Regents of the University of California (Oakland, CA)
Appl. No.:	18/130199
Filed:	April 03, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230399630 A1	Dec. 14, 2023

Related U.S. Application Data

division parent-doc US 16641950 US 11649442 WO PCT/US2018/049766 20180906 child-doc US 18130199
us-provisional-application US 62662043 20180424
us-provisional-application US 62556127 20170908

Publication Classification

Int. Cl.: C12N9/22 (20060101); C12N9/12 (20060101); C12N9/78 (20060101); C12N15/10 (20060101); C12N15/11 (20060101)

U.S. Cl.:

CPC C12N9/22 (20130101); C12N9/1252 (20130101); C12N9/78 (20130101); C12N15/1024 (20130101); C12N15/11 (20130101); C12Y207/07007 (20130101); C12Y305/04004 (20130101); C12Y305/04005 (20130101); C12N2800/80 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9161995	12/2014	Guschin et al.	N/A	N/A
10752920	12/2019	Doudna et al.	N/A	N/A
11193123	12/2020	Halperin	N/A	N/A
2002/0192788	12/2001	Ihlenfeldt et al.	N/A	N/A
2014/0010839	12/2013	Benkirane et al.	N/A	N/A
2015/0044772	12/2014	Zhao	N/A	N/A
2015/0166980	12/2014	Liu et al.	N/A	N/A
2016/0160193	12/2015	Hsieh et al.	N/A	N/A
2017/0058271	12/2016	Joung et al.	N/A	N/A
2017/0121693	12/2016	Liu et al.	N/A	N/A
2018/0073012	12/2017	Liu et al.	N/A	N/A
2018/0230494	12/2017	Joung et al.	N/A	N/A
2018/0237770	12/2017	May et al.	N/A	N/A
2018/0298414	12/2017	Janulaitis et al.	N/A	N/A
2018/0313843	12/2017	Keppler et al.	N/A	N/A
2019/0062409	12/2018	Luban	N/A	N/A
2019/0350978	12/2018	Beauchesne et al.	N/A	N/A
2020/0038424	12/2019	Hirano et al.	N/A	N/A
2020/0248155	12/2019	Halperin	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO 2011/116184	12/2010	WO	N/A
WO 2014/186686	12/2013	WO	N/A
WO 2016/028843	12/2015	WO	N/A
WO 2016/036754	12/2015	WO	N/A
WO 2016/183448	12/2015	WO	N/A
WO 2016/196805	12/2015	WO	N/A
WO 2017/106616	12/2016	WO	N/A
WO 2017/151719	12/2016	WO	N/A

WO 2017/189308	12/2016	WO	N/A
WO 2018/049168	12/2017	WO	N/A
WO 2018/165629	12/2017	WO	N/A
WO 2018/176009	12/2017	WO	N/A
WO 2018/027078	12/2017	WO	N/A
WO 2019/051097	12/2018	WO	N/A
WO 2019/070842	12/2018	WO	N/A
WO 2019/070843	12/2018	WO	N/A
WO 2020/012335	12/2019	WO	N/A
WO 2015/035139	12/2019	WO	N/A
WO 2020/047124	12/2019	WO	N/A
WO 2020/051561	12/2019	WO	N/A
WO 2020/191153	12/2019	WO	N/A
WO 2020/191171	12/2019	WO	N/A
WO 2020/191233	12/2019	WO	N/A
WO 2020/191234	12/2019	WO	N/A
WO 2020/191239	12/2019	WO	N/A
WO 2020/191241	12/2019	WO	N/A
WO 2020/191242	12/2019	WO	N/A
WO 2020/191243	12/2019	WO	N/A
WO 2020/191245	12/2019	WO	N/A
WO 2020/191246	12/2019	WO	N/A
WO 2020/191248	12/2019	WO	N/A
WO 2020/191249	12/2019	WO	N/A
WO 2021/188840	12/2020	WO	N/A

OTHER PUBLICATIONS

Antczak, et al. New functionality of RNAComposer: application to shape the axis of miR160 precursor structure. *Acta Biochim. Pol.* 63, 737-744 (2016). cited by applicant

Anzalone, et al. Search-and-replace genome editing without double-strand breaks or donor DNA. *Nature* 576, 149-157 (2019). cited by applicant

Bajaj, et al. An in vivo genome-wide CRISPR screen identifies the RNA-binding protein Stau2 as a key regulator of myeloid leukemia. *Nature Cancer* vol. 1 410-422 (2020). cited by applicant

Christensen, et al. Footprint of the retrotransposon R2Bm protein on its target site before and after cleavage. *Journal of molecular biology* vol. 336,5 (2004): 1035-45. cited by applicant

Co-pending U.S. Appl. No. 17/206,522, inventor Halperin; Schaked Omer, filed Mar. 19, 2021. cited by applicant

Co-pending U.S. Appl. No. 17/206,526, inventor Halperin; Schaked Omer, filed Mar. 19, 2021. cited by applicant

Co-pending U.S. Appl. No. 17/340,822, inventor Halperin; Schaked, filed Jun. 7, 2021. cited by applicant

Co-pending U.S. Appl. No. 17/506,419, inventors Halperin; Schaked Omer et al., filed Oct. 20, 2021. cited by applicant

Cox, et al. Therapeutic genome editing: prospects and challenges. *Nature Medicine* vol. 21 121-131 (2015). cited by applicant

Denli, et al. Primate-Specific ORF0 Contributes to Retrotransposon-Mediated Diversity, *Cell*, vol. 163, 3:1 (2015): 583-593. cited by applicant

Gaudelli, et al. Programmable base editing of A•T to G•C in genomic DNA without DNA cleavage. *Nature* 551, 464-471 (2017). cited by applicant

Grünewald, et al. Transcriptome-wide off-target RNA editing induced by CRISPR-guided DNA

base editors. *Nature* 569, 433-437 (2019). cited by applicant

Guilinger, et al.; “Fusion of catalytically inactive Cas9 to FokI nuclease improves the specificity of genome modification”; *Nat. Biotechnol.*; vol. 32, No. 6, pp. 577-582 (Jun. 2014). cited by applicant

Haapaniemi, et al. CRISPR-Cas9 genome editing induces a p53-mediated DNA damage response. *Nat. Med.* 24, 927-930 (2018). cited by applicant

Halperin, et al.; “CRISPR-guided DNA polymerases enable diversification of all nucleotides in a tunable window”; *Nature*; vol. 560, pp. 248-252 (Aug. 1, 2018). cited by applicant

Hrecka, et al. Vpx relieves inhibition of HIV-1 infection of macrophages mediated by the SAMHD1 protein. *Nature* 474, 658-661 (2011). cited by applicant

Ihry, et al. p53 inhibits CRISPR-Cas9 engineering in human pluripotent stem cells. *Nat. Med.* 24, 939-946 (2018). cited by applicant

Jinek, et al. A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity. *Science* 337, 816-821 (2012). cited by applicant

Komor, et al. Programmable editing of a target base in genomic DNA without double-stranded DNA cleavage. *Nature* 533, 420-424 (2016). cited by applicant

Krzywkowski, et al. Limited reverse transcriptase activity of phi29 DNA polymerase. *Nucleic acids research* vol. 46,7 (2018): 3625-3632. cited by applicant

Ledford, H. CRISPR Editing Wreaks Chromosomal Mayhem in Human Embryos. *Nature*. vol. 583. (Jul. 2, 2020): 17-18. cited by applicant

Ledford, H. CRISPR gene editing in human embryos wreaks chromosomal mayhem. *Nature* 583, 17-18 (2020). cited by applicant

Lemmon, et al. Rapid improvement of domestication traits in an orphan crop by genome editing. *Nature Plants* vol. 4 766-770 (2018). cited by applicant

Maeder, et al. Development of a gene-editing approach to restore vision loss in Leber congenital amaurosis type 10. *Nat. Med.* 25, 229-233 (2019). cited by applicant

Mali, et al. RNA-guided human genome engineering via Cas9. *Science* 339, 823-826 (2013). cited by applicant

Middleton, et al. Elexacaftor-Tezacaftor-Ivacaftor for Cystic Fibrosis with a Single Phe508del Allele. *New England Journal of Medicine* vol. 381 1809-1819 (2019). cited by applicant

Ostedgaard, et al. CFTR with a partially deleted R domain corrects the cystic fibrosis chloride transport defect in human airway epithelia in vitro and in mouse nasal mucosa in vivo. *Proc. Natl. Acad. Sci. U. S. A.* 99, 3093-3098 (2002). cited by applicant

Palikša, et al. Decreased Km to dNTPs is an essential M-MuLV reverse transcriptase adoption required to perform efficient cDNA synthesis in One-Step RT-PCR assay. *Protein Eng. Des. Sel.* 31, 79-89 (2018). cited by applicant

PCT Application No. PCT/US21/23041 International Search Report and Written Opinion dated Jul. 13, 2021. cited by applicant

Ribeiro, et al. Protein Engineering Strategies to Expand CRISPR-Cas9 Applications. *International Journal of Genomics*, vol. 2018, Aug. 2, 2018, pp. 1-12. cited by applicant

Silas, et al. Direct CRISPR spacer acquisition from RNA by a natural reverse transcriptase-Cas1 fusion protein. *Science*, New York, N.Y., vol. 351, issue 6276, Feb. 26, 2016. cited by applicant

St Gelais, et al.; “SAMHD1 restricts HIV-1 infection in dendritic cells (DCs) by dNTP depletion, but its expression in DCs and primary CD4+ T-lymphocytes cannot be upregulated by interferons”; *Retrovirology*; vol. 9, No. 105, 15 pages (2012). cited by applicant

Su, et al. Human DNA polymerase has reverse transcriptase activity in cellular environments. *The Journal of biological chemistry* vol. 294, 15 (2019): 6073-6081. cited by applicant

Toro, et al. The Reverse Transcriptases Associated with CRISPR-Cas Systems. *Scientific reports*, | vol. 7, issue 7089, Aug. 2, 2017. cited by applicant

U.S. Appl. No. 17/206,522 Examiner's Interview Summary dated Sep. 30, 2021. cited by applicant

U.S. Appl. No. 17/206,522 Final Office Action dated Sep. 15, 2021. cited by applicant

U.S. Appl. No. 17/206,522 Notice of Allowance dated Oct. 12, 2021. cited by applicant
U.S. Appl. No. 17/206,522 Office Action dated Jul. 22, 2021. cited by applicant
U.S. Appl. No. 17/206,526 Office Action dated Sep. 2, 2021. cited by applicant
Xie, et al. Effective and accurate gene silencing by a recombinant AAV-compatible microRNA scaffold. *Molecular Therapy* 28.2 (2020): 422-430. cited by applicant
Yan, et al. CyclinA2-Cyclin-dependent Kinase Regulates SAMHD1 Protein Phosphohydrolase Domain. *Journal of Biological Chemistry* vol. 290 13279-13292 (2015). cited by applicant
Yang, et al.; "Amino Acid Changes in a Unique Sequence of Bacteriophage T7 DNA Polymerase Alter the Processivity of Nucleotide Polymerization"; *The Journal of Biological Chemistry*; vol. 272, No. 10, pp. 6599-6606 (1997). cited by applicant
Yu, et al. The efficiency of Vpx-mediated SAMHD1 antagonism does not correlate with the potency of viral control in HIV-2-infected individuals. *Retrovirology* 10, 27 (2013). cited by applicant
Zhang, et al.; "Conserved Herpesvirus Protein Kinases Target SAMHD1 to Facilitate Virus Replication"; *Cell Reports*; vol. 27, pp. 449-459 (Jul. 9, 2019). cited by applicant
Zuo, et al. Cytosine base editor generates substantial off-target single-nucleotide variants in mouse embryos. *Science* 364, 289-292 (2019). cited by applicant

Primary Examiner: Noakes; Suzanne M

Attorney, Agent or Firm: Bozicevic, Field & Francis LLP

Background/Summary

CROSS-REFERENCE (1) This application is a division of U.S. patent application Ser. No. 16/641,950, now U.S. Pat. No. 11,649,442, filed Feb. 25, 2020, which is a U.S. National Phase Application of PCT Application No. PCT/US2018/049766, filed Sep. 6, 2018, which claims the benefit of U.S. Provisional Patent Application No. 62/556,127, filed Sep. 8, 2017, and U.S. Provisional Patent Application No. 62/662,043, filed Apr. 24, 2018, which applications are incorporated herein by reference in their entirety.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING PROVIDED AS A TEXT FILE SEQUENCE LISTING XML FILE

(1) A Sequence Listing is provided herewith as a Sequence Listing XML, "BERK-363DIV_SEQ_LIST" created on Aug. 28, 2023 and having a size of 2,173,363 bytes. The contents of the Sequence Listing XML are incorporated by reference herein in their entirety.

INTRODUCTION

(2) Directed evolution is a powerful discovery approach that isolates genetic material with desirable properties from a library of sequence variants. However, the sequence space that can be explored is limited by the efficiency of synthesizing and transforming a genetic library. This requirement for efficient transformation rates has confined directed evolution to only a few model organisms. The ability to program a cell to localize increased mutagenesis at user-defined loci would remove the need to transform a synthesized library of nucleic acids; unfortunately, current in vivo targeted mutagenesis platforms are either confined to targeting a set locus in a specific organism or have a narrow and biased editing window at user-defined loci.

(3) There is a need in the art for compositions and methods for mutagenizing a target DNA.

SUMMARY

(4) The present disclosure provides a fusion polypeptide comprising: a) an enzymatically active RNA-guided endonuclease that introduces a single-stranded break in a target DNA; and b) an error-

prone DNA polymerase. The present disclosure provides a system comprising: a) a fusion polypeptide of the present disclosure; and b) a guide RNA. The present disclosure provides a cell comprising a fusion polypeptide of the present disclosure, or a system of the present disclosure. The present disclosure provides a method of mutagenizing a target polynucleotide.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic depiction of an example of a fusion polypeptide of the present disclosure.
- (2) FIG. 2A-2C depict mutator designs (FIG. 2A), and mutational analysis of various mutators (FIGS. 2B and 2C).
- (3) FIG. 3 depicts the mutagenesis window length of various mutators.
- (4) FIG. 4A-4B depict characterization of the mutation rate, substitution bias, and mutagenesis window length in bacterial, fungal, and mammalian cells.
- (5) FIG. 5A-5F provides amino acid sequences of *Streptococcus pyogenes* Cas9 (FIG. 5A; SEQ ID NO: 1219) and variants of *Streptococcus pyogenes* Cas9 (FIG. 5B-5F). FIG. 5B: SEQ ID NO: 1220; FIG. 5C: SEQ ID NO:1221; FIG. 5D: SEQ ID NO:1222; FIG. 5E: SEQ ID NO:1223; and FIG. 5F: SEQ ID NO:1210.
- (6) FIG. 6 provides an amino acid sequence of *Staphylococcus aureus* Cas9 (SEQ ID No.: 1224).
- (7) FIG. 7A-7C provide amino acid sequences of *Francisella tularensis* Cpf1 (FIG. 7A; SEQ ID NO: 1225), Acidaminococcus sp. BV3L6 Cpf1 (FIG. 7B; SEQ ID NO:1226), and a variant Cpf1 (FIG. 7C; SEQ ID NO:1227).
- (8) FIG. 8A-8D provide amino acid sequences DNA polymerases PolI1M (FIG. 8A; SEQ ID NO: 1228), PolI2M (FIG. 8B; SEQ ID NO:1229), PolI3M (FIG. 8C; SEQ ID NO:1230), and PolI3M-TBD (FIG. 8D; SEQ ID NO:1231)
- (9) FIG. 9 provides an amino acid sequence of Phi29 DNA polymerase (SEQ ID NO:1232).
- (10) FIG. 10 provides an amino acid sequence of a T5 DNA polymerase (SEQ ID NO:1233).
- (11) FIG. 11A-11B provide amino acid sequences of T7 DNA polymerase (FIG. 11A; SEQ ID NO: 1234) and Sequenase (FIG. 11B; SEQ ID NO:1235).
- (12) FIG. 12 provides an amino acid sequence of a *Sulfolobus solfataricus* DNA-binding protein 7d (Sso7d) (SEQ ID NO:1236).
- (13) FIG. 13 provides an amino acid sequence of DNA polymerase Iota (SEQ ID NO:1237).
- (14) FIGS. 14A-14B provide an amino acid sequence of DNA polymerase n (SEQ ID NO:1238).
- (15) FIGS. 15A-15B provide an amino acid sequence of DNA polymerase K (SEQ ID NO:1239).
- (16) FIGS. 16A-16D provide an amino acid sequence of DNA polymerase 0 (SEQ ID NO:1240).
- (17) FIGS. 17A-17B provide an amino acid sequence of DNA polymerase v (SEQ ID NO:1241).
- (18) FIG. 18 provides an amino acid sequence of *E. coli* DNA polymerase IV (SEQ ID NO:1242).
- (19) FIG. 19A-19B provide an amino acid sequence of topoisomerase I (SEQ ID NO:1243).
- (20) FIG. 20 provides an amino acid sequence of a flap endonuclease (SEQ ID NO:1244).
- (21) FIG. 21 provides an amino acid sequence of a T4 DNA ligase (SEQ ID NO:1245).
- (22) FIGS. 22A-22E provide a schematic of an example of a fusion polypeptide of the present disclosure, a characterization of substitution frequency, and analysis of mutation rates.
- (23) FIGS. 23A-23H provide an analysis of mutation rates, mutagenesis window lengths, combinatorial mutations, multiplexed targeting, and continuous diversification of genomic loci.
- (24) FIGS. 24A-24D provide characterization of mutations to *E. coli*'s Ribosomal Subunit 5 gene, rpsE, that confer spectinomycin resistance.
- (25) FIG. 25 provides an analysis of the direction of mutagenesis relative to a gRNA.
- (26) FIG. 26 provides a characterization of the mutation rate of various mutators at a distance from a nick.

- (27) FIG. 27 provides a characterization of the target window length of a more processive DNA polymerase fused to enCas9.
- (28) FIG. 28 provides the off-target mutation rate of various mutators.
- (29) FIGS. 29A-29B illustrate coupling of mutagenesis by examples of fusion polypeptides of the present disclosure with a non-selectable genetic screen.
- (30) FIGS. 30A-30C illustrate the viability and growth rate of *E. coli* expressing examples of fusion polypeptides of the present disclosure.
- (31) FIG. 31 illustrates locations of gRNA targets and mutations relative to the *rpsE* gene.
- (32) FIGS. 32A-32B illustrate deletions in ribosomal protein S5 and the effect of the deletions on spectinomycin resistance.
- (33) FIG. 33 provides a comparison of *E. coli* diversification methods.
- (34) FIG. 34 provides oligonucleotide sequences.
- (35) FIG. 35 provides gRNA protospacer sequences.
- (36) FIG. 36 provides plasmid sequences (SEQ ID Nos: 1137-1141, respectively).
- (37) FIG. 37 provides amino acid sequences of various Cas9 and DNA polymerase polypeptides.
- (38) FIG. 38 is a schematic depiction of base editing.
- (39) FIG. 39A-39D depict use of a system of the present disclosure to introduce mutations into a target gene in eukaryotic cells.
- (40) FIG. 40A-40D depict fluorescence activated cell sorting (FACS) plots showing the analysis of a mutation into a target gene in eukaryotic cells.

DEFINITIONS

- (41) The terms “polynucleotide” and “nucleic acid,” used interchangeably herein, refer to a polymeric form of nucleotides of any length, either ribonucleotides or deoxynucleotides. Thus, this term includes, but is not limited to, single-, double-, or multi-stranded DNA or RNA, genomic DNA, cDNA, DNA-RNA hybrids, or a polymer comprising purine and pyrimidine bases or other natural, chemically or biochemically modified, non-natural, or derivatized nucleotide bases. The terms “polynucleotide” and “nucleic acid” should be understood to include, as applicable to the embodiment being described, single-stranded (such as sense or antisense) and double-stranded polynucleotides.
- (42) The terms “polypeptide,” “peptide,” and “protein”, are used interchangeably herein, refer to a polymeric form of amino acids of any length, which can include genetically coded and non-genetically coded amino acids, chemically or biochemically modified or derivatized amino acids, and polypeptides having modified peptide backbones. The term includes fusion proteins, including, but not limited to, fusion proteins with a heterologous amino acid sequence, fusions with heterologous and homologous leader sequences, with or without N-terminal methionine residues; immunologically tagged proteins; and the like.
- (43) The term “naturally-occurring” as used herein as applied to a nucleic acid, a protein, a cell, or an organism, refers to a nucleic acid, cell, protein, or organism that is found in nature.
- (44) As used herein the term “isolated” is meant to describe a polynucleotide, a polypeptide, or a cell that is in an environment different from that in which the polynucleotide, the polypeptide, or the cell naturally occurs. An isolated genetically modified host cell may be present in a mixed population of genetically modified host cells.
- (45) “Heterologous,” as used herein, refers to a nucleotide or amino acid sequence that is not found in the native nucleic acid or protein, respectively. For example, relative to a Cas9 polypeptide, a heterologous polypeptide comprises an amino acid sequence from a protein other than the Cas9 polypeptide. Thus, for example, a polymerase polypeptide is heterologous to a Cas9 polypeptide.
- (46) “Recombinant,” as used herein, means that a particular nucleic acid (DNA or RNA) is the product of various combinations of cloning, restriction, and/or ligation steps resulting in a construct having a structural coding or non-coding sequence distinguishable from endogenous nucleic acids found in natural systems. Generally, nucleotide sequences encoding the structural coding sequence

can be assembled from cDNA fragments and short oligonucleotide linkers, or from a series of synthetic oligonucleotides, to provide a synthetic nucleic acid which is capable of being expressed from a recombinant transcriptional unit contained in a cell or in a cell-free transcription and translation system. Such sequences can be provided in the form of an open reading frame uninterrupted by internal non-translated sequences, or introns, which are typically present in eukaryotic genes. Genomic DNA comprising the relevant nucleotide sequences can also be used in the formation of a recombinant gene or transcriptional unit. Sequences of non-translated DNA may be present 5' or 3' from the open reading frame, where such sequences do not interfere with manipulation or expression of the coding regions, and may indeed act to modulate production of a desired product by various mechanisms (see "DNA regulatory sequences", below).

(47) Thus, e.g., the term "recombinant" polynucleotide or "recombinant" nucleic acid refers to one which is not naturally occurring, e.g., is made by the artificial combination of two otherwise separated segments of sequence through human intervention. This artificial combination is often accomplished by either chemical synthesis means, or by the artificial manipulation of isolated segments of nucleic acids, e.g., by genetic engineering techniques. Such artificial combination can be carried out to join together nucleic acid segments of desired functions to generate a desired combination of functions.

(48) Similarly, the term "recombinant" polypeptide refers to a polypeptide which is not naturally occurring, e.g., is made by the artificial combination of two otherwise separated segments of amino acid sequence through human intervention. Thus, e.g., a polypeptide that comprises a heterologous amino acid sequence is recombinant.

(49) By "construct" or "vector" is meant a recombinant nucleic acid, generally recombinant DNA, which has been generated for the purpose of the expression and/or propagation of a specific nucleotide sequence(s), or is to be used in the construction of other recombinant nucleotide sequences.

(50) The terms "DNA regulatory sequences," "control elements," and "regulatory elements," used interchangeably herein, refer to transcriptional and translational control sequences, such as promoters, enhancers, polyadenylation signals, terminators, protein degradation signals, and the like, that provide for and/or regulate expression of a coding sequence and/or production of an encoded polypeptide in a host cell.

(51) The term "transformation" is used interchangeably herein with "genetic modification" and refers to a permanent or transient genetic change induced in a cell following introduction of new nucleic acid (e.g., DNA exogenous to the cell) into the cell. Genetic change ("modification") can be accomplished either by incorporation of the new nucleic acid into the genome of the host cell, or by transient or stable maintenance of the new nucleic acid as an episomal element. Where the cell is a eukaryotic cell, a permanent genetic change can be achieved by introduction of new DNA into the genome of the cell. In prokaryotic cells, permanent changes can be introduced into the chromosome or via extrachromosomal elements such as plasmids and expression vectors, which may contain one or more selectable markers to aid in their maintenance in the recombinant host cell. Suitable methods of genetic modification include viral infection, transfection, conjugation, protoplast fusion, electroporation, particle gun technology, calcium phosphate precipitation, direct microinjection, and the like. The choice of method is generally dependent on the type of cell being transformed and the circumstances under which the transformation is taking place (i.e. in vitro, ex vivo, or in vivo). A general discussion of these methods can be found in Ausubel, et al, Short Protocols in Molecular Biology, 3rd ed., Wiley & Sons, 1995.

(52) "Operably linked" refers to a juxtaposition wherein the components so described are in a relationship permitting them to function in their intended manner. For instance, a promoter is operably linked to a coding sequence if the promoter affects its transcription or expression. As used herein, the terms "heterologous promoter" and "heterologous control regions" refer to promoters and other control regions that are not normally associated with a particular nucleic acid in nature.

For example, a “transcriptional control region heterologous to a coding region” is a transcriptional control region that is not normally associated with the coding region in nature.

(53) A “host cell,” as used herein, denotes an *in vivo* or *in vitro* eukaryotic cell, a prokaryotic cell, or a cell from a multicellular organism (e.g., a cell line) cultured as a unicellular entity, which eukaryotic or prokaryotic cells can be, or have been, used as recipients for a nucleic acid (e.g., an expression vector), and include the progeny of the original cell which has been genetically modified by the nucleic acid. It is understood that the progeny of a single cell may not necessarily be completely identical in morphology or in genomic or total DNA complement as the original parent, due to natural, accidental, or deliberate mutation. A “recombinant host cell” (also referred to as a “genetically modified host cell”) is a host cell into which has been introduced a heterologous nucleic acid, e.g., an expression vector. For example, a prokaryotic host cell is a genetically modified prokaryotic host cell (e.g., a bacterium), by virtue of introduction into a suitable prokaryotic host cell of a heterologous nucleic acid, e.g., an exogenous nucleic acid that is foreign to (not normally found in nature in) the prokaryotic host cell, or a recombinant nucleic acid that is not normally found in the prokaryotic host cell; and a eukaryotic host cell is a genetically modified eukaryotic host cell, by virtue of introduction into a suitable eukaryotic host cell of a heterologous nucleic acid, e.g., an exogenous nucleic acid that is foreign to the eukaryotic host cell, or a recombinant nucleic acid that is not normally found in the eukaryotic host cell.

(54) The term “conservative amino acid substitution” refers to the interchangeability in proteins of amino acid residues having similar side chains. For example, a group of amino acids having aliphatic side chains consists of glycine, alanine, valine, leucine, and isoleucine; a group of amino acids having aliphatic-hydroxyl side chains consists of serine and threonine; a group of amino acids having amide-containing side chains consists of asparagine and glutamine; a group of amino acids having aromatic side chains consists of phenylalanine, tyrosine, and tryptophan; a group of amino acids having basic side chains consists of lysine, arginine, and histidine; and a group of amino acids having sulfur-containing side chains consists of cysteine and methionine. Exemplary conservative amino acid substitution groups are: valine-leucine-isoleucine, phenylalanine-tyrosine, lysine-arginine, alanine-valine, and asparagine-glutamine.

(55) A polynucleotide or polypeptide has a certain percent “sequence identity” to another polynucleotide or polypeptide, meaning that, when aligned, that percentage of bases or amino acids are the same, and in the same relative position, when comparing the two sequences. Sequence similarity can be determined in a number of different manners. To determine sequence identity, sequences can be aligned using the methods and computer programs, including BLAST, available over the world wide web at ncbi.nlm.nih.gov/BLAST. See, e.g., Altschul et al. (1990), *J. Mol. Biol.* 215:403-10. Another alignment algorithm is FASTA, available in the Genetics Computing Group (GCG) package, from Madison, Wisconsin, USA, a wholly owned subsidiary of Oxford Molecular Group, Inc. Other techniques for alignment are described in *Methods in Enzymology*, vol. 266: Computer Methods for Macromolecular Sequence Analysis (1996), ed. Doolittle, Academic Press, Inc., a division of Harcourt Brace & Co., San Diego, California, USA. Of particular interest are alignment programs that permit gaps in the sequence. The Smith-Waterman is one type of algorithm that permits gaps in sequence alignments. See *Meth. Mol. Biol.* 70:173-187 (1997). Also, the GAP program using the Needleman and Wunsch alignment method can be utilized to align sequences. See *J. Mol. Biol.* 48:443-453 (1970).

(56) Before the present invention is further described, it is to be understood that this invention is not limited to particular embodiments described, as such may, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting, since the scope of the present invention will be limited only by the appended claims.

(57) Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and

lower limit of that range and any other stated or intervening value in that stated range, is encompassed within the invention. The upper and lower limits of these smaller ranges may independently be included in the smaller ranges, and are also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the invention.

(58) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs.

Although any methods and materials similar or equivalent to those described herein can also be used in the practice or testing of the present invention, the preferred methods and materials are now described. All publications mentioned herein are incorporated herein by reference to disclose and describe the methods and/or materials in connection with which the publications are cited.

(59) It must be noted that as used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a guide RNA” includes a plurality of such guide RNAs and reference to “the RNA-guided endonuclease” includes reference to one or more RNA-guided endonucleases and equivalents thereof known to those skilled in the art, and so forth. It is further noted that the claims may be drafted to exclude any optional element. As such, this statement is intended to serve as antecedent basis for use of such exclusive terminology as “solely,” “only” and the like in connection with the recitation of claim elements, or use of a “negative” limitation.

(60) It is appreciated that certain features of the invention, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the invention, which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable sub-combination. All combinations of the embodiments pertaining to the invention are specifically embraced by the present invention and are disclosed herein just as if each and every combination was individually and explicitly disclosed. In addition, all sub-combinations of the various embodiments and elements thereof are also specifically embraced by the present invention and are disclosed herein just as if each and every such sub-combination was individually and explicitly disclosed herein.

(61) The publications discussed herein are provided solely for their disclosure prior to the filing date of the present application. Nothing herein is to be construed as an admission that the present invention is not entitled to antedate such publication by virtue of prior invention. Further, the dates of publication provided may be different from the actual publication dates which may need to be independently confirmed.

DETAILED DESCRIPTION

(62) The present disclosure provides a fusion polypeptide comprising: a) an enzymatically active RNA-guided endonuclease that introduces a single-stranded break (a nick) in a target DNA; and b) an error-prone DNA polymerase. The present disclosure provides a system comprising: a) a fusion polypeptide of the present disclosure; and b) a guide RNA. The present disclosure provides a cell comprising a fusion polypeptide of the present disclosure, or a system of the present disclosure. The present disclosure provides a method of mutagenizing a target polynucleotide.

(63) Fusion Polypeptides

(64) The present disclosure provides a fusion polypeptide comprising: a) an enzymatically active RNA-guided endonuclease that introduces a single-stranded break in a target DNA; and b) an error-prone DNA polymerase. A fusion polypeptide of the present disclosure is also referred to herein as a “mutator.”

(65) In some cases, a fusion polypeptide of the present disclosure comprises, in order from N-terminus to C-terminus: a) an enzymatically active RNA-guided endonuclease that introduces a single-stranded break in a target DNA; and b) an error-prone DNA polymerase. In some cases, a fusion polypeptide of the present disclosure comprises, in order from N-terminus to C-terminus: a)

an error-prone DNA polymerase; and b) an enzymatically active RNA-guided endonuclease that introduces a single-stranded break in a target DNA.

(66) In some cases, a fusion polypeptide of the present disclosure comprises, in order from N-terminus to C-terminus: a) an enzymatically active RNA-guided endonuclease that introduces a single-stranded break in a target DNA; b) a peptide linker; and c) an error-prone DNA polymerase. In some instances, the fusion polypeptide comprises one or more nuclear localization signals (NLSs). For example, in some cases, the fusion polypeptide comprises a single NLS at the N-terminus of the fusion polypeptide. In some cases, the fusion polypeptide comprises 2, 3, or 4 NLSs at the N-terminus of the fusion polypeptide. In other instances, in some cases, the fusion polypeptide comprises a single NLS at the C-terminus of the fusion polypeptide. In some cases, the fusion polypeptide comprises 2, 3, or 4 NLSs at the C-terminus of the fusion polypeptide.

(67) In some cases, a fusion polypeptide of the present disclosure comprises, in order from N-terminus to C-terminus: a) an error-prone DNA polymerase; b) a peptide linker; and c) an enzymatically active RNA-guided endonuclease that introduces a single-stranded break in a target DNA. In some instances, the fusion polypeptide comprises one or more NLSs. For example, in some cases, the fusion polypeptide comprises a single NLS at the N-terminus of the fusion polypeptide. In some cases, the fusion polypeptide comprises 2, 3, or 4 NLSs at the N-terminus of the fusion polypeptide. In other instances, in some cases, the fusion polypeptide comprises a single NLS at the C-terminus of the fusion polypeptide. In some cases, the fusion polypeptide comprises 2, 3, or 4 NLSs at the C-terminus of the fusion polypeptide.

(68) The linker polypeptide may have any of a variety of amino acid sequences. Proteins can be joined by a spacer peptide, generally of a flexible nature, although other chemical linkages are not excluded. Suitable linkers include polypeptides of between 4 amino acids and 40 amino acids in length, or between 4 amino acids and 25 amino acids in length. These linkers can be produced by using synthetic, linker-encoding oligonucleotides to couple the proteins, or can be encoded by a nucleic acid sequence encoding the fusion protein. Peptide linkers with a degree of flexibility can be used. The linking peptides may have virtually any amino acid sequence, bearing in mind that the preferred linkers will have a sequence that results in a generally flexible peptide. The use of small amino acids, such as glycine and alanine, are of use in creating a flexible peptide. The creation of such sequences is routine to those of skill in the art. A variety of different linkers are commercially available and are considered suitable for use.

(69) Examples of linker polypeptides include glycine polymers (G)_n, glycine-serine polymers (including, for example, (GS)_n, (GSGGS)_n, (SEQ ID NO:1154), (GGSGGS)_n (SEQ ID NO:1155), and (GGGS)_n (SEQ ID NO:1156), where n is an integer of at least one); glycine-alanine polymers; and alanine-serine polymers. Exemplary linkers can comprise amino acid sequences including, but not limited to, GGSG (SEQ ID NO:1157), GGSGG (SEQ ID NO:1158), GSGSG (SEQ ID NO:1159), GSGGG (SEQ ID NO:1160), GGGSG (SEQ ID NO:1161), GSSSG (SEQ ID NO: 1162), and the like. Also suitable is a linker having the sequence (GGGGS)_n, where n is an integer of from 1 to 10 (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10). The ordinarily skilled artisan will recognize that design of a peptide conjugated to any desired element can include linkers that are all or partially flexible, such that the linker can include a flexible linker as well as one or more portions that confer less flexible structure.

(70) A fusion polypeptide of the present disclosure exhibits a high degree of processivity. The processivity of DNA synthesis by a DNA polymerase is defined as the number of nucleotides that a polymerase can incorporate into DNA during a single template-binding event, before dissociating from a DNA template.

(71) In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, exhibits a target mutation rate of from 10^{sup.}-8 to 10^{sup.}-2 mutations per nucleotide per genome replication event. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, exhibits a target mutation rate of greater than 10^{sup.}-8 mutations

per nucleotide per genome replication event, e.g., greater than 10^{-8} , greater than 10^{-7} , greater than 10^{-6} , greater than 10^{-5} , greater than 10^{-4} , or greater than 10^{-3} , mutations per nucleotide per genome replication event. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, exhibits a target mutation rate of from 10^{-8} to 10^{-7} mutations per nucleotide per genome replication event. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, exhibits a target mutation rate of from 10^{-7} to 10^{-6} mutations per nucleotide per genome replication event. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, exhibits a target mutation rate of from 10^{-7} to 10^{-5} mutations per nucleotide per genome replication event. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, exhibits a target mutation rate of from 10^{-5} to 10^{-4} mutations per nucleotide per genome replication event. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, exhibits a target mutation rate of from 10^{-4} to 10^{-3} mutations per nucleotide per genome replication event. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, exhibits a target mutation rate of from 10^{-3} to 10^{-2} mutations per nucleotide per genome replication event.

(72) In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, exhibits a target mutation rate of 1 mutation per nucleotide per genome replication event.

(73) In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, exhibits a ratio of target mutation rate to global mutation rate of at least 1.5:1, at least 2:1, at least 5:1, at least 10:1, at least 25:1, at least 50:1, at least $10^{2.1}$:1, at least $5 \times 10^{2.1}$:1, at least $10^{3.1}$:1, at least $5 \times 10^{3.1}$:1, at least $10^{4.1}$:1, or more than $10^{4.1}$:1. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, exhibits a ratio of target mutation rate to global mutation rate of from about 1.5:1 to $10^{4.1}$:1, e.g., from about 1.5:1 to 2:1, from 2:1 to 5:1, from 5:1 to 10:1, from 10:1 to 25:1, from 25:1 to 50:1, from 50:1 to $10^{2.1}$:1, from $10^{2.1}$:1 to $5 \times 10^{2.1}$:1, from $5 \times 10^{2.1}$:1 to $10^{3.1}$:1, from $10^{3.1}$:1 to $5 \times 10^{3.1}$:1, from $5 \times 10^{3.1}$:1 to $10^{4.1}$:1, or more than $10^{4.1}$:1.

(74) In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, exhibits a target mutation rate that is at least 2-fold higher than the target mutation rate exhibited by the error-prone DNA polymerase present in the fusion polypeptide when the error-prone DNA polymerase is not fused to the RNA-guided endonuclease present in the fusion polypeptide. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, exhibits a target mutation rate that is at least 2-fold, at least 5-fold, at least 10-fold, at least 50-fold, at least $10^{2.1}$ -fold, at least $5 \times 10^{2.1}$ -fold, at least $10^{3.1}$ -fold, at least $5 \times 10^{3.1}$ -fold, or at least $10^{4.1}$ -fold, higher than the target mutation rate exhibited by the error-prone DNA polymerase present in the fusion polypeptide when the error-prone DNA polymerase is not fused to the RNA-guided endonuclease present in the fusion polypeptide. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, exhibits a target mutation rate that is more than $10^{4.1}$ -fold higher than the target mutation rate exhibited by the error-prone DNA polymerase present in the fusion polypeptide when the error-prone DNA polymerase is not fused to the RNA-guided endonuclease present in the fusion polypeptide.

(75) In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, introduces mutations at a distance of from 1 nucleotide to $10^{4.4}$ nucleotides from a nick in a target DNA introduced by the RNA-guided endonuclease. For example, in some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, introduces mutations at a distance of from 1 nucleotide (nt) to 10 nucleotides (nt), from 10 nt to 50 nt, from 50 nt to 100 nt, from 100 nt to 500 nt, from 500 nt to $10^{3.3}$ nt, from $10^{3.3}$ nt to $5 \times 10^{3.3}$ nt, or from $5 \times 10^{3.3}$ nt to $10^{4.4}$ nt from a nick in a target DNA introduced by the RNA-guided

endonuclease. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, introduces mutations at a distance of from 1 nt to 10 nt from a nick in a target DNA introduced by the RNA-guided endonuclease. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, introduces mutations at a distance of from 1 nt to 25 nt from a nick in a target DNA introduced by the RNA-guided endonuclease. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, introduces mutations at a distance of from 10 nt to 25 nt from a nick in a target DNA introduced by the RNA-guided endonuclease. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, introduces mutations at a distance of from 1 nt to 50 nt from a nick in a target DNA introduced by the RNA-guided endonuclease. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, introduces mutations at a distance of from 10 nt to 50 nt from a nick in a target DNA introduced by the RNA-guided endonuclease. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, introduces mutations at a distance of from 25 nt to 50 nt from a nick in a target DNA introduced by the RNA-guided endonuclease. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, introduces mutations at a distance of from 1 nt to 100 nt from a nick in a target DNA introduced by the RNA-guided endonuclease. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, introduces mutations at a distance of from 10 nt to 100 nt from a nick in a target DNA introduced by the RNA-guided endonuclease. In some cases, a fusion polypeptide of the present disclosure, when complexed with a guide RNA, introduces mutations at a distance of from 50 nt to 100 nt from a nick in a target DNA introduced by the RNA-guided endonuclease.

(76) RNA-Guided Endonucleases

(77) A fusion polypeptide of the present disclosure comprises: a) an enzymatically active RNA-guided endonuclease that introduces a single-stranded break in a target DNA; and b) an error-prone DNA polymerase. An RNA-guided endonuclease is also referred to herein as a “genome-editing nuclease.”

(78) Examples of RNA-guided endonucleases are CRISPR/Cas endonucleases (e.g., class 2 CRISPR/Cas endonucleases such as a type II, type V, or type VI CRISPR/Cas endonucleases). A CRISPR/Cas endonuclease is also referred to as a CRISPR/Cas effector polypeptide. A suitable genome editing nuclease is a CRISPR/Cas endonuclease (e.g., a class 2 CRISPR/Cas endonuclease such as a type II, type V, or type VI CRISPR/Cas endonuclease). In some cases, a suitable RNA-guided endonuclease is a class 2 CRISPR/Cas endonuclease. In some cases, a suitable RNA-guided endonuclease is a class 2 type II CRISPR/Cas endonuclease (e.g., a Cas9 protein). In some cases, a genome targeting composition includes a class 2 type V CRISPR/Cas endonuclease (e.g., a Cpf1 protein, a C2c1 protein, or a C2c3 protein). In some cases, a suitable RNA-guided endonuclease is a class 2 type VI CRISPR/Cas endonuclease (e.g., a C2c2 protein; also referred to as a “Cas13a” protein). Also suitable for use is a CasX protein. Also suitable for use is a CasY protein.

(79) In some cases, the genome-editing endonuclease is a Type II CRISPR/Cas endonuclease. In some cases, the genome-editing endonuclease is a Cas9 polypeptide. The Cas9 protein is guided to a target site (e.g., stabilized at a target site) within a target nucleic acid sequence (e.g., a chromosomal sequence or an extrachromosomal sequence, e.g., an episomal sequence, a minicircle sequence, a mitochondrial sequence, a chloroplast sequence, etc.) by virtue of its association with the protein-binding segment of the Cas9 guide RNA. In some cases, a Cas9 polypeptide comprises an amino acid sequence having at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or more than 99%, amino acid sequence identity to the *Streptococcus pyogenes* Cas9 depicted in FIG. 5A. In some cases, a Cas9 polypeptide comprises the amino acid sequence depicted in one of FIG. 5A-5F.

(80) In some cases, the Cas9 polypeptide used in a composition or method of the present disclosure is a *Staphylococcus aureus* Cas9 (saCas9) polypeptide. In some cases, the saCas9 polypeptide

comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the saCas9 amino acid sequence depicted in FIG. 6.

(81) In some cases, the Cas9 polypeptide used in a composition or method of the present disclosure is a *Campylobacter jejuni* Cas9 (CjCas9) polypeptide. CjCas9 recognizes the 5'-NNNVR YM-3' as the protospacer-adjacent motif (PAM). The amino acid sequence of CjCas9 is set forth in SEQ ID NO: 50. In some cases, a Cas9 polypeptide suitable for use in a composition or method of the present disclosure comprises an amino acid sequence having at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or more than 99%, amino acid sequence identity to the CjCas9 amino acid sequence set forth in SEQ ID NO:50.

(82) In some cases, a suitable Cas9 polypeptide is a high-fidelity (HF) Cas9 polypeptide. Kleinstiver et al. (2016) *Nature* 529:490. For example, amino acids N497, R661, Q695, and Q926 of the amino acid sequence depicted in FIG. 5A are substituted, e.g., with alanine. For example, an HF Cas9 polypeptide can comprise an amino acid sequence having at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the amino acid sequence depicted in FIG. 5A, where amino acids N497, R661, Q695, and Q926 are substituted, e.g., with alanine.

(83) In some cases, a suitable Cas9 polypeptide exhibits altered PAM specificity. See, e.g., Kleinstiver et al. (2015) *Nature* 523:481.

(84) In some cases, the genome-editing endonuclease is a type V CRISPR/Cas endonuclease. In some cases a type V CRISPR/Cas endonuclease is a Cpf1 protein. In some cases, a Cpf1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the Cpf1 amino acid sequence depicted in FIG. 7A, FIG. 7B, or FIG. 7C.

(85) In some cases, the genome-editing endonuclease is a CasX or a CasY polypeptide. CasX and CasY polypeptides are described in Burstein et al. (2017) *Nature* 542:237.

(86) RNA-Guided Endonucleases

(87) An RNA-guided endonuclease is also referred to herein as a “genome editing nuclease.” Examples of suitable genome editing nucleases are CRISPR/Cas endonucleases (e.g., class 2 CRISPR/Cas endonucleases such as a type II, type V, or type VI CRISPR/Cas endonucleases). A suitable genome editing nuclease is a CRISPR/Cas endonuclease (e.g., a class 2 CRISPR/Cas endonuclease such as a type II, type V, or type VI CRISPR/Cas endonuclease). In some cases, a genome targeting composition includes a class 2 CRISPR/Cas endonuclease. In some cases, a genome targeting composition includes a class 2 type II CRISPR/Cas endonuclease (e.g., a Cas9 protein). In some cases, a genome targeting composition includes a class 2 type V CRISPR/Cas endonuclease (e.g., a Cpf1 protein, a C2c1 protein, or a C2c3 protein). In some cases, a genome targeting composition includes a class 2 type VI CRISPR/Cas endonuclease (e.g., a C2c2 protein; also referred to as a “Cas13a” protein). Also suitable for use is a CasX protein. Also suitable for use is a CasY protein.

(88) In some cases, a genome editing nuclease is a fusion protein that is fused to a heterologous polypeptide (also referred to as a “fusion partner”). In some cases, a genome editing nuclease is fused to an amino acid sequence (a fusion partner) that provides for subcellular localization, i.e., the fusion partner is a subcellular localization sequence (e.g., one or more nuclear localization signals (NLSs) for targeting to the nucleus, two or more NLSs, three or more NLSs, etc.).

(89) In some cases, the genome-editing endonuclease is a Type II CRISPR/Cas endonuclease. In some cases, the genome-editing endonuclease is a Cas9 polypeptide. The Cas9 protein is guided to a target site (e.g., stabilized at a target site) within a target nucleic acid sequence (e.g., a chromosomal sequence or an extrachromosomal sequence, e.g., an episomal sequence, a minicircle sequence, a mitochondrial sequence, a chloroplast sequence, etc.) by virtue of its association with

the protein-binding segment of the Cas9 guide RNA. In some cases, a Cas9 polypeptide comprises an amino acid sequence having at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, or more than 99%, amino acid sequence identity to the *Streptococcus pyogenes* Cas9 depicted in FIG. 5A. In some cases, the Cas9 polypeptide used in a composition or method of the present disclosure is a *Staphylococcus aureus* Cas9 (saCas9) polypeptide. In some cases, the saCas9 polypeptide comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the saCas9 amino acid sequence depicted in FIG. 6.

(90) In some cases, a suitable Cas9 polypeptide is a high-fidelity (HF) Cas9 polypeptide. Kleinstiver et al. (2016) *Nature* 529:490. For example, amino acids N497, R661, Q695, and Q926 of the amino acid sequence depicted in FIG. 5A are substituted, e.g., with alanine. For example, an HF Cas9 polypeptide can comprise an amino acid sequence having at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the amino acid sequence depicted in FIG. 5A, where amino acids N497, R661, Q695, and Q926 are substituted, e.g., with alanine.

(91) In some cases, a suitable Cas9 polypeptide exhibits altered PAM specificity. See, e.g., Kleinstiver et al. (2015) *Nature* 523:481.

(92) In some cases, the genome-editing endonuclease is a type V CRISPR/Cas endonuclease. In some cases, a type V CRISPR/Cas endonuclease is a Cpf1 protein. In some cases, a Cpf1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the Cpf1 amino acid sequence depicted in FIG. 7A. In some cases, a Cpf1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the Cpf1 amino acid sequence depicted in FIG. 7B. In some cases, a Cpf1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the Cpf1 amino acid sequence depicted in FIG. 7C.

(93) A nucleic acid that binds to a class 2 CRISPR/Cas endonuclease (e.g., a Cas9 protein; a type V or type VI CRISPR/Cas protein; a Cpf1 protein; etc.) and targets the complex to a specific location within a target nucleic acid is referred to herein as a “guide RNA” or “CRISPR/Cas guide nucleic acid” or “CRISPR/Cas guide RNA.” A guide RNA provides target specificity to the complex (the RNP complex) by including a targeting segment, which includes a guide sequence (also referred to herein as a targeting sequence), which is a nucleotide sequence that is complementary to a sequence of a target nucleic acid.

(94) In some cases, a guide RNA includes two separate nucleic acid molecules: an “activator” and a “targeter” and is referred to herein as a “dual guide RNA”, a “double-molecule guide RNA”, a “two-molecule guide RNA”, or a “dgRNA.” In some cases, the guide RNA is one molecule (e.g., for some class 2 CRISPR/Cas proteins, the corresponding guide RNA is a single molecule; and in some cases, an activator and targeter are covalently linked to one another, e.g., via intervening nucleotides), and the guide RNA is referred to as a “single guide RNA”, a “single-molecule guide RNA,” a “one-molecule guide RNA”, or simply “sgRNA.”

(95) In some cases, a composition of the present disclosure comprises an RNA-guided endonuclease, or both an RNA-guided endonuclease and a guide RNA. In some cases, e.g., where a target nucleic acid comprises a deleterious mutation in a defective allele (e.g., a deleterious mutation in a retinal cell target nucleic acid), the RNA-guided endonuclease/guide RNA complex, together with a donor nucleic acid comprising a nucleotide sequence that corrects the deleterious mutation (e.g., a donor nucleic acid comprising a nucleotide sequence that encodes a functional

copy of the protein encoded by the defective allele), can be used to correct the deleterious mutation, e.g., via homology-directed repair (HDR).

(96) In some cases, a composition of the present disclosure comprises: i) an RNA-guided endonuclease; and ii) one guide RNA. In some cases, the guide RNA is a single-molecule (or “single guide”) guide RNA (an “sgRNA”). In some cases, the guide RNA is a dual-molecule (or “dual-guide”) guide RNA (“dgRNA”).

(97) In some cases, a composition of the present disclosure comprises: i) an RNA-guided endonuclease; and ii) 2 separate sgRNAs, where the 2 separate sgRNAs provide for deletion of a target nucleic acid via non-homologous end joining (NHEJ). In some cases, the guide RNAs are sgRNAs. In some cases, the guide RNAs are dgRNAs.

(98) In some cases, a composition of the present disclosure comprises: i) a Cpf1 polypeptide; and ii) a guide RNA precursor; in these cases, the precursor can be cleaved by the Cpf1 polypeptide to generate 2 or more guide RNAs.

(99) Class 2 CRISPR/Cas Endonucleases

(100) RNA-mediated adaptive immune systems in bacteria and archaea rely on Clustered Regularly Interspaced Short Palindromic Repeat (CRISPR) genomic loci and CRISPR-associated (Cas) proteins that function together to provide protection from invading viruses and plasmids. In class 2 CRISPR systems, the functions of the effector complex (e.g., the cleavage of target DNA) are carried out by a single endonuclease (e.g., see Zetsche et al., *Cell*. 2015 Oct. 22; 163 (3): 759-71; Makarova et al., *Nat Rev Microbiol*. 2015 November; 13 (11): 722-36; Shmakov et al., *Mol Cell*. 2015 Nov. 5; 60 (3): 385-97); and Shmakov et al. (2017) *Nature Reviews Microbiology* 15:169. As such, the term “class 2 CRISPR/Cas protein” is used herein to encompass the endonuclease (the target nucleic acid cleaving protein) from class 2 CRISPR systems. Thus, the term “class 2 CRISPR/Cas endonuclease” as used herein encompasses type II CRISPR/Cas proteins (e.g., Cas9); type V-A CRISPR/Cas proteins (e.g., Cpf1 (also referred to as “Cas12a”)); type V-B CRISPR/Cas proteins (e.g., C2c1 (also referred to as “Cas12b”)); type V-C CRISPR/Cas proteins (e.g., C2c3 (also referred to as “Cas12c”)); type V-U1 CRISPR/Cas proteins (e.g., C2c4); type V-U2 CRISPR/Cas proteins (e.g., C2c8); type V-U5 CRISPR/Cas proteins (e.g., C2c5); type V-U4 CRISPR/Cas proteins (e.g., C2c9); type V-U3 CRISPR/Cas proteins (e.g., C2c10); type VI-A CRISPR/Cas proteins (e.g., C2c2 (also known as “Cas13a”)); type VI-B CRISPR/Cas proteins (e.g., Cas13b (also known as C2c4)); and type VI-C CRISPR/Cas proteins (e.g., Cas13c (also known as C2c7)). To date, class 2 CRISPR/Cas proteins encompass type II, type V, and type VI CRISPR/Cas proteins, but the term is also meant to encompass any class 2 CRISPR/Cas protein suitable for binding to a corresponding guide RNA and forming an RNP complex.

(101) Type II CRISPR/Cas Endonucleases (e.g., Cas 9)

(102) In natural Type II CRISPR/Cas systems, Cas9 functions as an RNA-guided endonuclease that uses a dual-guide RNA having a crRNA and trans-activating crRNA (tracrRNA) for target recognition and cleavage by a mechanism involving two nuclease active sites in Cas9 that together generate double-stranded DNA breaks (DSBs), or can individually generate single-stranded DNA breaks (SSBs). The Type II CRISPR endonuclease Cas9 and engineered dual-(dgRNA) or single guide RNA (sgRNA) form a ribonucleoprotein (RNP) complex that can be targeted to a desired DNA sequence. Guided by a dual-RNA complex or a chimeric single-guide RNA, Cas9 generates site-specific DSBs or SSBs within double-stranded DNA (dsDNA) target nucleic acids, which are repaired either by non-homologous end joining (NHEJ) or homology-directed recombination (HDR).

(103) A type II CRISPR/Cas endonuclease is a type of class 2 CRISPR/Cas endonuclease. In some cases, the type II CRISPR/Cas endonuclease is a Cas9 protein. A Cas9 protein forms a complex with a Cas9 guide RNA. The guide RNA provides target specificity to a Cas9-guide RNA complex by having a nucleotide sequence (a guide sequence) that is complementary to a sequence (the target site) of a target nucleic acid (as described elsewhere herein). The Cas9 protein of the complex

provides the site-specific activity. In other words, the Cas9 protein is guided to a target site (e.g., stabilized at a target site) within a target nucleic acid sequence (e.g. a chromosomal sequence or an extrachromosomal sequence, e.g., an episomal sequence, a minicircle sequence, a mitochondrial sequence, a chloroplast sequence, etc.) by virtue of its association with the protein-binding segment of the Cas9 guide RNA.

(104) A Cas9 protein can bind and/or modify (e.g., cleave, nick, methylate, demethylate, etc.) a target nucleic acid and/or a polypeptide associated with target nucleic acid (e.g., methylation or acetylation of a histone tail) (e.g., when the Cas9 protein includes a fusion partner with an activity). In some cases, the Cas9 protein is a naturally-occurring protein (e.g., naturally occurs in bacterial and/or archaeal cells). In other cases, the Cas9 protein is not a naturally-occurring polypeptide (e.g., the Cas9 protein is a variant Cas9 protein, a chimeric protein, and the like).

(105) Examples of suitable Cas9 proteins include, but are not limited to, those set forth in SEQ ID NOs: 5-816. Naturally occurring Cas9 proteins bind a Cas9 guide RNA, are thereby directed to a specific sequence within a target nucleic acid (a target site), and cleave the target nucleic acid (e.g., cleave dsDNA to generate a double strand break, cleave ssDNA, cleave ssRNA, etc.). A chimeric Cas9 protein is a fusion protein comprising a Cas9 polypeptide that is fused to a heterologous protein (referred to as a fusion partner), where the heterologous protein provides an activity (e.g., one that is not provided by the Cas9 protein). The fusion partner can provide an activity, e.g., enzymatic activity (e.g., nuclease activity, activity for DNA and/or RNA methylation, activity for DNA and/or RNA cleavage, activity for histone acetylation, activity for histone methylation, activity for RNA modification, activity for RNA-binding, activity for RNA splicing etc.). In some cases, a portion of the Cas9 protein (e.g., the RuvC domain and/or the HNH domain) exhibits reduced nuclease activity relative to the corresponding portion of a wild type Cas9 protein (e.g., in some cases the Cas9 protein is a nickase). In some cases, the Cas9 protein is enzymatically inactive, or has reduced enzymatic activity relative to a wild-type Cas9 protein (e.g., relative to *Streptococcus pyogenes* Cas9).

(106) In some cases, a fusion protein comprises: a) a catalytically inactive Cas9 protein (or other catalytically inactive CRISPR effector polypeptide); and b) a catalytically active endonuclease. For example, in some cases, the catalytically active endonuclease is a FokI polypeptide. As one non-limiting example, in some cases, a fusion protein comprises: a) a catalytically inactive Cas9 protein (or other catalytically inactive CRISPR effector polypeptide); and b) is a FokI nuclease comprising an amino acid sequence having at least at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the FokI amino acid sequence provided below; where the FokI nuclease has a length of from about 195 amino acids to about 200 amino acids.

(107) FokI nuclease amino acid sequence:

(108) TABLE-US-00001 (SEQ ID NO: 1214)

QLVKSELEEKKSELRHKLKYVPHEYIELIEIARNSTQDRILEMKV
MEFFMKVYGYRGKHLGGSRKPDGAIYTVGSPIDYGVIVDTKAYSG
GYNLPIGQADEMQRYVEENQTRNKHINPNEWVKVYPSSVTEFKFL
FVSGHFKGNYKAQLTRLNHITNCNGAVLSVEELLIGGEMIKAGTL TLEEVRRKFNNGEINF.

(109) Assays to determine whether given protein interacts with a Cas9 guide RNA can be any convenient binding assay that tests for binding between a protein and a nucleic acid. Suitable binding assays (e.g., gel shift assays) will be known to one of ordinary skill in the art (e.g., assays that include adding a Cas9 guide RNA and a protein to a target nucleic acid).

(110) Assays to determine whether a protein has an activity (e.g., to determine if the protein has nuclease activity that cleaves a target nucleic acid and/or some heterologous activity) can be any convenient assay (e.g., any convenient nucleic acid cleavage assay that tests for nucleic acid cleavage). Suitable assays (e.g., cleavage assays) will be known to one of ordinary skill in the art and can include adding a Cas9 guide RNA and a protein to a target nucleic acid.

(111) Many Cas9 orthologs from a wide variety of species have been identified and in some cases the proteins share only a few identical amino acids. Identified Cas9 orthologs have similar domain architecture with a central HNH endonuclease domain and a split RuvC/RNaseH domain (e.g., RuvCI, RuvCII, and RuvCIII) (e.g., see Table 1). For example, a Cas9 protein can have 3 different regions (sometimes referred to as RuvC-I, RuvC-II, and RucC-III), that are not contiguous with respect to the primary amino acid sequence of the Cas9 protein, but fold together to form a RuvC domain once the protein is produced and folds. Thus, Cas9 proteins can be said to share at least 4 key motifs with a conserved architecture. Motifs 1, 2, and 4 are RuvC like motifs while motif 3 is an HNH-motif. The motifs set forth in Table 1 may not represent the entire RuvC-like and/or HNH domains as accepted in the art, but Table 1 does present motifs that can be used to help determine whether a given protein is a Cas9 protein.

(112) TABLE-US-00002 TABLE 1 Table 1 lists 4 motifs that are present in Cas9 sequences from various species. The amino acids listed in Table 1 are from the Cas9 from *S. pyogenes* (SEQ ID NO: 5). Motif Amino acids Highly # Motif (residue #s) conserved 1 RuvC- IGLDIGTNSVG D10, G12, G17 like I WAVI (7-21) (SEQ ID NO: 1) 2 RuvC- IVIEMARE E762 like II (759-766) (SEQ ID NO: 2) 3 HNH- DVDHIVPQS H840, N854, motif FLKDDSIDN N863 KVLTRSDKN (837-863) (SEQ ID NO: 3) 4 RuvC- HHAHDAYL H982, H983, like (982-989) A984, D986, III (SEQ ID A987 NO: 4)

(113) In some cases, a suitable Cas9 protein comprises an amino acid sequence having 4 motifs, each of motifs 1-4 having 60% or more, 70% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, 99% or more or 100% amino acid sequence identity to motifs 1-4 as set forth in SEQ ID NOs: 1-4, respectively (e.g., see Table 1), or to the corresponding portions in any of the amino acid sequences set forth in SEQ ID NOs: 5-816.

(114) In other words, in some cases, a suitable Cas9 polypeptide comprises an amino acid sequence having 4 motifs, each of motifs 1-4 having 60% or more, 70% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, 99% or more or 100% amino acid sequence identity to motifs 1-4 of the Cas9 amino acid sequence set forth in SEQ ID NO: 5 (e.g., the sequences set forth in SEQ ID NOs: 1-4, e.g., see Table 1), or to the corresponding portions in any of the amino acid sequences set forth in SEQ ID NOs: 6-816.

(115) In some cases, a suitable Cas9 protein comprises an amino acid sequence having 4 motifs, each of motifs 1-4 having 60% or more amino acid sequence identity to motifs 1-4 of the Cas9 amino acid sequence set forth as SEQ ID NO: 5 (the motifs are in Table 1, and are set forth as SEQ ID NOs: 1-4, respectively), or to the corresponding portions in any of the amino acid sequences set forth in SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 4 motifs, each of motifs 1-4 having 70% or more amino acid sequence identity to motifs 1-4 of the Cas9 amino acid sequence set forth as SEQ ID NO: 5 (the motifs are in Table 1, and are set forth as SEQ ID NOs: 1-4, respectively), or to the corresponding portions in any of the amino acid sequences set forth in SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 4 motifs, each of motifs 1-4 having 75% or more amino acid sequence identity to motifs 1-4 of the Cas9 amino acid sequence set forth as SEQ ID NO: 5 (the motifs are in Table 1, and are set forth as SEQ ID NOs: 1-4, respectively), or to the corresponding portions in any of the amino acid sequences set forth in SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 4 motifs, each of motifs 1-4 having 80% or more amino acid sequence identity to motifs 1-4 of the Cas9 amino acid sequence set forth as SEQ ID NO: 5 (the motifs are in Table 1, and are set forth as SEQ ID NOs: 1-4, respectively), or to the corresponding portions in any of the amino acid sequences set forth in SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 4 motifs, each of motifs 1-4 having 85% or more amino acid sequence identity to motifs 1-4 of the Cas9 amino acid sequence set forth as SEQ ID NO: 5 (the motifs are in Table 1, and are set

forth as SEQ ID NOs: 1-4, respectively), or to the corresponding portions in any of the amino acid sequences set forth in SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 4 motifs, each of motifs 1-4 having 90% or more amino acid sequence identity to motifs 1-4 of the Cas9 amino acid sequence set forth as SEQ ID NO: 5 (the motifs are in Table 1, and are set forth as SEQ ID NOs: 1-4, respectively), or to the corresponding portions in any of the amino acid sequences set forth in SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 4 motifs, each of motifs 1-4 having 95% or more amino acid sequence identity to motifs 1-4 of the Cas9 amino acid sequence set forth as SEQ ID NO: 5 (the motifs are in Table 1, and are set forth as SEQ ID NOs: 1-4, respectively), or to the corresponding portions in any of the amino acid sequences set forth in SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 4 motifs, each of motifs 1-4 having 99% or more amino acid sequence identity to motifs 1-4 of the Cas9 amino acid sequence set forth as SEQ ID NO: 5 (the motifs are in Table 1, and are set forth as SEQ ID NOs: 1-4, respectively), or to the corresponding portions in any of the amino acid sequences set forth in SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 4 motifs, each of motifs 1-4 having 100% amino acid sequence identity to motifs 1-4 of the Cas9 amino acid sequence set forth as SEQ ID NO: 5 (the motifs are in Table 1, and are set forth as SEQ ID NOs: 1-4, respectively), or to the corresponding portions in any of the amino acid sequences set forth in SEQ ID NOs: 6-816. Any Cas9 protein as defined above can be used as a Cas9 polypeptide, as part of a chimeric Cas9 polypeptide (e.g., a Cas9 fusion protein), any of which can be used in an RNP of the present disclosure.

(116) In some cases, a suitable Cas9 protein comprises an amino acid sequence having 60% or more, 70% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, 99% or more or 100% amino acid sequence identity to amino acids 7-166 or 731-1003 of the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to the corresponding portions in any of the amino acid sequences set forth as SEQ ID NOs: 6-816.

(117) In some cases, a suitable Cas9 protein comprises an amino acid sequence having 60% or more amino acid sequence identity to amino acids 7-166 or 731-1003 of the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to the corresponding portions in any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 70% or more amino acid sequence identity to amino acids 7-166 or 731-1003 of the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to the corresponding portions in any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 75% or more amino acid sequence identity to amino acids 7-166 or 731-1003 of the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to the corresponding portions in any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 80% or more amino acid sequence identity to amino acids 7-166 or 731-1003 of the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to the corresponding portions in any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 85% or more amino acid sequence identity to amino acids 7-166 or 731-1003 of the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to the corresponding portions in any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 90% or more amino acid sequence identity to amino acids 7-166 or 731-1003 of the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to the corresponding portions in any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 95% or more amino acid sequence identity to amino acids 7-166 or 731-1003 of the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to the corresponding portions in any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an

amino acid sequence having 99% or more amino acid sequence identity to amino acids 7-166 or 731-1003 of the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to the corresponding portions in any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 100% amino acid sequence identity to amino acids 7-166 or 731-1003 of the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to the corresponding portions in any of the amino acid sequences set forth as SEQ ID NOs: 6-816. Any Cas9 protein as defined above can be used as a Cas9 polypeptide, as part of a chimeric Cas9 polypeptide (e.g., a Cas9 fusion protein), any of which can be used in an RNP of the present disclosure.

(118) In some cases, a suitable Cas9 protein comprises an amino acid sequence having 60% or more, 70% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, 99% or more or 100% amino acid sequence identity to the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to any of the amino acid sequences set forth as SEQ ID NOs: 6-816.

(119) In some cases, a suitable Cas9 protein comprises an amino acid sequence having 60% or more amino acid sequence identity to the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 70% or more amino acid sequence identity to the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 75% or more amino acid sequence identity to the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 80% or more amino acid sequence identity to the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 85% or more amino acid sequence identity to the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 90% or more amino acid sequence identity to the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 95% or more amino acid sequence identity to the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 99% or more amino acid sequence identity to the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to any of the amino acid sequences set forth as SEQ ID NOs: 6-816. In some cases, a suitable Cas9 protein comprises an amino acid sequence having 100% amino acid sequence identity to the Cas9 amino acid sequence set forth in SEQ ID NO: 5, or to any of the amino acid sequences set forth as SEQ ID NOs: 6-816. Any Cas9 protein as defined above can be used as a Cas9 polypeptide, as part of a chimeric Cas9 polypeptide (e.g., a Cas9 fusion protein), any of which can be used in an RNP of the present disclosure.

(120) In some cases, a Cas9 protein comprises 4 motifs (as listed in Table 1), at least one with (or each with) amino acid sequences having 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, 99% or more or 100% amino acid sequence identity to each of the 4 motifs listed in Table 1 (SEQ ID NOs: 1-4), or to the corresponding portions in any of the amino acid sequences set forth as SEQ ID NOs: 6-816.

(121) Examples of various Cas9 proteins (and Cas9 domain structure) and Cas9 guide RNAs (as well as information regarding requirements related to protospacer adjacent motif (PAM) sequences present in targeted nucleic acids) can be found in the art, for example, see Jinek et al., *Science*. 2012 Aug. 17; 337 (6096): 816-21; Chylinski et al., *RNA Biol.* 2013 May; 10 (5): 726-37; Ma et al., *Biomed Res Int.* 2013; 2013:270805; Hou et al., *Proc Natl Acad Sci USA.* 2013 Sep. 24; 110 (39): 15644-9; Jinek et al., *Elife.* 2013; 2: e00471; Pattanayak et al., *Nat Biotechnol.* 2013

September; 31 (9): 839-43; Qi et al., *Cell*. 2013 Feb. 28; 152 (5): 1173-83; Wang et al., *Cell*. 2013 May 9; 153 (4): 910.sup.-8; Auer et al., *Genome Res*. 2013 Oct. 31; Chen et al., *Nucleic Acids Res*. 2013 Nov. 1; 41 (20): e19; Cheng et al., *Cell Res*. 2013 October; 23 (10): 1163-71; Cho et al., *Genetics*. 2013 November; 195 (3): 1177-80; DiCarlo et al., *Nucleic Acids Res*. 2013 April; 41 (7): 4336-43; Dickinson et al., *Nat Methods*. 2013 October; 10 (10): 1028-34; Ebina et al., *Sci Rep*. 2013; 3:2510; Fujii et al., *Nucleic Acids Res*. 2013 Nov. 1; 41 (20): e187; Hu et al., *Cell Res*. 2013 November; 23 (11): 1322-5; Jiang et al., *Nucleic Acids Res*. 2013 Nov. 1; 41 (20): e188; Larson et al., *Nat Protoc*. 2013 November; 8 (11): 2180-96; Mali et al., *Nat Methods*. 2013 October; 10 (10): 957-63; Nakayama et al., *Genesis*. 2013 December; 51 (12): 835-43; Ran et al., *Nat Protoc*. 2013 November; 8 (11): 2281-308; Ran et al., *Cell*. 2013 Sep. 12; 154 (6): 1380-9; Upadhyay et al., *G3 (Bethesda)*. 2013 Dec. 9; 3 (12): 2233-8; Walsh et al., *Proc Natl Acad Sci USA*. 2013 Sep. 24; 110 (39): 15514-5; Xie et al., *Mol Plant*. 2013 Oct. 9; Yang et al., *Cell*. 2013 Sep. 12; 154 (6): 1370-9; Briner et al., *Mol Cell*. 2014 Oct. 23; 56 (2): 333-9; Shmakov et al., *Nat Rev Microbiol*. 2017 March; 15 (3): 169-182; and U.S. patents and patent applications: U.S. Pat. Nos. 8,906,616; 8,895,308; 8,889,418; 8,889,356; 8,871,445; 8,865,406; 8,795,965; 8,771,945; 8,697,359; 20140068797; 20140170753; 20140179006; 20140179770; 20140186843; 20140186919; 20140186958; 20140189896; 20140227787; 20140234972; 20140242664; 20140242699; 20140242700; 20140242702; 20140248702; 20140256046; 20140273037; 20140273226; 20140273230; 20140273231; 20140273232; 20140273233; 20140273234; 20140273235; 20140287938; 20140295556; 20140295557; 20140298547; 20140304853; 20140309487; 20140310828; 20140310830; 20140315985; 20140335063; 20140335620; 20140342456; 20140342457; 20140342458; 20140349400; 20140349405; 20140356867; 20140356956; 20140356958; 20140356959; 20140357523; 20140357530; 20140364333; and 20140377868; each of which is hereby incorporated by reference in its entirety.

(122) Variant Cas9 Proteins—Nickases and dCas9

(123) In some cases, a Cas9 protein is a variant Cas9 protein. A variant Cas9 protein has an amino acid sequence that is different by at least one amino acid (e.g., has a deletion, insertion, substitution, fusion) when compared to the amino acid sequence of a corresponding wild type Cas9 protein. In some instances, the variant Cas9 protein has an amino acid change (e.g., deletion, insertion, or substitution) that reduces the nuclease activity of the Cas9 protein. For example, in some instances, the variant Cas9 protein has 50% or less, 40% or less, 30% or less, 20% or less, 10% or less, 5% or less, or 1% or less of the nuclease activity of the corresponding wild-type Cas9 protein. In some cases, the variant Cas9 protein has no substantial nuclease activity. When a Cas9 protein is a variant Cas9 protein that has no substantial nuclease activity, it can be referred to as a nuclease defective Cas9 protein or “dCas9” for “dead” Cas9. A protein (e.g., a class 2 CRISPR/Cas protein, e.g., a Cas9 protein) that cleaves one strand but not the other of a double stranded target nucleic acid is referred to herein as a “nickase” (e.g., a “nickase Cas9”).

(124) In some cases, a variant Cas9 protein can cleave the complementary strand (sometimes referred to in the art as the target strand) of a target nucleic acid but has reduced ability to cleave the non-complementary strand (sometimes referred to in the art as the non-target strand) of a target nucleic acid. For example, the variant Cas9 protein can have a mutation (amino acid substitution) that reduces the function of the RuvC domain. Thus, the Cas9 protein can be a nickase that cleaves the complementary strand, but does not cleave the non-complementary strand. As a non-limiting example, in some embodiments, a variant Cas9 protein has a mutation at an amino acid position corresponding to residue D10 (e.g., D10A, aspartate to alanine) of SEQ ID NO: 5 (or the corresponding position of any of the proteins set forth in SEQ ID NOs: 6-261 and 264-816) and can therefore cleave the complementary strand of a double stranded target nucleic acid but has reduced ability to cleave the non-complementary strand of a double stranded target nucleic acid (thus resulting in a single strand break (SSB) instead of a double strand break (DSB) when the variant Cas9 protein cleaves a double stranded target nucleic acid) (see, for example, Jinek et al., *Science*.

2012 Aug. 17; 337 (6096): 816-21). See, e.g., SEQ ID NO: 262.

(125) In some cases, a variant Cas9 protein can cleave the non-complementary strand of a target nucleic acid but has reduced ability to cleave the complementary strand of the target nucleic acid. For example, the variant Cas9 protein can have a mutation (amino acid substitution) that reduces the function of the HNH domain. Thus, the Cas9 protein can be a nickase that cleaves the non-complementary strand, but does not cleave the complementary strand. As a non-limiting example, in some embodiments, the variant Cas9 protein has a mutation at an amino acid position corresponding to residue H840 (e.g., an H840A mutation, histidine to alanine) of SEQ ID NO: 5 (or the corresponding position of any of the proteins set forth as SEQ ID NOs: 6-261 and 264-816) and can therefore cleave the non-complementary strand of the target nucleic acid but has reduced ability to cleave (e.g., does not cleave) the complementary strand of the target nucleic acid. Such a Cas9 protein has a reduced ability to cleave a target nucleic acid (e.g., a single stranded target nucleic acid) but retains the ability to bind a target nucleic acid (e.g., a single stranded target nucleic acid). See, e.g., SEQ ID NO: 263.

(126) In some cases, a variant Cas9 protein has a reduced ability to cleave both the complementary and the non-complementary strands of a double stranded target nucleic acid. As a non-limiting example, in some cases, the variant Cas9 protein harbors mutations at amino acid positions corresponding to residues D10 and H840 (e.g., D10A and H840A) of SEQ ID NO: 5 (or the corresponding residues of any of the proteins set forth as SEQ ID NOs: 6-261 and 264-816) such that the polypeptide has a reduced ability to cleave (e.g., does not cleave) both the complementary and the non-complementary strands of a target nucleic acid. Such a Cas9 protein has a reduced ability to cleave a target nucleic acid (e.g., a single stranded or double stranded target nucleic acid) but retains the ability to bind a target nucleic acid. A Cas9 protein that cannot cleave target nucleic acid (e.g., due to one or more mutations, e.g., in the catalytic domains of the RuvC and HNH domains) is referred to as a “dead” Cas9 or simply “dCas9.” See, e.g., SEQ ID NO: 264.

(127) Other residues can be mutated to achieve the above effects (i.e. inactivate one or the other nuclease portions). As non-limiting examples, residues D10, G12, G17, E762, H840, N854, N863, H982, H983, A984, D986, and/or A987 of SEQ ID NO: 5 (or the corresponding mutations of any of the proteins set forth as SEQ ID NOs: 6-816) can be altered (i.e., substituted). Also, mutations other than alanine substitutions are suitable.

(128) In some embodiments, a variant Cas9 protein that has reduced catalytic activity (e.g., when a Cas9 protein has a D10, G12, G17, E762, H840, N854, N863, H982, H983, A984, D986, and/or a A987 mutation of SEQ ID NO: 5 or the corresponding mutations of any of the proteins set forth as SEQ ID NOs: 6-816, e.g., D10A, G12A, G17A, E762A, H840A, N854A, N863A, H982A, H983A, A984A, and/or D986A), the variant Cas9 protein can still bind to target nucleic acid in a site-specific manner (because it is still guided to a target nucleic acid sequence by a Cas9 guide RNA) as long as it retains the ability to interact with the Cas9 guide RNA.

(129) In addition to the above, a variant Cas9 protein can have the same parameters for sequence identity as described above for Cas9 proteins. Thus, in some cases, a suitable variant Cas9 protein comprises an amino acid sequence having 4 motifs, each of motifs 1-4 having 60% or more, 70% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, 99% or more or 100% amino acid sequence identity to motifs 1-4 of the Cas9 amino acid sequence set forth as SEQ ID NO: 5 (the motifs are in Table 1, above, and are set forth as SEQ ID NOs: 1-4, respectively), or to the corresponding portions in any of the amino acid sequences set forth in SEQ ID NOs: 6-816.

(130) In some cases, a suitable variant Cas9 protein comprises an amino acid sequence having 4 motifs, each of motifs 1-4 having 60% or more amino acid sequence identity to motifs 1-4 of the Cas9 amino acid sequence set forth as SEQ ID NO: 5 (the motifs are in Table 1, above, and are set forth as SEQ ID NOs: 1-4, respectively), or to the corresponding portions in any of the amino acid sequences set forth in SEQ ID NOs: 6-816. In some cases, a suitable variant Cas9 protein comprises an amino acid sequence having 4 motifs, each of motifs 1-4 having 70% or more amino

(134) Type V and Type VI CRISPR/Cas Endonucleases

(135) In some cases, a genome targeting composition of the present disclosure includes a type V or type VI CRISPR/Cas endonuclease (i.e., the genome editing endonuclease is a type V or type VI CRISPR/Cas endonuclease) (e.g., Cpf1, C2c1, C2c2, C2c3). Type V and type VI CRISPR/Cas endonucleases are a type of class 2 CRISPR/Cas endonuclease. Examples of type V CRISPR/Cas endonucleases include but are not limited to: Cpf1, C2c1, and C2c3. An example of a type VI CRISPR/Cas endonuclease is C2c2. In some cases, a subject genome targeting composition includes a type V CRISPR/Cas endonuclease (e.g., Cpf1, C2c1, C2c3). In some cases, a Type V CRISPR/Cas endonuclease is a Cpf1 protein. In some cases, a subject genome targeting composition includes a type VI CRISPR/Cas endonuclease (e.g., Cas13a).

(136) Like type II CRISPR/Cas endonucleases, type V and VI CRISPR/Cas endonucleases form a complex with a corresponding guide RNA. The guide RNA provides target specificity to an endonuclease-guide RNA RNP complex by having a nucleotide sequence (a guide sequence) that is complementary to a sequence (the target site) of a target nucleic acid (as described elsewhere herein). The endonuclease of the complex provides the site-specific activity. In other words, the endonuclease is guided to a target site (e.g., stabilized at a target site) within a target nucleic acid sequence (e.g. a chromosomal sequence or an extrachromosomal sequence, e.g., an episomal sequence, a minicircle sequence, a mitochondrial sequence, a chloroplast sequence, etc.) by virtue of its association with the protein-binding segment of the guide RNA.

(137) Examples and guidance related to type V and type VI CRISPR/Cas proteins (e.g., Cpf1, C2c1, C2c2, and C2c3 guide RNAs) can be found in the art, for example, see Zetsche et al., *Cell*. 2015 Oct. 22; 163 (3): 759-71; Makarova et al., *Nat Rev Microbiol*. 2015 November; 13 (11): 722-36; Shmakov et al., *Mol Cell*. 2015 Nov. 5; 60 (3): 385-97; and Shmakov et al. (2017) *Nature Reviews Microbiology* 15:169.

(138) In some cases, the Type V or type VI CRISPR/Cas endonuclease (e.g., Cpf1, C2c1, C2c2, C2c3) is enzymatically active, e.g., the Type V or type VI CRISPR/Cas polypeptide, when bound to a guide RNA, cleaves a target nucleic acid. In some cases, the Type V or type VI CRISPR/Cas endonuclease (e.g., Cpf1, C2c1, C2c2, C2c3) exhibits reduced enzymatic activity relative to a corresponding wild-type a Type V or type VI CRISPR/Cas endonuclease (e.g., Cpf1, C2c1, C2c2, C2c3), and retains DNA binding activity.

(139) In some cases a type V CRISPR/Cas endonuclease is a Cpf1 protein. In some cases, a Cpf1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the Cpf1 amino acid sequence set forth in any of SEQ ID NOs: 818-822. In some cases, a Cpf1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to a contiguous stretch of from 100 amino acids to 200 amino acids (aa), from 200 aa to 400 aa, from 400 aa to 600 aa, from 600 aa to 800 aa, from 800 aa to 1000 aa, from 1000 aa to 1100 aa, from 1100 aa to 1200 aa, or from 1200 aa to 1300 aa, of the Cpf1 amino acid sequence set forth in any of SEQ ID NOs: 818-822.

(140) In some cases, a Cpf1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCI domain of the Cpf1 amino acid sequence set forth in any of SEQ ID NOs: 818-822. In some cases, a Cpf1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCII domain of the Cpf1 amino acid sequence set

forth in any of SEQ ID NOs: 818-822. In some cases, a Cpf1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCIII domain of the Cpf1 amino acid sequence set forth in any of SEQ ID NOs: 818-822. In some cases, a Cpf1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCI, RuvCII, and RuvCIII domains of the Cpf1 amino acid sequence set forth in any of SEQ ID NOs: 818-822.

(141) In some cases, the Cpf1 protein exhibits reduced enzymatic activity relative to a wild-type Cpf1 protein (e.g., relative to a Cpf1 protein comprising the amino acid sequence set forth in any of SEQ ID NOs: 818-822), and retains DNA binding activity. In some cases, a Cpf1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the Cpf1 amino acid sequence set forth in any of SEQ ID NOs: 818-822; and comprises an amino acid substitution (e.g., a D.fwdarw.A substitution) at an amino acid residue corresponding to amino acid 917 of the Cpf1 amino acid sequence set forth in SEQ ID NO: 818. In some cases, a Cpf1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the Cpf1 amino acid sequence set forth in any of SEQ ID NOs: 818-822; and comprises an amino acid substitution (e.g., an E.fwdarw.A substitution) at an amino acid residue corresponding to amino acid 1006 of the Cpf1 amino acid sequence set forth in SEQ ID NO: 818. In some cases, a Cpf1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the Cpf1 amino acid sequence set forth in any of SEQ ID NOs: 818-822; and comprises an amino acid substitution (e.g., a D.fwdarw.A substitution) at an amino acid residue corresponding to amino acid 1255 of the Cpf1 amino acid sequence set forth in SEQ ID NO: 818.

(142) In some cases, a suitable Cpf1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the Cpf1 amino acid sequence set forth in any of SEQ ID NOs: 818-822.

(143) In some cases a type V CRISPR/Cas endonuclease is a C2c1 protein (examples include those set forth as SEQ ID NOs: 823-830). In some cases, a C2c1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the C2c1 amino acid sequence set forth in any of SEQ ID NOs: 823-830. In some cases, a C2c1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to a contiguous stretch of from 100 amino acids to 200 amino acids (aa), from 200 aa to 400 aa, from 400 aa to 600 aa, from 600 aa to 800 aa, from 800 aa to 1000 aa, from 1000 aa to 1100 aa, from 1100 aa to 1200 aa, or from 1200 aa to 1300 aa, of the C2c1 amino acid sequence set forth in any of SEQ ID NOs: 823-830.

(144) In some cases, a C2c1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%,

at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCI domain of the C2c1 amino acid sequences set forth in any of SEQ ID NOs: 823-830). In some cases, a C2c1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCII domain of the C2c1 amino acid sequence set forth in any of SEQ ID NOs: 823-830. In some cases, a C2c1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCIII domain of the C2c1 amino acid sequence set forth in any of SEQ ID NOs: 823-830. In some cases, a C2c1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCI, RuvCII, and RuvCIII domains of the C2c1 amino acid sequence set forth in any of SEQ ID NOs: 823-830.

(145) In some cases, the C2c1 protein exhibits reduced enzymatic activity relative to a wild-type C2c1 protein (e.g., relative to a C2c1 protein comprising the amino acid sequence set forth in any of SEQ ID NOs: 823-830), and retains DNA binding activity. In some cases, a suitable C2c1 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the C2c1 amino acid sequence set forth in any of SEQ ID NOs: 823-830.

(146) In some cases a type V CRISPR/Cas endonuclease is a C2c3 protein (examples include those set forth as SEQ ID NOs: 831-834). In some cases, a C2c3 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the C2c3 amino acid sequence set forth in any of SEQ ID NOs: 831-834. In some cases, a C2c3 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to a contiguous stretch of from 100 amino acids to 200 amino acids (aa), from 200 aa to 400 aa, from 400 aa to 600 aa, from 600 aa to 800 aa, from 800 aa to 1000 aa, from 1000 aa to 1100 aa, from 1100 aa to 1200 aa, or from 1200 aa to 1300 aa, of the C2c3 amino acid sequence set forth in any of SEQ ID NOs: 831-834.

(147) In some cases, a C2c3 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCI domain of the C2c3 amino acid sequence set forth in any of SEQ ID NOs: 831-834. In some cases, a C2c3 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCII domain of the C2c3 amino acid sequence set forth in any of SEQ ID NOs: 831-834. In some cases, a C2c3 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCIII domain of the C2c3 amino acid sequence set forth in any of SEQ ID NOs: 831-834. In some cases, a C2c3 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least

85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCI, RuvCII, and RuvCIII domains of the C2c3 amino acid sequence set forth in any of SEQ ID NOs: 831-834.

(148) In some cases, the C2c3 protein exhibits reduced enzymatic activity relative to a wild-type C2c3 protein (e.g., relative to a C2c3 protein comprising the amino acid sequence set forth in any of SEQ ID NOs: 831-834), and retains DNA binding activity. In some cases, a suitable C2c3 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the C2c3 amino acid sequence set forth in any of SEQ ID NOs: 831-834.

(149) In some cases a type VI CRISPR/Cas endonuclease is a C2c2 protein (examples include those set forth as SEQ ID NOs: 835-846). In some cases, a C2c2 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the C2c2 amino acid sequence set forth in any of SEQ ID NOs: 835-846. In some cases, a C2c2 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to a contiguous stretch of from 100 amino acids to 200 amino acids (aa), from 200 aa to 400 aa, from 400 aa to 600 aa, from 600 aa to 800 aa, from 800 aa to 1000 aa, from 1000 aa to 1100 aa, from 1100 aa to 1200 aa, or from 1200 aa to 1300 aa, of the C2c2 amino acid sequence set forth in any of SEQ ID NOs: 835-846.

(150) In some cases, a C2c2 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCI domain of the C2c2 amino acid sequence set forth in any of SEQ ID NOs: 835-846. In some cases, a C2c2 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCII domain of the C2c2 amino acid sequence set forth in any of SEQ ID NOs: 835-846. In some cases, a C2c2 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCIII domain of the C2c2 amino acid sequence set forth in any of SEQ ID NOs: 835-846. In some cases, a C2c2 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the RuvCI, RuvCII, and RuvCIII domains of the C2c2 amino acid sequence set forth in any of SEQ ID NOs: 835-846.

(151) In some cases, the C2c2 protein exhibits reduced enzymatic activity relative to a wild-type C2c2 protein (e.g., relative to a C2c2 protein comprising the amino acid sequence set forth in any of SEQ ID NOs: 835-846), and retains DNA binding activity. In some cases, a suitable C2c2 protein comprises an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the C2c2 amino acid sequence set forth in any of SEQ ID NOs: 835-846.

(152) Examples and guidance related to type V or type VI CRISPR/Cas endonucleases (including domain structure) and guide RNAs (as well as information regarding requirements related to protospacer adjacent motif (PAM) sequences present in targeted nucleic acids) can be found in the

art, for example, see Zetsche et al., *Cell*. 2015 Oct. 22; 163 (3): 759-71; Makarova et al., *Nat Rev Microbiol*. 2015 November; 13 (11): 722-36; Shmakov et al., *Mol Cell*. 2015 Nov. 5; 60 (3): 385-97; and Shmakov et al., *Nat Rev Microbiol*. 2017 March; 15 (3): 169-182; and U.S. patents and patent applications: 9,580,701; 20170073695, 20170058272, 20160362668, 20160362667, 20160298078, 20160289637, 20160215300, 20160208243, and 20160208241, each of which is hereby incorporated by reference in its entirety.

(153) CasX and Cas Y Proteins

(154) Suitable RNA-guided endonucleases include CasX and CasY proteins. See, e.g., Burstein et al. (2017) *Nature* 542:237.

(155) Cas9 Guide RNA

(156) A nucleic acid molecule that binds to a Cas9 protein and targets the complex to a specific location within a target nucleic acid is referred to herein as a “Cas9 guide RNA.”

(157) A Cas9 guide RNA (can be said to include two segments, a first segment (referred to herein as a “targeting segment”); and a second segment (referred to herein as a “protein-binding segment”). By “segment” it is meant a segment/section/region of a molecule, e.g., a contiguous stretch of nucleotides in a nucleic acid molecule. A segment can also mean a region/section of a complex such that a segment may comprise regions of more than one molecule.

(158) The first segment (targeting segment) of a Cas9 guide RNA includes a nucleotide sequence (a guide sequence) that is complementary to (and therefore hybridizes with) a specific sequence (a target site) within a target nucleic acid (e.g., a target ssRNA, a target ssDNA, the complementary strand of a double stranded target DNA, etc.). The protein-binding segment (or “protein-binding sequence”) interacts with (binds to) a Cas9 polypeptide. The protein-binding segment of a subject Cas9 guide RNA includes two complementary stretches of nucleotides that hybridize to one another to form a double stranded RNA duplex (dsRNA duplex). Site-specific binding and/or cleavage of a target nucleic acid (e.g., genomic DNA) can occur at locations (e.g., target sequence of a target locus) determined by base-pairing complementarity between the Cas9 guide RNA (the guide sequence of the Cas9 guide RNA) and the target nucleic acid.

(159) A Cas9 guide RNA and a Cas9 protein form a complex (e.g., bind via non-covalent interactions). The Cas9 guide RNA provides target specificity to the complex by including a targeting segment, which includes a guide sequence (a nucleotide sequence that is complementary to a sequence of a target nucleic acid). The Cas9 protein of the complex provides the site-specific activity (e.g., cleavage activity or an activity provided by the Cas9 protein when the Cas9 protein is a Cas9 fusion polypeptide, i.e., has a fusion partner). In other words, the Cas9 protein is guided to a target nucleic acid sequence (e.g. a target sequence in a chromosomal nucleic acid, e.g., a chromosome; a target sequence in an extrachromosomal nucleic acid, e.g. an episomal nucleic acid, a minicircle, an ssRNA, an ssDNA, etc.; a target sequence in a mitochondrial nucleic acid; a target sequence in a chloroplast nucleic acid; a target sequence in a plasmid; a target sequence in a viral nucleic acid; etc.) by virtue of its association with the Cas9 guide RNA.

(160) The “guide sequence” also referred to as the “targeting sequence” of a Cas9 guide RNA can be modified so that the Cas9 guide RNA can target a Cas9 protein to any desired sequence of any desired target nucleic acid, with the exception that the protospacer adjacent motif (PAM) sequence can be taken into account. Thus, for example, a Cas9 guide RNA can have a targeting segment with a sequence (a guide sequence) that has complementarity with (e.g., can hybridize to) a sequence in a nucleic acid in a eukaryotic cell, e.g., a viral nucleic acid, a eukaryotic nucleic acid (e.g., a eukaryotic chromosome, chromosomal sequence, a eukaryotic RNA, etc.), and the like.

(161) In some embodiments, a Cas9 guide RNA includes two separate nucleic acid molecules: an “activator” and a “targeter” and is referred to herein as a “dual Cas9 guide RNA”, a “double-molecule Cas9 guide RNA”, or a “two-molecule Cas9 guide RNA” a “dual guide RNA”, or a “dgRNA.” In some embodiments, the activator and targeter are covalently linked to one another (e.g., via intervening nucleotides) and the guide RNA is referred to as a “single guide RNA”, a

“Cas9 single guide RNA”, a “single-molecule Cas9 guide RNA,” or a “one-molecule Cas9 guide RNA”, or simply “sgRNA.”

(162) A Cas9 guide RNA comprises a crRNA-like (“CRISPR RNA”/“targeter”/“crRNA”/“crRNA repeat”) molecule and a corresponding tracrRNA-like (“trans-acting CRISPR RNA”/“activator”/“tracrRNA”) molecule. A crRNA-like molecule (targeter) comprises both the targeting segment (single stranded) of the Cas9 guide RNA and a stretch (“duplex-forming segment”) of nucleotides that forms one half of the dsRNA duplex of the protein-binding segment of the Cas9 guide RNA. A corresponding tracrRNA-like molecule (activator/tracrRNA) comprises a stretch of nucleotides (duplex-forming segment) that forms the other half of the dsRNA duplex of the protein-binding segment of the guide nucleic acid. In other words, a stretch of nucleotides of a crRNA-like molecule are complementary to and hybridize with a stretch of nucleotides of a tracrRNA-like molecule to form the dsRNA duplex of the protein-binding domain of the Cas9 guide RNA. As such, each targeter molecule can be said to have a corresponding activator molecule (which has a region that hybridizes with the targeter). The targeter molecule additionally provides the targeting segment. Thus, a targeter and an activator molecule (as a corresponding pair) hybridize to form a Cas9 guide RNA. The exact sequence of a given crRNA or tracrRNA molecule is characteristic of the species in which the RNA molecules are found. A subject dual Cas9 guide RNA can include any corresponding activator and targeter pair.

(163) The term “activator” or “activator RNA” is used herein to mean a tracrRNA-like molecule (tracrRNA: “trans-acting CRISPR RNA”) of a Cas9 dual guide RNA (and therefore of a Cas9 single guide RNA when the “activator” and the “targeter” are linked together by, e.g., intervening nucleotides). Thus, for example, a Cas9 guide RNA (dgRNA or sgRNA) comprises an activator sequence (e.g., a tracrRNA sequence). A tracr molecule (a tracrRNA) is a naturally existing molecule that hybridizes with a CRISPR RNA molecule (a crRNA) to form a Cas9 dual guide RNA. The term “activator” is used herein to encompass naturally existing tracrRNAs, but also to encompass tracrRNAs with modifications (e.g., truncations, sequence variations, base modifications, backbone modifications, linkage modifications, etc.) where the activator retains at least one function of a tracrRNA (e.g., contributes to the dsRNA duplex to which Cas9 protein binds). In some cases the activator provides one or more stem loops that can interact with Cas9 protein. An activator can be referred to as having a tracr sequence (tracrRNA sequence) and in some cases is a tracrRNA, but the term “activator” is not limited to naturally existing tracrRNAs.

(164) The term “targeter” or “targeter RNA” is used herein to refer to a crRNA-like molecule (crRNA: “CRISPR RNA”) of a Cas9 dual guide RNA (and therefore of a Cas9 single guide RNA when the “activator” and the “targeter” are linked together, e.g., by intervening nucleotides). Thus, for example, a Cas9 guide RNA (dgRNA or sgRNA) comprises a targeting segment (which includes nucleotides that hybridize with (are complementary to) a target nucleic acid, and a duplex-forming segment (e.g., a duplex forming segment of a crRNA, which can also be referred to as a crRNA repeat). Because the sequence of a targeting segment (the segment that hybridizes with a target sequence of a target nucleic acid) of a targeter is modified by a user to hybridize with a desired target nucleic acid, the sequence of a targeter will often be a non-naturally occurring sequence. However, the duplex-forming segment of a targeter (described in more detail below), which hybridizes with the duplex-forming segment of an activator, can include a naturally existing sequence (e.g., can include the sequence of a duplex-forming segment of a naturally existing crRNA, which can also be referred to as a crRNA repeat). Thus, the term targeter is used herein to distinguish from naturally occurring crRNAs, despite the fact that part of a targeter (e.g., the duplex-forming segment) often includes a naturally occurring sequence from a crRNA. However, the term “targeter” encompasses naturally occurring crRNAs.

(165) A Cas9 guide RNA can also be said to include 3 parts: (i) a targeting sequence (a nucleotide sequence that hybridizes with a sequence of the target nucleic acid); (ii) an activator sequence (as described above) (in some cases, referred to as a tracr sequence); and (iii) a sequence that

hybridizes to at least a portion of the activator sequence to form a double stranded duplex. A targeter has (i) and (iii); while an activator has (ii).

(166) A Cas9 guide RNA (e.g. a dual guide RNA or a single guide RNA) can be comprised of any corresponding activator and targeter pair. In some cases, the duplex forming segments can be swapped between the activator and the targeter. In other words, in some cases, the targeter includes a sequence of nucleotides from a duplex forming segment of a tracrRNA (which sequence would normally be part of an activator) while the activator includes a sequence of nucleotides from a duplex forming segment of a crRNA (which sequence would normally be part of a targeter).

(167) As noted above, a targeter comprises both the targeting segment (single stranded) of the Cas9 guide RNA and a stretch (“duplex-forming segment”) of nucleotides that forms one half of the dsRNA duplex of the protein-binding segment of the Cas9 guide RNA. A corresponding tracrRNA-like molecule (activator) comprises a stretch of nucleotides (a duplex-forming segment) that forms the other half of the dsRNA duplex of the protein-binding segment of the Cas9 guide RNA. In other words, a stretch of nucleotides of the targeter is complementary to and hybridizes with a stretch of nucleotides of the activator to form the dsRNA duplex of the protein-binding segment of a Cas9 guide RNA. As such, each targeter can be said to have a corresponding activator (which has a region that hybridizes with the targeter). The targeter molecule additionally provides the targeting segment. Thus, a targeter and an activator (as a corresponding pair) hybridize to form a Cas9 guide RNA. The particular sequence of a given naturally existing crRNA or tracrRNA molecule is characteristic of the species in which the RNA molecules are found. Examples of suitable activator and targeter are well known in the art.

(168) A Cas9 guide RNA (e.g. a dual guide RNA or a single guide RNA) can be comprised of any corresponding activator and targeter pair. Non-limiting examples of nucleotide sequences that can be included in a Cas9 guide RNA (dgRNA or sgRNA) include sequences set forth in SEQ ID NOs: 847-1095, or complements thereof. For example, in some cases, sequences from SEQ ID NOs: 847-977 (which are from tracrRNAs) or complements thereof, can pair with sequences from SEQ ID NOs: 867-1095 (which are from crRNAs), or complements thereof, to form a dsRNA duplex of a protein binding segment.

(169) Targeting Segment of a Cas9 Guide RNA

(170) The first segment of a subject guide nucleic acid includes a guide sequence (i.e., a targeting sequence) (a nucleotide sequence that is complementary to a sequence (a target site) in a target nucleic acid). In other words, the targeting segment of a subject guide nucleic acid can interact with a target nucleic acid (e.g., double stranded DNA (dsDNA)) in a sequence-specific manner via hybridization (i.e., base pairing). As such, the nucleotide sequence of the targeting segment may vary (depending on the target) and can determine the location within the target nucleic acid that the Cas9 guide RNA and the target nucleic acid will interact. The targeting segment of a Cas9 guide RNA can be modified (e.g., by genetic engineering)/designed to hybridize to any desired sequence (target site) within a target nucleic acid (e.g., a eukaryotic target nucleic acid such as genomic DNA).

(171) The targeting segment can have a length of 7 or more nucleotides (nt) (e.g., 8 or more, 9 or more, 10 or more, 12 or more, 15 or more, 20 or more, 25 or more, 30 or more, or 40 or more nucleotides). In some cases, the targeting segment can have a length of from 7 to 100 nucleotides (nt) (e.g., from 7 to 80 nt, from 7 to 60 nt, from 7 to 40 nt, from 7 to 30 nt, from 7 to 25 nt, from 7 to 22 nt, from 7 to 20 nt, from 7 to 18 nt, from 8 to 80 nt, from 8 to 60 nt, from 8 to 40 nt, from 8 to 30 nt, from 8 to 25 nt, from 8 to 22 nt, from 8 to 20 nt, from 8 to 18 nt, from 10 to 100 nt, from 10 to 80 nt, from 10 to 60 nt, from 10 to 40 nt, from 10 to 30 nt, from 10 to 25 nt, from 10 to 22 nt, from 10 to 20 nt, from 10 to 18 nt, from 12 to 100 nt, from 12 to 80 nt, from 12 to 60 nt, from 12 to 40 nt, from 12 to 30 nt, from 12 to 25 nt, from 12 to 22 nt, from 12 to 20 nt, from 12 to 18 nt, from 14 to 100 nt, from 14 to 80 nt, from 14 to 60 nt, from 14 to 40 nt, from 14 to 30 nt, from 14 to 25 nt, from 14 to 22 nt, from 14 to 20 nt, from 14 to 18 nt, from 16 to 100 nt, from 16 to 80 nt, from

16 to 60 nt, from 16 to 40 nt, from 16 to 30 nt, from 16 to 25 nt, from 16 to 22 nt, from 16 to 20 nt, from 16 to 18 nt, from 18 to 100 nt, from 18 to 80 nt, from 18 to 60 nt, from 18 to 40 nt, from 18 to 30 nt, from 18 to 25 nt, from 18 to 22 nt, or from 18 to 20 nt).

(172) The nucleotide sequence (the targeting sequence) of the targeting segment that is complementary to a nucleotide sequence (target site) of the target nucleic acid can have a length of 10 nt or more. For example, the targeting sequence of the targeting segment that is complementary to a target site of the target nucleic acid can have a length of 12 nt or more, 15 nt or more, 18 nt or more, 19 nt or more, or 20 nt or more. In some cases, the nucleotide sequence (the targeting sequence) of the targeting segment that is complementary to a nucleotide sequence (target site) of the target nucleic acid has a length of 12 nt or more. In some cases, the nucleotide sequence (the targeting sequence) of the targeting segment that is complementary to a nucleotide sequence (target site) of the target nucleic acid has a length of 18 nt or more.

(173) For example, the targeting sequence of the targeting segment that is complementary to a target sequence of the target nucleic acid can have a length of from 10 to 100 nucleotides (nt) (e.g., from 10 to 90 nt, from 10 to 75 nt, from 10 to 60 nt, from 10 to 50 nt, from 10 to 35 nt, from 10 to 30 nt, from 10 to 25 nt, from 10 to 22 nt, from 10 to 20 nt, from 12 to 100 nt, from 12 to 90 nt, from 12 to 75 nt, from 12 to 60 nt, from 12 to 50 nt, from 12 to 35 nt, from 12 to 30 nt, from 12 to 25 nt, from 12 to 22 nt, from 12 to 20 nt, from 15 to 100 nt, from 15 to 90 nt, from 15 to 75 nt, from 15 to 60 nt, from 15 to 50 nt, from 15 to 35 nt, from 15 to 30 nt, from 15 to 25 nt, from 15 to 22 nt, from 15 to 20 nt, from 17 to 100 nt, from 17 to 90 nt, from 17 to 75 nt, from 17 to 60 nt, from 17 to 50 nt, from 17 to 35 nt, from 17 to 30 nt, from 17 to 25 nt, from 17 to 22 nt, from 17 to 20 nt, from 18 to 100 nt, from 18 to 90 nt, from 18 to 75 nt, from 18 to 60 nt, from 18 to 50 nt, from 18 to 35 nt, from 18 to 30 nt, from 18 to 25 nt, from 18 to 22 nt, or from 18 to 20 nt). In some cases, the targeting sequence of the targeting segment that is complementary to a target sequence of the target nucleic acid has a length of from 15 nt to 30 nt. In some cases, the targeting sequence of the targeting segment that is complementary to a target sequence of the target nucleic acid has a length of from 15 nt to 25 nt. In some cases, the targeting sequence of the targeting segment that is complementary to a target sequence of the target nucleic acid has a length of from 18 nt to 30 nt. In some cases, the targeting sequence of the targeting segment that is complementary to a target sequence of the target nucleic acid has a length of from 18 nt to 25 nt. In some cases, the targeting sequence of the targeting segment that is complementary to a target sequence of the target nucleic acid has a length of from 18 nt to 22 nt. In some cases, the targeting sequence of the targeting segment that is complementary to a target site of the target nucleic acid is 20 nucleotides in length. In some cases, the targeting sequence of the targeting segment that is complementary to a target site of the target nucleic acid is 19 nucleotides in length.

(174) The percent complementarity between the targeting sequence (guide sequence) of the targeting segment and the target site of the target nucleic acid can be 60% or more (e.g., 65% or more, 70% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, 97% or more, 98% or more, 99% or more, or 100%). In some cases, the percent complementarity between the targeting sequence of the targeting segment and the target site of the target nucleic acid is 100% over the seven contiguous 5'-most nucleotides of the target site of the target nucleic acid. In some cases, the percent complementarity between the targeting sequence of the targeting segment and the target site of the target nucleic acid is 60% or more over about 20 contiguous nucleotides. In some cases, the percent complementarity between the targeting sequence of the targeting segment and the target site of the target nucleic acid is 100% over the fourteen contiguous 5'-most nucleotides of the target site of the target nucleic acid and as low as 0% or more over the remainder. In such a case, the targeting sequence can be considered to be 14 nucleotides in length. In some cases, the percent complementarity between the targeting sequence of the targeting segment and the target site of the target nucleic acid is 100% over the seven contiguous 5'-most nucleotides of the target site of the target nucleic acid and as low as 0% or more over the remainder. In such a case, the targeting

sequence can be considered to be 12 nucleotides in length. In some cases, the percent complementarity between the targeting sequence of the targeting segment and the target site of the target nucleic acid is 100% over the 13 contiguous 5'-most nucleotides of the target site of the target nucleic acid and as low as 0% or more over the remainder. In such a case, the targeting sequence can be considered to be 13 nucleotides in length. In some cases, the percent complementarity between the targeting sequence of the targeting segment and the target site of the target nucleic acid is 100% over the 14 contiguous 5'-most nucleotides of the target site of the target nucleic acid and as low as 0% or more over the remainder. In such a case, the targeting sequence can be considered to be 14 nucleotides in length. In some cases, the percent complementarity between the targeting sequence of the targeting segment and the target site of the target nucleic acid is 100% over the 17 contiguous 5'-most nucleotides of the target site of the target nucleic acid and as low as 0% or more over the remainder. In such a case, the targeting sequence can be considered to be 17 nucleotides in length. In some cases, the percent complementarity between the targeting sequence of the targeting segment and the target site of the target nucleic acid is 100% over the 18 contiguous 5'-most nucleotides of the target site of the target nucleic acid and as low as 0% or more over the remainder. In such a case, the targeting sequence can be considered to be 18 nucleotides in length.

(177) Examples of various Cas9 proteins and Cas9 guide RNAs (as well as information regarding requirements related to protospacer adjacent motif (PAM) sequences present in targeted nucleic acids) can be found in the art, for example, see Jinek et al., *Science*. 2012 August 17; 337 (6096): 816-21; Chylinski et al., *RNA Biol.* 2013 May; 10 (5): 726-37; Ma et al., *Biomed Res Int.* 2013; 2013:270805; Hou et al., *Proc Natl Acad Sci USA.* 2013 Sep. 24; 110 (39): 15644-9; Jinek et al., *Elife.* 2013; 2: e00471; Pattanayak et al., *Nat Biotechnol.* 2013 September; 31 (9): 839-43; Qi et al., *Cell.* 2013 Feb. 28; 152 (5): 1173-83; Wang et al., *Cell.* 2013 May 9; 153 (4): 910.sup.-8; Auer et al., *Genome Res.* 2013 Oct. 31; Chen et al., *Nucleic Acids Res.* 2013 Nov. 1; 41 (20): e19; Cheng et al., *Cell Res.* 2013 October; 23 (10): 1163-71; Cho et al., *Genetics.* 2013 November; 195 (3): 1177-80; DiCarlo et al., *Nucleic Acids Res.* 2013 April; 41 (7): 4336-43; Dickinson et al., *Nat Methods.* 2013 October; 10 (10): 1028-34; Ebina et al., *Sci Rep.* 2013; 3:2510; Fujii et al., *Nucleic Acids Res.* 2013 Nov. 1; 41 (20): e187; Hu et al., *Cell Res.* 2013 November; 23 (11): 1322-5; Jiang et al., *Nucleic Acids Res.* 2013 Nov. 1; 41 (20): e188; Larson et al., *Nat Protoc.* 2013 November; 8 (11): 2180-96; Mali et al., *Nat Methods.* 2013 October; 10 (10): 957-63; Nakayama et al., *Genesis.* 2013 December; 51 (12): 835-43; Ran et al., *Nat Protoc.* 2013 November; 8 (11): 2281-308; Ran et al., *Cell.* 2013 Sep. 12; 154 (6): 1380-9; Upadhyay et al., *G3 (Bethesda).* 2013 Dec. 9; 3 (12): 2233-8; Walsh et al., *Proc Natl Acad Sci USA.* 2013 Sep. 24; 110 (39): 15514-5; Xie et al., *Mol Plant.* 2013 Oct. 9; Yang et al., *Cell.* 2013 Sep. 12; 154 (6): 1370-9; Briner et al., *Mol Cell.* 2014 Oct. 23; 56 (2): 333-9; and U.S. patents and patent applications: U.S. Pat. Nos. 8,906,616; 8,895,308; 8,889,418; 8,889,356; 8,871,445; 8,865,406; 8,795,965; 8,771,945; 8,697,359; 20140068797; 20140170753; 20140179006; 20140179770; 20140186843; 20140186919; 20140186958; 20140189896; 20140227787; 20140234972; 20140242664; 20140242699; 20140242700; 20140242702; 20140248702; 20140256046; 20140273037; 20140273226; 20140273230; 20140273231; 20140273232; 20140273233; 20140273234; 20140273235; 20140287938; 20140295556; 20140295557; 20140298547; 20140304853; 20140309487; 20140310828; 20140310830; 20140315985; 20140335063; 20140335620; 20140342456; 20140342457; 20140342458; 20140349400; 20140349405; 20140356867; 20140356956; 20140356958; 20140356959; 20140357523; 20140357530; 20140364333; and 20140377868; all of which are hereby incorporated by reference in their entirety.

(178) Guide RNAs Corresponding to Type V and Type VI CRISPR/Cas Endonucleases (e.g., Cpf1 Guide RNA)

(179) A guide RNA that binds to a type V or type VI CRISPR/Cas protein (e.g., Cpf1, C2c1, C2c2, C2c3), and targets the complex to a specific location within a target nucleic acid is referred to

herein generally as a “type V or type VI CRISPR/Cas guide RNA”. An example of a more specific term is a “Cpf1 guide RNA.”

(180) A type V or type VI CRISPR/Cas guide RNA (e.g., cpf1 guide RNA) can have a total length of from 30 nucleotides (nt) to 200 nt, e.g., from 30 nt to 180 nt, from 30 nt to 160 nt, from 30 nt to 150 nt, from 30 nt to 125 nt, from 30 nt to 100 nt, from 30 nt to 90 nt, from 30 nt to 80 nt, from 30 nt to 70 nt, from 30 nt to 60 nt, from 30 nt to 50 nt, from 50 nt to 200 nt, from 50 nt to 180 nt, from 50 nt to 160 nt, from 50 nt to 150 nt, from 50 nt to 125 nt, from 50 nt to 100 nt, from 50 nt to 90 nt, from 50 nt to 80 nt, from 50 nt to 70 nt, from 50 nt to 60 nt, from 70 nt to 200 nt, from 70 nt to 180 nt, from 70 nt to 160 nt, from 70 nt to 150 nt, from 70 nt to 125 nt, from 70 nt to 100 nt, from 70 nt to 90 nt, or from 70 nt to 80 nt). In some cases, a type V or type VI CRISPR/Cas guide RNA (e.g., cpf1 guide RNA) has a total length of at least 30 nt (e.g., at least 40 nt, at least 50 nt, at least 60 nt, at least 70 nt, at least 80 nt, at least 90 nt, at least 100 nt, or at least 120 nt,).

(181) In some cases, a Cpf1 guide RNA has a total length of 35 nt, 36 nt, 37 nt, 38 nt, 39 nt, 40 nt, 41 nt, 42 nt, 43 nt, 44 nt, 45 nt, 46 nt, 47 nt, 48 nt, 49 nt, or 50 nt.

(182) Like a Cas9 guide RNA, a type V or type VI CRISPR/Cas guide RNA (e.g., cpf1 guide RNA) can include a target nucleic acid-binding segment and a duplex-forming region (e.g., in some cases formed from two duplex-forming segments, i.e., two stretches of nucleotides that hybridize to one another to form a duplex).

(183) The target nucleic acid-binding segment of a type V or type VI CRISPR/Cas guide RNA (e.g., cpf1 guide RNA) can have a length of from 15 nt to 30 nt, e.g., 15 nt, 16 nt, 17 nt, 18 nt, 19 nt, 20 nt, 21 nt, 22 nt, 23 nt, 24 nt, 25 nt, 26 nt, 27 nt, 28 nt, 29 nt, or 30 nt. In some cases, the target nucleic acid-binding segment has a length of 23 nt. In some cases, the target nucleic acid-binding segment has a length of 24 nt. In some cases, the target nucleic acid-binding segment has a length of 25 nt.

(184) The guide sequence of a type V or type VI CRISPR/Cas guide RNA (e.g., cpf1 guide RNA) can have a length of from 15 nt to 30 nt (e.g., 15 to 25 nt, 15 to 24 nt, 15 to 23 nt, 15 to 22 nt, 15 to 21 nt, 15 to 20 nt, 15 to 19 nt, 15 to 18 nt, 17 to 30 nt, 17 to 25 nt, 17 to 24 nt, 17 to 23 nt, 17 to 22 nt, 17 to 21 nt, 17 to 20 nt, 17 to 19 nt, 17 to 18 nt, 18 to 30 nt, 18 to 25 nt, 18 to 24 nt, 18 to 23 nt, 18 to 22 nt, 18 to 21 nt, 18 to 20 nt, 18 to 19 nt, 19 to 30 nt, 19 to 25 nt, 19 to 24 nt, 19 to 23 nt, 19 to 22 nt, 19 to 21 nt, 19 to 20 nt, 20 to 30 nt, 20 to 25 nt, 20 to 24 nt, 20 to 23 nt, 20 to 22 nt, 20 to 21 nt, 15 nt, 16 nt, 17 nt, 18 nt, 19 nt, 20 nt, 21 nt, 22 nt, 23 nt, 24 nt, 25 nt, 26 nt, 27 nt, 28 nt, 29 nt, or 30 nt). In some cases, the guide sequence has a length of 17 nt. In some cases, the guide sequence has a length of 18 nt. In some cases, the guide sequence has a length of 19 nt. In some cases, the guide sequence has a length of 20 nt. In some cases, the guide sequence has a length of 21 nt. In some cases, the guide sequence has a length of 22 nt. In some cases, the guide sequence has a length of 23 nt. In some cases, the guide sequence has a length of 24 nt.

(185) The guide sequence of a type V or type VI CRISPR/Cas guide RNA (e.g., cpf1 guide RNA) can have 100% complementarity with a corresponding length of target nucleic acid sequence. The guide sequence can have less than 100% complementarity with a corresponding length of target nucleic acid sequence. For example, the guide sequence of a type V or type VI CRISPR/Cas guide RNA (e.g., cpf1 guide RNA) can have 1, 2, 3, 4, or 5 nucleotides that are not complementary to the target nucleic acid sequence. For example, in some cases, where a guide sequence has a length of 25 nucleotides, and the target nucleic acid sequence has a length of 25 nucleotides, in some cases, the target nucleic acid-binding segment has 100% complementarity to the target nucleic acid sequence. As another example, in some cases, where a guide sequence has a length of 25 nucleotides, and the target nucleic acid sequence has a length of 25 nucleotides, in some cases, the target nucleic acid-binding segment has 1 non-complementary nucleotide and 24 complementary nucleotides with the target nucleic acid sequence. As another example, in some cases, where a guide sequence has a length of 25 nucleotides, and the target nucleic acid sequence has a length of 25 nucleotides, in some cases, the target nucleic acid-binding segment has 2 non-complementary

nucleotides and 23 complementary nucleotides with the target nucleic acid sequence.

(186) The duplex-forming segment of a type V or type VI CRISPR/Cas guide RNA (e.g., cpf1 guide RNA) (e.g., of a targeter RNA or an activator RNA) can have a length of from 15 nt to 25 nt (e.g., 15 nt, 16 nt, 17 nt, 18 nt, 19 nt, 20 nt, 21 nt, 22 nt, 23 nt, 24 nt, or 25 nt).

(187) The RNA duplex of a type V or type VI CRISPR/Cas guide RNA (e.g., cpf1 guide RNA) can have a length of from 5 base pairs (bp) to 40 bp (e.g., from 5 to 35 bp, 5 to 30 bp, 5 to 25 bp, 5 to 20 bp, 5 to 15 bp, 5-12 bp, 5-10 bp, 5-8 bp, 6 to 40 bp, 6 to 35 bp, 6 to 30 bp, 6 to 25 bp, 6 to 20 bp, 6 to 15 bp, 6 to 12 bp, 6 to 10 bp, 6 to 8 bp, 7 to 40 bp, 7 to 35 bp, 7 to 30 bp, 7 to 25 bp, 7 to 20 bp, 7 to 15 bp, 7 to 12 bp, 7 to 10 bp, 8 to 40 bp, 8 to 35 bp, 8 to 30 bp, 8 to 25 bp, 8 to 20 bp, 8 to 15 bp, 8 to 12 bp, 8 to 10 bp, 9 to 40 bp, 9 to 35 bp, 9 to 30 bp, 9 to 25 bp, 9 to 20 bp, 9 to 15 bp, 9 to 12 bp, 9 to 10 bp, 10 to 40 bp, 10 to 35 bp, 10 to 30 bp, 10 to 25 bp, 10 to 20 bp, 10 to 15 bp, or 10 to 12 bp).

(188) As an example, a duplex-forming segment of a Cpf1 guide RNA can comprise a nucleotide sequence selected from (5' to 3'): AAUUUCUACUGUUGUAGAU (SEQ ID NO: 1096), AAUUUCUGCUGUUGCAGAU (SEQ ID NO: 1097), AAUUUCCACUGUUGUGGAU (SEQ ID NO: 1098), AAUUCCUACUGUUGUAGGU (SEQ ID NO: 1099), AAUUUCUACUAUUGUAGAU (SEQ ID NO: 1100), AAUUUCUACUGCUGUAGAU (SEQ ID NO: 1101), AAUUUCUACUUUGUAGAU (SEQ ID NO: 1102), and AAUUUCUACUUGUAGAU (SEQ ID NO: 1103). The guide sequence can then follow (5' to 3') the duplex forming segment.

(189) A non-limiting example of an activator RNA (e.g. tracrRNA) of a C2c1 guide RNA (dual guide or single guide) is an RNA that includes the nucleotide sequence GAAUUUUUCAACGGGUGUGCCAAUGGCCACUUUCCAGGUGGCAAAGCCCGUUGA GCUUCUCAAAAAG (SEQ ID NO: 1104). In some cases, a C2c1 guide RNA (dual guide or single guide) is an RNA that includes the nucleotide sequence In some cases, a C2c1 guide RNA (dual guide or single guide) is an RNA that includes the nucleotide sequence GUCUAGAGGACAGAAUUUUUCAACGGGUGUGCCAAUGGCCACUUUCCAGGUGGC AAAGCCCGUUGAGCUUCUCAAAAAG (SEQ ID NO: 1105). In some cases, a C2c1 guide RNA (dual guide or single guide) is an RNA that includes the nucleotide sequence UCUAGAGGACAGAAUUUUUCAACGGGUGUGCCAAUGGCCACUUUCCAGGUGGCA AAGCCCGUUGAGCUUCUCAAAAAG (SEQ ID NO: 1106). A non-limiting example of an activator RNA (e.g. tracrRNA) of a C2c1 guide RNA (dual guide or single guide) is an RNA that includes the nucleotide sequence ACUUUCCAGGCAAAGCCCGUUGAGCUUCUCAAAAAG (SEQ ID NO: 1107). In some cases, a duplex forming segment of a C2c1 guide RNA (dual guide or single guide) of an activator RNA (e.g. tracrRNA) includes the nucleotide sequence AGCUUCUCA or the nucleotide sequence GCUUCUCA (the duplex forming segment from a naturally existing tracrRNA).

(190) A non-limiting example of a targeter RNA (e.g. crRNA) of a C2c1 guide RNA (dual guide or single guide) is an RNA with the nucleotide sequence

(191) CUGAGAAGUGGCACNNNNNNNNNNNNNNNNNNNNNNNN (SEQ ID NO: 1110), where the Ns represent the guide sequence, which will vary depending on the target sequence, and although 20 Ns are depicted a range of different lengths are acceptable. In some cases, a duplex forming segment of a C2c1 guide RNA (dual guide or single guide) of a targeter RNA (e.g. crRNA) includes the nucleotide sequence CUGAGAAGUGGCAC (SEQ ID NO: 1111) or includes the nucleotide sequence CUGAGAAGU or includes the nucleotide sequence UGAGAAGUGGCAC (SEQ ID NO: 1113) or includes the nucleotide sequence UGAGAAGU.

(192) Examples and guidance related to type V or type VI CRISPR/Cas endonucleases and guide RNAs (as well as information regarding requirements related to protospacer adjacent motif (PAM) sequences present in targeted nucleic acids) can be found in the art, for example, see Zetsche et al., *Cell*. 2015 Oct. 22; 163 (3): 759-71; Makarova et al., *Nat Rev Microbiol*. 2015 November; 13 (11):

(193) Nucleic Acid Modifications

(194) In some embodiments, a subject nucleic acid (e.g., a DNA or an RNA encoding a fusion polypeptide of the present disclosure; a DNA or RNA encoding an RNA guided endonuclease; a guide RNA, etc.) has one or more modifications, e.g., a base modification, a backbone modification, a sugar modification, etc., to provide the nucleic acid with a new or enhanced feature (e.g., improved stability). A nucleoside is a base-sugar combination. The base portion of the nucleoside is normally a heterocyclic base. The two most common classes of such heterocyclic bases are the purines and the pyrimidines. Nucleotides are nucleosides that further include a phosphate group covalently linked to the sugar portion of the nucleoside. For those nucleosides that include a pentofuranosyl sugar, the phosphate group can be linked to the 2', the 3', or the 5' hydroxyl moiety of the sugar. In forming oligonucleotides, the phosphate groups covalently link adjacent nucleosides to one another to form a linear polymeric compound. In turn, the respective ends of this linear polymeric compound can be further joined to form a circular compound, however, linear compounds are suitable. In addition, linear compounds may have internal nucleotide base complementarity and may therefore fold in a manner as to produce a fully or partially double-stranded compound. Within oligonucleotides, the phosphate groups are commonly referred to as forming the internucleoside backbone of the oligonucleotide. The normal linkage or backbone of RNA and DNA is a 3' to 5' phosphodiester linkage.

(195) Suitable nucleic acid modifications include, but are not limited to: 2' O-methyl modified nucleotides, 2' Fluoro modified nucleotides, locked nucleic acid (LNA) modified nucleotides, peptide nucleic acid (PNA) modified nucleotides, nucleotides with phosphorothioate linkages, and a 5' cap (e.g., a 7-methylguanylate cap (m7G)). Additional details and additional modifications are described below.

(196) In some cases, 2% or more of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) are modified (e.g., 3% or more, 5% or more, 7.5% or more, 10% or more, 15% or more, 20% or more, 25% or more, 30% or more, 35% or more, 40% or more, 45% or more, 50% or more, 55% or more, 60% or more, 65% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, or 100% of the nucleotides of a subject nucleic acid are modified). In some cases, 2% or more of the nucleotides of a subject guide RNA are modified (e.g., 3% or more, 5% or more, 7.5% or more, 10% or more, 15% or more, 20% or more, 25% or more, 30% or more, 35% or more, 40% or more, 45% or more, 50% or more, 55% or more, 60% or more, 65% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, or 100% of the nucleotides of a subject guide RNA are modified). In some cases, 2% or more of the nucleotides of a guide RNA are modified (e.g., 3% or more, 5% or more, 7.5% or more, 10% or more, 15% or more, 20% or more, 25% or more, 30% or more, 35% or more, 40% or more, 45% or more, 50% or more, 55% or more, 60% or more, 65% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, or 100% of the nucleotides of a guide RNA are modified).

(197) In some cases, the number of nucleotides of a subject nucleic acid (e.g., a guide RNA, etc.) that are modified is in a range of from 3% to 100% (e.g., 3% to 100%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 100%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 100%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to 75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%). In some cases, the number of nucleotides of a subject that are modified is in a range of from 3% to 100% (e.g., 3% to 100%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 100%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 100%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to

75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%). In some cases, the number of nucleotides of a guide RNA that are modified is in a range of from 3% to 100% (e.g., 3% to 100%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 100%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 100%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to 75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%).

(198) In some cases, one or more of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) are modified (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more, 14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a subject nucleic acid are modified). In some cases, one or more of the nucleotides of a subject guide RNA are modified (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more, 14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a subject guide RNA are modified). In some cases, one or more of the nucleotides of a guide RNA are modified (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more, 14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a guide RNA are modified).

(199) In some cases, 99% or less of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) are modified (e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or less, 50% or less, or 45% or less of the nucleotides of a subject nucleic acid are modified). In some cases, 99% or less of the nucleotides of a subject guide RNA are modified (e.g., e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or less, 50% or less, or 45% or less of the nucleotides of a subject guide RNA are modified). In some cases, 99% or less of the nucleotides of a guide RNA are modified (e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or less, 50% or less, or 45% or less of the nucleotides of a guide RNA are modified).

(200) In some cases, the number of nucleotides of a nucleic acid nucleic acid (e.g., a guide RNA, etc.) that are modified is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20, 2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10). In some cases, the number of nucleotides of a subject guide RNA that are modified is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20, 2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10). In some cases, the number of nucleotides of a guide RNA that are modified is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20, 2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10).

(201) In some cases, 20 or fewer of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) are modified (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or fewer, or one, of the nucleotides of a subject nucleic acid are modified). In some cases, 20 or fewer of the nucleotides of a subject guide RNA are modified (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or fewer, or one, of the nucleotides of a subject guide RNA are modified). In some cases, 20 or fewer of the nucleotides of a guide RNA are modified (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or

fewer, or one, of the nucleotides of a guide RNA are modified).

(202) A 2'-O-Methyl modified nucleotide (also referred to as 2'-O-Methyl RNA) is a naturally occurring modification of RNA found in tRNA and other small RNAs that arises as a post-transcriptional modification. Oligonucleotides can be directly synthesized that contain 2'-O-Methyl RNA. This modification increases T_m of RNA: RNA duplexes but results in only small changes in RNA: DNA stability. It is stable with respect to attack by single-stranded ribonucleases and is typically 5 to 10-fold less susceptible to DNases than DNA. It is commonly used in antisense oligos as a means to increase stability and binding affinity to the target message.

(203) In some cases, 2% or more of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) are 2'-O-Methyl modified (e.g., 3% or more, 5% or more, 7.5% or more, 10% or more, 15% or more, 20% or more, 25% or more, 30% or more, 35% or more, 40% or more, 45% or more, 50% or more, 55% or more, 60% or more, 65% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, or 100% of the nucleotides of a subject nucleic acid are 2'-O-Methyl modified). In some cases, 2% or more of the nucleotides of a subject guide RNA are 2'-O-Methyl modified (e.g., 3% or more, 5% or more, 7.5% or more, 10% or more, 15% or more, 20% or more, 25% or more, 30% or more, 35% or more, 40% or more, 45% or more, 50% or more, 55% or more, 60% or more, 65% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, or 100% of the nucleotides of a subject guide RNA are 2'-O-Methyl modified). In some cases, 2% or more of the nucleotides of a guide RNA are 2'-O-Methyl modified (e.g., 3% or more, 5% or more, 7.5% or more, 10% or more, 15% or more, 20% or more, 25% or more, 30% or more, 35% or more, 40% or more, 45% or more, 50% or more, 55% or more, 60% or more, 65% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, or 100% of the nucleotides of a guide RNA are 2'-O-Methyl modified).

(204) In some cases, the number of nucleotides of a nucleic acid nucleic acid (e.g., a guide RNA, etc.) that are 2'-O-Methyl modified is in a range of from 3% to 100% (e.g., 3% to 100%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 100%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 100%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to 75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%). In some cases, the number of nucleotides of a guide RNA that are 2'-O-Methyl modified is in a range of from 3% to 100% (e.g., 3% to 100%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 100%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 100%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to 75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%). In some cases, the number of nucleotides of a guide RNA that are 2'-O-Methyl modified is in a range of from 3% to 100% (e.g., 3% to 100%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 100%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 100%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to 75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%).

(205) In some cases, one or more of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) are 2'-O-Methyl modified (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more, 14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a subject nucleic acid are 2'-O-Methyl modified). In some cases, one or more of the nucleotides of a guide RNA are 2'-O-Methyl modified (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more,

14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a subject guide RNA are 2'-O-Methyl modified). In some cases, one or more of the nucleotides of a guide RNA are 2'-O-Methyl modified (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more, 14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a guide RNA are 2'-O-Methyl modified).

(206) In some cases, 99% or less of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) are 2'-O-Methyl modified (e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or less, 50% or less, or 45% or less of the nucleotides of a subject nucleic acid are 2'-O-Methyl modified). In some cases, 99% or less of the nucleotides of a subject guide RNA are 2'-O-Methyl modified (e.g., e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or less, 50% or less, or 45% or less of the nucleotides of a subject guide RNA are 2'-O-Methyl modified). In some cases, 99% or less of the nucleotides of a guide RNA are 2'-O-Methyl modified (e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or less, 50% or less, or 45% or less of the nucleotides of a guide RNA are 2'-O-Methyl modified).

(207) In some cases, the number of nucleotides of a nucleic acid nucleic acid (e.g., a guide RNA, etc.) that are 2'-O-Methyl modified is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20, 2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10). In some cases, the number of nucleotides of a subject guide RNA that are 2'-O-Methyl modified is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20, 2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10). In some cases, the number of nucleotides of a guide RNA that are 2'-O-Methyl modified is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20, 2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10).

(208) In some cases, 20 or fewer of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) are 2'-O-Methyl modified (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or fewer, or one, of the nucleotides of a subject nucleic acid are 2'-O-Methyl modified). In some cases, 20 or fewer of the nucleotides of a subject guide RNA are 2'-O-Methyl modified (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or fewer, or one, of the nucleotides of a subject guide RNA are 2'-O-Methyl modified). In some cases, 20 or fewer of the nucleotides of a guide RNA are 2'-O-Methyl modified (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or fewer, or one, of the nucleotides of a guide RNA are 2'-O-Methyl modified).

(209) 2' Fluoro modified nucleotides (e.g., 2' Fluoro bases) have a fluorine modified ribose which increases binding affinity (T_m) and also confers some relative nuclease resistance when compared to native RNA. These modifications are commonly employed in ribozymes and siRNAs to improve stability in serum or other biological fluids.

(210) In some cases, 2% or more of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) are 2' Fluoro modified (e.g., 3% or more, 5% or more, 7.5% or more, 10% or more, 15% or more, 20% or more, 25% or more, 30% or more, 35% or more, 40% or more, 45% or more, 50% or more, 55% or more, 60% or more, 65% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, or 100% of the nucleotides of a subject nucleic acid are 2' Fluoro modified). In some cases, 2% or more of the nucleotides of a subject guide RNA are 2' Fluoro modified (e.g., 3% or more,

5% or more, 7.5% or more, 10% or more, 15% or more, 20% or more, 25% or more, 30% or more, 35% or more, 40% or more, 45% or more, 50% or more, 55% or more, 60% or more, 65% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, or 100% of the nucleotides of a subject guide RNA are 2' Fluoro modified). In some cases, 2% or more of the nucleotides of a guide RNA are 2' Fluoro modified (e.g., 3% or more, 5% or more, 7.5% or more, 10% or more, 15% or more, 20% or more, 25% or more, 30% or more, 35% or more, 40% or more, 45% or more, 50% or more, 55% or more, 60% or more, 65% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, or 100% of the nucleotides of a guide RNA are 2' Fluoro modified).

(211) In some cases, the number of nucleotides of a nucleic acid (e.g., a guide RNA, etc.) that are 2' Fluoro modified is in a range of from 3% to 100% (e.g., 3% to 100%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 100%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 100%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to 75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%). In some cases, the number of nucleotides of a guide RNA that are 2' Fluoro modified is in a range of from 3% to 100% (e.g., 3% to 100%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 100%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 100%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to 75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%). In some cases, the number of nucleotides of a guide RNA that are 2' Fluoro modified is in a range of from 3% to 100% (e.g., 3% to 100%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 100%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 100%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to 75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%).

(212) In some cases, one or more of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) are 2' Fluoro modified (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more, 14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a subject nucleic acid are 2' Fluoro modified). In some cases, one or more of the nucleotides of a subject guide RNA are 2' Fluoro modified (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more, 14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a guide RNA are 2' Fluoro modified). In some cases, one or more of the nucleotides of a guide RNA are 2' Fluoro modified (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more, 14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a guide RNA are 2' Fluoro modified).

(213) In some cases, 99% or less of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) are 2' Fluoro modified (e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or less, 50% or less, or 45% or less of the nucleotides of a subject nucleic acid are 2' Fluoro modified). In some cases, 99% or less of the nucleotides of a subject guide RNA are 2' Fluoro modified (e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or less, 50% or less, or 45% or less of the nucleotides of a subject guide RNA are 2' Fluoro modified). In some cases, 99% or less of the nucleotides of a guide RNA are 2' Fluoro modified (e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or

less, 50% or less, or 45% or less of the nucleotides of a guide RNA are 2' Fluoro modified).

(214) In some cases, the number of nucleotides of a nucleic acid nucleic acid (e.g., a guide RNA, etc.) that are 2' Fluoro modified is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20, 2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10). In some cases, the number of nucleotides of a subject guide RNA that are 2' Fluoro modified is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20, 2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10). In some cases, the number of nucleotides of a guide RNA that are 2' Fluoro modified is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20, 2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10).

(215) In some cases, 20 or fewer of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) are 2' Fluoro modified (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or fewer, or one, of the nucleotides of a subject nucleic acid are 2' Fluoro modified). In some cases, 20 or fewer of the nucleotides of a subject guide RNA are 2' Fluoro modified (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or fewer, or one, of the nucleotides of a subject guide RNA are 2' Fluoro modified). In some cases, 20 or fewer of the nucleotides of a guide RNA are 2' Fluoro modified (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or fewer, or one, of the nucleotides of a guide RNA are 2' Fluoro modified).

(216) LNA bases have a modification to the ribose backbone that locks the base in the C3'-endo position, which favors RNA A-type helix duplex geometry. This modification significantly increases T_m and is also very nuclease resistant. Multiple LNA insertions can be placed in an oligo at any position except the 3'-end. Applications have been described ranging from antisense oligos to hybridization probes to SNP detection and allele specific PCR. Due to the large increase in T_m conferred by LNAs, they also can cause an increase in primer dimer formation as well as self-hairpin formation. In some cases, the number of LNAs incorporated into a single oligo is 10 bases or less.

(217) In some cases, the number of nucleotides of a nucleic acid nucleic acid (e.g., a guide RNA, etc.) that have an LNA base is in a range of from 3% to 99% (e.g., 3% to 99%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 99%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 99%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to 75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%). In some cases, the number of nucleotides of a guide RNA that have an LNA base is in a range of from 3% to 99% (e.g., 3% to 99%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 99%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 99%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to 75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%). In some cases, the number of nucleotides of a guide RNA that have an LNA base is in a range of from 3% to 99% (e.g., 3% to 99%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 99%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 99%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to 75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%).

55%, 10% to 50%, 10% to 45%, or 10% to 40%).

(218) In some cases, one or more of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) have an LNA base (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more, 14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a subject nucleic acid have an LNA base). In some cases, one or more of the nucleotides of a subject guide RNA have an LNA base (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more, 14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a subject guide RNA have an LNA base). In some cases, one or more of the nucleotides of a guide RNA have an LNA base (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more, 14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a guide RNA have an LNA base).

(219) In some cases, 99% or less of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) have an LNA base (e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or less, 50% or less, or 45% or less of the nucleotides of a subject nucleic acid have an LNA base). In some cases, 99% or less of the nucleotides of a guide RNA have an LNA base (e.g., e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or less, 50% or less, or 45% or less of the nucleotides of a guide RNA have an LNA base). In some cases, 99% or less of the nucleotides of a guide RNA have an LNA base (e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or less, 50% or less, or 45% or less of the nucleotides of a guide RNA have an LNA base).

(220) In some cases, the number of nucleotides of a nucleic acid nucleic acid (e.g., a guide RNA, etc.) that have an LNA base is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20, 2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10). In some cases, the number of nucleotides of a guide RNA that have an LNA base is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20, 2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10). In some cases, the number of nucleotides of a guide RNA that have an LNA base is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20, 2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10).

(221) In some cases, 20 or fewer of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) have an LNA base (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or fewer, or one, of the nucleotides of a subject nucleic acid have an LNA base). In some cases, 20 or fewer of the nucleotides of a subject guide RNA have an LNA base (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or fewer, or one, of the nucleotides of a subject guide RNA have an LNA base). In some cases, 20 or fewer of the nucleotides of a guide RNA have an LNA base (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or fewer, or one, of the nucleotides of a guide RNA have an LNA base).

(222) The phosphorothioate (PS) bond (i.e., a phosphorothioate linkage) substitutes a sulfur atom for a non-bridging oxygen in the phosphate backbone of a nucleic acid (e.g., an oligo). This modification renders the internucleotide linkage resistant to nuclease degradation.

Phosphorothioate bonds can be introduced between the last 3-5 nucleotides at the 5'- or 3'-end of the oligo to inhibit exonuclease degradation. Including phosphorothioate bonds within the oligo (e.g., throughout the entire oligo) can help reduce attack by endonucleases as well.

(223) In some cases, the number of nucleotides of a nucleic acid (e.g., a guide RNA, etc.) that have a phosphorothioate linkage is in a range of from 3% to 99% (e.g., 3% to 99%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 99%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 99%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to 75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%). In some cases, the number of nucleotides of a guide RNA that have a phosphorothioate linkage is in a range of from 3% to 99% (e.g., 3% to 99%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 99%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 99%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to 75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%). In some cases, the number of nucleotides of a guide RNA that have a phosphorothioate linkage is in a range of from 3% to 99% (e.g., 3% to 99%, 3% to 95%, 3% to 90%, 3% to 85%, 3% to 80%, 3% to 75%, 3% to 70%, 3% to 65%, 3% to 60%, 3% to 55%, 3% to 50%, 3% to 45%, 3% to 40%, 5% to 99%, 5% to 95%, 5% to 90%, 5% to 85%, 5% to 80%, 5% to 75%, 5% to 70%, 5% to 65%, 5% to 60%, 5% to 55%, 5% to 50%, 5% to 45%, 5% to 40%, 10% to 99%, 10% to 95%, 10% to 90%, 10% to 85%, 10% to 80%, 10% to 75%, 10% to 70%, 10% to 65%, 10% to 60%, 10% to 55%, 10% to 50%, 10% to 45%, or 10% to 40%).

(224) In some cases, one or more of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) have a phosphorothioate linkage (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more, 14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a subject nucleic acid have a phosphorothioate linkage). In some cases, one or more of the nucleotides of a subject guide RNA have a phosphorothioate linkage (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more, 14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a subject guide RNA have a phosphorothioate linkage). In some cases, one or more of the nucleotides of a guide RNA have a phosphorothioate linkage (e.g., 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, 12 or more, 13 or more, 14 or more, 15 or more, 16 or more, 17 or more, 18 or more, 19 or more, 20 or more, 21 or more, 22 or more, or all of the nucleotides of a guide RNA have a phosphorothioate linkage).

(225) In some cases, 99% or less of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) have a phosphorothioate linkage (e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or less, 50% or less, or 45% or less of the nucleotides of a subject nucleic acid have a phosphorothioate linkage). In some cases, 99% or less of the nucleotides of a subject guide RNA have a phosphorothioate linkage (e.g., e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or less, 50% or less, or 45% or less of the nucleotides of a guide RNA have a phosphorothioate linkage). In some cases, 99% or less of the nucleotides of a guide RNA have a phosphorothioate linkage (e.g., 99% or less, 95% or less, 90% or less, 85% or less, 80% or less, 75% or less, 70% or less, 65% or less, 60% or less, 55% or less, 50% or less, or 45% or less of the nucleotides of a guide RNA have a phosphorothioate linkage).

(226) In some cases, the number of nucleotides of a nucleic acid (e.g., a guide RNA, etc.) that have a phosphorothioate linkage is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20, 2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10). In some cases, the number of nucleotides of a guide RNA that have a phosphorothioate linkage is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20,

2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10). In some cases, the number of nucleotides of a guide RNA that have a phosphorothioate linkage is in a range of from 1 to 30 (e.g., 1 to 25, 1 to 20, 1 to 18, 1 to 15, 1 to 10, 2 to 25, 2 to 20, 2 to 18, 2 to 15, 2 to 10, 3 to 25, 3 to 20, 3 to 18, 3 to 15, or 3 to 10).

(227) In some cases, 20 or fewer of the nucleotides of a nucleic acid (e.g., a guide RNA, etc.) have a phosphorothioate linkage (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or fewer, or one, of the nucleotides of a subject nucleic acid have a phosphorothioate linkage). In some cases, 20 or fewer of the nucleotides of a guide RNA have a phosphorothioate linkage (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or fewer, or one, of the nucleotides of a subject guide RNA have a phosphorothioate linkage). In some cases, 20 or fewer of the nucleotides of a guide RNA have a phosphorothioate linkage (e.g., 19 or fewer, 18 or fewer, 17 or fewer, 16 or fewer, 15 or fewer, 14 or fewer, 13 or fewer, 12 or fewer, 11 or fewer, 10 or fewer, 9 or fewer, 8 or fewer, 7 or fewer, 6 or fewer, 5 or fewer, 4 or fewer, 3 or fewer, 2 or fewer, or one, of the nucleotides of a guide RNA have a phosphorothioate linkage).

(228) In some cases, a nucleic acid (e.g., a guide RNA, etc.) has one or more nucleotides that are 2'-O-Methyl modified nucleotides. In some embodiments, a subject nucleic acid (e.g., a guide RNA, etc.) has one or more 2' Fluoro modified nucleotides. In some cases, a subject nucleic acid (e.g., a guide RNA, etc.) has one or more LNA bases. In some cases, a subject nucleic acid (e.g., a guide RNA, etc.) has one or more nucleotides that are linked by a phosphorothioate bond (i.e., the subject nucleic acid has one or more phosphorothioate linkages). In some embodiments, a subject nucleic acid (e.g., a guide RNA, etc.) has a 5' cap (e.g., a 7-methylguanylate cap (m7G)).

(229) In some cases, a nucleic acid (e.g., a DNA or RNA encoding an RNA guided endonuclease, a guide RNA, etc.) has a combination of modified nucleotides. For example, a nucleic acid can have a 5' cap (e.g., a 7-methylguanylate cap (m7G)) in addition to having one or more nucleotides with other modifications (e.g., a 2'-O-Methyl nucleotide and/or a 2' Fluoro modified nucleotide and/or a LNA base and/or a phosphorothioate linkage). A nucleic acid can have any combination of modifications. For example, a subject nucleic acid can have any combination of the above described modifications.

(230) In some cases, a guide RNA has one or more nucleotides that are 2'-O-Methyl modified nucleotides. In some embodiments, a guide RNA has one or more 2' Fluoro modified nucleotides. In some embodiments, a guide RNA has one or more LNA bases. In some embodiments, a guide RNA has one or more nucleotides that are linked by a phosphorothioate bond (i.e., the subject nucleic acid has one or more phosphorothioate linkages). In some embodiments, a guide RNA has a 5' cap (e.g., a 7-methylguanylate cap (m7G)).

(231) In some cases, a guide RNA has a combination of modified nucleotides. For example, a guide RNA can have a 5' cap (e.g., a 7-methylguanylate cap (m7G)) in addition to having one or more nucleotides with other modifications (e.g., a 2'-O-Methyl nucleotide and/or a 2' Fluoro modified nucleotide and/or a LNA base and/or a phosphorothioate linkage). A guide RNA can have any combination of modifications. For example, a guide RNA can have any combination of the above described modifications.

(232) Modified Backbones and Modified Internucleoside Linkages

(233) Examples of suitable nucleic acids containing modifications include nucleic acids containing modified backbones or non-natural internucleoside linkages. Nucleic acids having modified backbones include those that retain a phosphorus atom in the backbone and those that do not have a phosphorus atom in the backbone.

(234) Suitable modified oligonucleotide backbones containing a phosphorus atom therein include, for example, phosphorothioates, chiral phosphorothioates, phosphorodithioates, phosphotriesters,

aminoalkylphosphotriesters, methyl and other alkyl phosphonates including 3'-alkylene phosphonates, 5'-alkylene phosphonates and chiral phosphonates, phosphinates, phosphoramidates including 3'-amino phosphoramidate and aminoalkylphosphoramidates, phosphorodiamidates, thionophosphoramidates, thionoalkylphosphonates, thionoalkylphosphotriesters, selenophosphates and boranophosphates having normal 3'-5' linkages, 2'-5' linked analogs of these, and those having inverted polarity wherein one or more internucleotide linkages is a 3' to 3', 5' to 5' or 2' to 2' linkage. Suitable oligonucleotides having inverted polarity comprise a single 3' to 3' linkage at the 3'-most internucleotide linkage i.e. a single inverted nucleoside residue which may be a basic (the nucleobase is missing or has a hydroxyl group in place thereof). Various salts (such as, for example, potassium or sodium), mixed salts and free acid forms are also included.

(235) In some cases, a nucleic acid comprises one or more phosphorothioate and/or heteroatom internucleoside linkages, in particular $\text{—CH.sub.2—NH—O—CH.sub.2—}$, $\text{—CH.sub.2—N(CH.sub.3)—O—CH.sub.2—}$ (known as a methylene (methylimino) or MMI backbone), $\text{—CH.sub.2—O—N(CH.sub.3)—CH.sub.2—}$, $\text{—CH.sub.2—N(CH.sub.3)—N(CH.sub.3)—CH.sub.2—}$ and $\text{—O—N(CH.sub.3)—CH.sub.2—CH.sub.2—}$ (wherein the native phosphodiester internucleotide linkage is represented as $\text{—O—P(=O)(OH)—O—CH.sub.2—}$). MMI type internucleoside linkages are disclosed in the above referenced U.S. Pat. No. 5,489,677. Suitable amide internucleoside linkages are disclosed in U.S. Pat. No. 5,602,240.

(236) Also suitable are nucleic acids having morpholino backbone structures as described in, e.g., U.S. Pat. No. 5,034,506. For example, in some embodiments, a subject nucleic acid comprises a 6-membered morpholino ring in place of a ribose ring. In some of these embodiments, a phosphorodiamidate or other non-phosphodiester internucleoside linkage replaces a phosphodiester linkage.

(237) Suitable modified polynucleotide backbones that do not include a phosphorus atom therein have backbones that are formed by short chain alkyl or cycloalkyl internucleoside linkages, mixed heteroatom and alkyl or cycloalkyl internucleoside linkages, or one or more short chain heteroatomic or heterocyclic internucleoside linkages. These include those having morpholino linkages (formed in part from the sugar portion of a nucleoside); siloxane backbones; sulfide, sulfoxide and sulfone backbones; formacetyl and thioformacetyl backbones; methylene formacetyl and thioformacetyl backbones; riboacetyl backbones; alkene containing backbones; sulfamate backbones; methyleneimino and methylenehydrazino backbones; sulfonate and sulfonamide backbones; amide backbones; and others having mixed N, O, S and CH₂ component parts.

(238) Mimetics

(239) A nucleic acid can be a nucleic acid mimetic. The term “mimetic” as it is applied to polynucleotides is intended to include polynucleotides wherein only the furanose ring or both the furanose ring and the internucleotide linkage are replaced with non-furanose groups, replacement of only the furanose ring is also referred to in the art as being a sugar surrogate. The heterocyclic base moiety or a modified heterocyclic base moiety is maintained for hybridization with an appropriate target nucleic acid. One such nucleic acid, a polynucleotide mimetic that has been shown to have excellent hybridization properties, is referred to as a peptide nucleic acid (PNA). In PNA, the sugar-backbone of a polynucleotide is replaced with an amide containing backbone, in particular an aminoethylglycine backbone. The nucleotides are retained and are bound directly or indirectly to aza nitrogen atoms of the amide portion of the backbone.

(240) One polynucleotide mimetic that has been reported to have excellent hybridization properties is a peptide nucleic acid (PNA). The backbone in PNA compounds is two or more linked aminoethylglycine units which gives PNA an amide containing backbone. The heterocyclic base moieties are bound directly or indirectly to aza nitrogen atoms of the amide portion of the backbone. Representative U.S. patents that describe the preparation of PNA compounds include, but are not limited to: U.S. Pat. Nos. 5,539,082; 5,714,331; and 5,719,262.

(241) Another class of polynucleotide mimetic that has been studied is based on linked morpholino

units (morpholino nucleic acid) having heterocyclic bases attached to the morpholino ring. A number of linking groups have been reported that link the morpholino monomeric units in a morpholino nucleic acid. One class of linking groups has been selected to give a non-ionic oligomeric compound. The non-ionic morpholino-based oligomeric compounds are less likely to have undesired interactions with cellular proteins. Morpholino-based polynucleotides are non-ionic mimics of oligonucleotides which are less likely to form undesired interactions with cellular proteins (Dwayne A. Braasch and David R. Corey, *Biochemistry*, 2002, 41 (14), 4503-4510). Morpholino-based polynucleotides are disclosed in U.S. Pat. No. 5,034,506. A variety of compounds within the morpholino class of polynucleotides have been prepared, having a variety of different linking groups joining the monomeric subunits.

(242) A further class of polynucleotide mimetic is referred to as cyclohexenyl nucleic acids (CeNA). The furanose ring normally present in a DNA/RNA molecule is replaced with a cyclohexenyl ring. CeNA DMT protected phosphoramidite monomers have been prepared and used for oligomeric compound synthesis following classical phosphoramidite chemistry. Fully modified CeNA oligomeric compounds and oligonucleotides having specific positions modified with CeNA have been prepared and studied (see Wang et al., *J. Am. Chem. Soc.*, 2000, 122, 8595-8602). In general the incorporation of CeNA monomers into a DNA chain increases its stability of a DNA/RNA hybrid. CeNA oligoadenylates formed complexes with RNA and DNA complements with similar stability to the native complexes. The study of incorporating CeNA structures into natural nucleic acid structures was shown by NMR and circular dichroism to proceed with easy conformational adaptation.

(243) A further modification includes Locked Nucleic Acids (LNAs) in which the 2'-hydroxyl group is linked to the 4' carbon atom of the sugar ring thereby forming a 2'-C,4'-C-oxymethylene linkage thereby forming a bicyclic sugar moiety. The linkage can be a methylene ($\text{---CH}_2\text{---}$), group bridging the 2' oxygen atom and the 4' carbon atom wherein n is 1 or 2 (Singh et al., *Chem. Commun.*, 1998, 4, 455-456). LNA and LNA analogs display very high duplex thermal stabilities with complementary DNA and RNA ($T_m = \pm 3$ to $+10^\circ \text{C.}$), stability towards 3'-exonucleolytic degradation and good solubility properties. Potent and nontoxic antisense oligonucleotides containing LNAs have been described (e.g., Wahlestedt et al., *Proc. Natl. Acad. Sci. U.S.A.*, 2000, 97, 5633-5638).

(244) The synthesis and preparation of the LNA monomers adenine, cytosine, guanine, 5-methylcytosine, thymine and uracil, along with their oligomerization, and nucleic acid recognition properties have been described (e.g., Koshkin et al., *Tetrahedron*, 1998, 54, 3607-3630). LNAs and preparation thereof are also described in WO 98/39352 and WO 99/14226, as well as U.S. applications 20120165514, 20100216983, 20090041809, 20060117410, 20040014959, 20020094555, and 20020086998.

(245) Modified Sugar Moieties

(246) A nucleic acid can also include one or more substituted sugar moieties. Suitable polynucleotides comprise a sugar substituent group selected from: OH; F; O-, S-, or N-alkyl; O-, S-, or N-alkenyl; O-, S- or N-alkynyl; or O-alkyl-O-alkyl, wherein the alkyl, alkenyl and alkynyl may be substituted or unsubstituted C₁ to C₁₀ alkyl or C₂ to C₁₀ alkenyl and alkynyl. Particularly suitable are O((CH₂)_nO)_mCH₃, O(CH₂)_nOCH₃, O(CH₂)_nNH₂, O(CH₂)_nCH₃, O(CH₂)_nONH₂, and O(CH₂)_nON((CH₂)_nCH₃)₂, where n and m are from 1 to about 10. Other suitable polynucleotides comprise a sugar substituent group selected from: C₁ to C₁₀ lower alkyl, substituted lower alkyl, alkenyl, alkynyl, alkaryl, aralkyl, O-alkaryl or O-aralkyl, SH, SCH₃, OCN, Cl, Br, CN, CF₃, OCF₃, SOCH₃, SO₂CH₃, ONO₂, NO₂, N₃, NH₂, heterocycloalkyl, heterocycloalkaryl, aminoalkylamino, polyalkylamino, substituted silyl, an RNA cleaving group, a reporter group, an intercalator, a group for improving the pharmacokinetic properties of an

oligonucleotide, or a group for improving the pharmacodynamic properties of an oligonucleotide, and other substituents having similar properties. A suitable modification includes 2'-methoxyethoxy (2'-O—CH₂CH₂OCH₃, also known as 2'-O-(2-methoxyethyl) or 2'-MOE) (Martin et al., *Helv. Chim. Acta*, 1995, 78, 486-504) i.e., an alkoxyalkoxy group. A further suitable modification includes 2'-dimethylaminoethoxy, i.e., a O(CH₂)₂ON(CH₃)₂ group, also known as 2'-DMAOE, as described in examples herein below, and 2'-dimethylaminoethoxyethoxy (also known in the art as 2'-O-dimethyl-amino-ethoxy-ethyl or 2'-DMAEOE), i.e., 2'-O—CH₂—O—CH₂—N(CH₃)₂.

(247) Other suitable sugar substituent groups include methoxy (—O—CH₃), aminopropoxy (—OCH₂CH₂NH₂), allyl (—CH₂—CH=CH₂), —O-allyl (—O—CH₂—CH=CH₂) and fluoro (F). 2'-sugar substituent groups may be in the arabino (up) position or ribo (down) position. A suitable 2'-arabino modification is 2'-F. Similar modifications may also be made at other positions on the oligomeric compound, particularly the 3' position of the sugar on the 3' terminal nucleoside or in 2'-5' linked oligonucleotides and the 5' position of 5' terminal nucleotide. Oligomeric compounds may also have sugar mimetics such as cyclobutyl moieties in place of the pentofuranosyl sugar.

(248) Base Modifications and Substitutions

(249) A nucleic acid may also include nucleobase (often referred to in the art simply as "base") modifications or substitutions. As used herein, "unmodified" or "natural" nucleobases include the purine bases adenine (A) and guanine (G), and the pyrimidine bases thymine (T), cytosine (C) and uracil (U). Modified nucleobases include other synthetic and natural nucleobases such as 5-methylcytosine (5-me-C), 5-hydroxymethyl cytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-methyl and other alkyl derivatives of adenine and guanine, 2-propyl and other alkyl derivatives of adenine and guanine, 2-thiouracil, 2-thiothymine and 2-thiocytosine, 5-halouracil and cytosine, 5-propynyl (—C≡CH) uracil and cytosine and other alkynyl derivatives of pyrimidine bases, 6-azo uracil, cytosine and thymine, 5-uracil (pseudouracil), 4-thiouracil, 8-halo, 8-amino, 8-thiol, 8-thioalkyl, 8-hydroxyl and other 8-substituted adenines and guanines, 5-halo particularly 5-bromo, 5-trifluoromethyl and other 5-substituted uracils and cytosines, 7-methylguanine and 7-methyladenine, 2-F-adenine, 2-amino-adenine, 8-azaguanine and 8-azaadenine, 7-deazaguanine and 7-deazaadenine and 3-deazaguanine and 3-deazaadenine. Further modified nucleobases include tricyclic pyrimidines such as phenoxazine cytidine(1H-pyrimido(5,4-b)(1,4)benzoxazin-2(3H)-one), phenothiazine cytidine (1H-pyrimido(5,4-b)(1,4)benzothiazin-2(3H)-one), G-clamps such as a substituted phenoxazine cytidine (e.g. 9-(2-aminoethoxy)-H-pyrimido(5,4-b)(1,4)benzoxazin-2(3H)-one), carbazole cytidine (2H-pyrimido(4,5-b)indol-2-one), pyridoindole cytidine (H-pyrido(3',2':4,5)pyrrolo(2,3-d)pyrimidin-2-one).

(250) Heterocyclic base moieties may also include those in which the purine or pyrimidine base is replaced with other heterocycles, for example 7-deaza-adenine, 7-deazaguanosine, 2-aminopyridine and 2-pyridone. Further nucleobases include those disclosed in U.S. Pat. No. 3,687,808, those disclosed in *The Concise Encyclopedia Of Polymer Science And Engineering*, pages 858-859, Kroschwitz, J. I., ed. John Wiley & Sons, 1990, those disclosed by Englisch et al., *Angewandte Chemie, International Edition*, 1991, 30, 613, and those disclosed by Sanghvi, Y. S., Chapter 15, *Antisense Research and Applications*, pages 289-302, Crooke, S. T. and Lebleu, B., ed., CRC Press, 1993. Certain of these nucleobases are useful for increasing the binding affinity of an oligomeric compound. These include 5-substituted pyrimidines, 6-azapyrimidines and N-2, N-6 and O-6 substituted purines, including 2-aminopropyladenine, 5-propynyluracil and 5-propynylcytosine. 5-methylcytosine substitutions have been shown to increase nucleic acid duplex stability by 0.6-1.2° C. (Sanghvi et al., eds., *Antisense Research and Applications*, CRC Press, Boca Raton, 1993, pp. 276-278) and are suitable base substitutions, e.g., when combined with 2'-O-methoxyethyl sugar modifications.

(251) Error-Prone DNA Polymerases

(252) A number of error-prone DNA polymerases are known in the art, and any known error-prone DNA polymerase is suitable for use in a fusion polypeptide of the present disclosure. A suitable error-prone DNA polymerase possesses nick translating activity.

(253) Suitable error-prone DNA polymerases include, but are not limited to, Taq polymerase, *Thermus flavus* DNA polymerase I, *Thermus thermophilus* HB-8 DNA polymerase I, *Thermophilus ruber* DNA polymerase I, *Thermophilus brokianus* DNA polymerase I, *Thermophilus caldophilus* GK14 DNA polymerase I, *Thermophilus filoformis* DNA polymerase I, *Bacillus stearothermophilus* DNA polymerase I, *Bacillus caldotonex* YT-G DNA polymerase I, and *Bacillus caldovelox* YT-F DNA polymerase I. Suitable error-prone DNA polymerases include, but are not limited to, a *Niastella koreensis* error-prone DNA polymerase, a *Mucilaginibacter paludis* error-prone DNA polymerase, a *Methylobacterium extorquens* error-prone DNA polymerase, and a *Stenotrophomonas maltophilia* error-prone DNA polymerase.

(254) In some cases, a suitable error-prone DNA polymerase is *Escherichia coli* DNA polymerase I, with three fidelity-reducing mutations; this error-prone DNA polymerase is referred to as Poll3M. Poll3M comprises D424A, I709N, and A759R substitutions relative to wild-type *E. coli* DNA polymerase I. In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the DNA polymerase I amino acid sequence depicted in FIG. 8; where the DNA polymerase has an Ala at amino acid position 424, an Asn at amino acid position 709, and an Arg at amino acid position 759 of the amino acid sequence depicted in FIG. 8, or a corresponding amino acid in another DNA polymerase.

(255) In some cases, a suitable error-prone DNA polymerase is *Escherichia coli* DNA polymerase I, with five fidelity-reducing mutations: D242A, I709N, A759R, F742Y, and P796H. In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the DNA polymerase I amino acid sequence depicted in FIG. 8; where the DNA polymerase has an Ala at amino acid position 242, an Asn at amino acid position 709, an Arg at amino acid position 759, a Tyr at amino acid position 742, and a His at amino acid position 796; or corresponding amino acids in another DNA polymerase.

(256) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, or at least 99%, amino acid sequence identity to the DNA polymerase I amino acid sequence depicted in FIG. 8; where the DNA polymerase has an Ala at amino acid position 424, and an Asn at amino acid position 709.

(257) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the DNA polymerase I amino acid sequence depicted in FIG. 9.

(258) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the DNA polymerase I amino acid sequence depicted in FIG. 10.

(259) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the DNA polymerase I amino acid sequence depicted in FIG. 11.

(260) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the DNA polymerase iota amino acid sequence depicted in FIG. 13.

(261) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the DNA polymerase q amino acid sequence depicted in FIGS. 14A-14B.

(262) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid

sequence identity to the DNA polymerase K amino acid sequence depicted in FIGS. 15A-15B.

(263) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the DNA polymerase 0 amino acid sequence depicted in FIGS. 16A-16D.

(264) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the DNA polymerase v (nu) amino acid sequence depicted in FIGS. 17A-17B.

(265) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the *E. coli* DNA polymerase IV amino acid sequence depicted in FIG. 18.

(266) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase R having the following amino acid sequence:

(267) TABLE-US-00003 (SEQ ID NO: 1163)

MSKRKAPQETLNGGITDMLTELANFEKNVSQAIHKYNAYR
KAASVIAKYPHKIKSGAEAKKLPGVGT KIAEKIDEFLATG
KLRKLEKIRQDDTSSINFLTRVSGIGPSAARKFVDEGIK
TLEDLRKNEDKLNHHQRIGLKYFGDFEK RIPREEM LQM QD
IVLNEVKKVDSEYIATVCGSFRRGAESSGDM DVLLTHPSF
TSESTKQPKLLHQVVEQLQKVHFITDTLSKGETKFMGVCQ
LPSKNDEKEYPHRRIDIRLIPKDQYYCGVLYFTGSDIFNK

NMRAHALEKGFTINEYTIRPLGVTGVAGEPLPVDSEKDIF DYIQWKYREPKDRSE.

(268) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase iota having the following amino acid sequence:

MEKLGVEPEEEGGGDDDEEDAEAWAMELADVGA AASSQGVHDQVLPTPNASSRVIVH
VDLDCFYAQVEMISNPELKDKPLGVQQKYL VVTCN YEARKLG VKKLMNVRDAKEKCP
QLVLVNGEDLTRYREMSYKVTELLEEFSPVVERLGF DENFVDLTEMVEKRLQQLQSDE
LSAVTVSGHVYNNQSINLLDVLHIRLLVGSQIAAEMREAMYNQLGLTGCAGVASNKLL
AKLVSGVFKPNQQT VLLPESCQH LIHSLNHIKEIPGIGYKTAKCLEALGINSVRDLQTFSP
KILEKELGISVAQRIQKLSFGEDNSPVILSGPPQSFSEEDSFKKCSSEVEAKNKIEELLASLL
NRVCQDGRKPHTVRLIIRRYSSSEKH YGRESRQCPIPSHVIQKLGTGNYDVMTPMVDILM
KLFRNMVNVKMPFHLTLLSVCFCNLKALNTAKKGLIDYYLMPSLSTTSRSGKHSFKMK
DTHMEDFPKDKETNRDFLPSGRIESTRTRESPLDTTNFSKEKDINEFPLCSLPEGVDQE VF
KQLPVDIQEEILSGKSREKFQKGGSVSCPLHASRGVLSFFSKKQM QDIPINPRDHLSSSKQ
VSSVSPCEPGTSGFNSSSSSYMSSQKDYSYYLDNRLKDERISQGPKEPQGFHFTNSNPAV
SAFHSFPNLQSEQLFSRNHTTDSHKQTVATDSHEGLTENREPDSVDEKITFPSDIDPQVFY
ELPEAVQKELLA EWKRAGSDFHIGHK (SEQ ID NO:1164).

In some cases, such a DNA polymerase generates T.fwdarw.G substitutions.

(269) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase iota having the following amino acid sequence (amino acids 1-445 of DNA polymerase iota):

MEKLGVEPEEEGGGDDDEEDAEAWAMELADVGA AASSQGVHDQVLPTPNASSRVIVH
VDLDCFYAQVEMISNPELKDKPLGVQQKYL VVTCN YEARKLG VKKLMNVRDAKEKCP
QLVLVNGEDLTRYREMSYKVTELLEEFSPVVERLGF DENFVDLTEMVEKRLQQLQSDE
LSAVTVSGHVYNNQSINLLDVLHIRLLVGSQIAAEMREAMYNQLGLTGCAGVASNKLL
AKLVSGVFKPNQQT VLLPESCQH LIHSLNHIKEIPGIGYKTAKCLEALGINSVRDLQTFSP
KILEKELGISVAQRIQKLSFGEDNSPVILSGPPQSFSEEDSFKKCSSEVEAKNKIEELLASLL
NRVCQDGRKPHTVRLIIRRYSSSEKH YGRESRQCPIPSHVIQKLGTGNYDVMTPMVDILM

KLFRNMVNVKMPFHLTLTLLSVCFCNLKALNTAK (SEQ ID NO:1165); and having a length of 445 amino acids. In some cases, such a DNA polymerase generates T.fwdarw.G substitutions. In some cases, such a DNA polymerase has a T.fwdarw.G error rate approaching 1.

(270) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase iota having the following amino acid sequence (amino acids 26-445 of DNA polymerase iota):

ELADVGAASSQGVHDQVLPTPNASSRVIVHVDLDCFYAQVEMISNPDKPLGVQQ
KYLVTTCNYEARKLGVKMLMNVDAKEKCPQLVLVNGEDLTRYREMSYKVTELEEF
SPVVERLGFDFENFVDLTEMVEKRLQQLQSDLSAVTVSGHVYNNQSINLLDVLHIRLLV
GSQIAAEMREAMYNQLGLTGCAGVASNKLLAKLVSGVFKPNQQTVLLPESCQHLSL
NHIKEIPGIGYKTAKCLEALGINSVRDLQTFSPKILEKELGISVAQRIQKLSFGEDNSPVILS
GPPQSFSEEDSFKKCSSEVEAKNKIEELLASLLNRVCQDGRKPHTVRLIIRRYSSSEKHYGR
ESRQCPIPSHVIQKLGTGNYDVMTPMVDILMKLFRNMVNVKMPFHLTLTLLSVCFCNLKA
LNTAK (SEQ ID NO:1166); and having a length of 419 amino acids. In some cases, such a DNA polymerase generates T.fwdarw.G substitutions. In some cases, such a DNA polymerase has a T.fwdarw.G error rate approaching 1.

(271) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase nu (v) having the following amino acid sequence:

ENYEALVGFDLCNTPLSSVAQKIMSAMHSGDLVDSKTWKGSTETMEVINKSSVKYSVQ
LEDKRTQSPEKKDLKSLRSQTSRGSACLSPQSFSVRLTDQLSADQKQKSSISLTLSSCLIP
QYNQEASVLQKKKGHKRKHFLMENINNENKGSINLKRKHITYNNLSEKTSKQMALEEDT
DDAEGYLNSGNSGALKKKHFCDIRHLDDWAKSQLIEMLKQAAALVITVMYTDGSTQLG
ADQTPVSSVRGIVVLVKRQAEGGHGCPDAPACGPVLEGFVSDDPICIYQIEHSAIWDQEQ
EAHQQFARNVLFQTMKCKCPVICFNAKDFVRIVLQFFGNDGSWKHVADFIGLDPRIAA
WLIDPSDATPSFEDLVEKYCEKSITVKVNSTYGNSSRNIVNQNVRENKTLTYRLTMDLCS
KLKDYGLWQLFRTLELPLIPILAVMESHAIQVNKEEMEKTSAALLGARLKELEQEAHFVA
GERFLITSNNQLREILFGKLKLHLLSQRNSLPRTGLQKYPSTSEAVLNALRDLHPLPKIILE
YRQVHKIKSTFVDGLLACMKKGSISSTWNQTGTVTGRLSAKHPNIQGISKHPIQITTPKN
FKGKEDKILTISPRAMFVSSKGHTFLAADFSQIELRILTHLSGDPELLKLFQESERDDVFST
LTSQWKDVPVEQVTHADREQTKKVYAVVYGAGKERLAACLGVPIQEAQAQFLESFLQ
KYKKIKDFARAAIAQCHQTGCVVSIMGRRRPLPRIHAHDQQLRAQAERQAVNFVVQGS
AADLCKLAMIHVFTAVAASHTLTARLVAQIHDELLFEVEDPQIPECAALVRRRTMESLEQ
VQALELQLQVPLKVSLSAGRSWGHVPLQEAWGPPGPCRTESPSNSLAAPGSPASTQP
PPLHFSPSFCL (SEQ ID NO:1167). In some cases, such a DNA polymerase generates G.fwdarw.T substitutions.

(272) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase eta (η) having the following amino acid sequence:

(273) TABLE-US-00004 (SEQ ID NO: 1168)

MATGQDRVVALVDMDCFFVQVEQRQNPHLRNKPCAVVQYK
SWKGGGIIAVSYEARAFGVTRSMWADDAKKLCPDLLLAQV
RESRGKANLTKYREASVEVMEIMSRFAVIERASIDEAYVD
LTSVQERLQKLQGQPISADLLPSTYIEGLPQGPTTAEET
VQKEGMRKQGLFQWLDLQIDNLTSPDLQLTVGAVIVEEM
RAAIERETGFQCSAGISHNKVLAKLACGLNKPQRQTLVSH
GSVPQLFSQMPIRKIRSLGGKLGASVIEILGIEYMGELTQ
FTESQLQSHFGEKNGSWLYAMCRGIEHDPVKPRQLPKTIG
CSKNFPGKTALATREQVQWWLLQLAQELEERLTKDRNDND

RVATQLVLRVQGDRLSSLRCCALTRYDAHKMSHDAF
TVIKNCNTSGIQTEWSPPLTMLFLCATKFSASAPSSSTDI
TSFLSSDPSSLPKVPVTSSEAKTQGS GPAVTATKKATTSL
ESFFQKAAERQKVKEASLSSLTAPTQAPMSNSPSKPSLPF
QTSQSTGTEPFFKQKSLLLKQKQLNNSVSSPQQNPWSNC
KALPNSLPTEYPGCVPVCEGVSKLEESSKATPAEMDLAHN
SQSMHASSASKSVLEVTQKATPNPSLLAAEDQVPCEKCGS
LVPVWDMPEHMDYHFALELQKSFLQPHSSNPQVVSAVSHQ
GKRNPKSPLACTNKRPRPEGMQTLESFFKPLTH.

(274) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase eta (η) having the following amino acid sequence:

MATGQDRVVALVDMDCFFVQVEQRQNPHLRNKPCAVVQYKSWKGGGIIAVSYEARAF
GVTRSMWADDAKKLCPDLLLAQVRESRGKANLTKYREASVEVMEIMSRFAVIERASID
EAYVDLTSVQERLQKLQGQPISADLLPSTYIEGLPQGPTTAEETVQKEGMRKQGLFQW
LDSLQIDNLTSPDLQLTVGAVIVEEMRAAIERETGFQCSAGISHNKVLAKLACGLNKP
NRTLVSHGSVPQLFSQMPIRKIRSLGGKLGASVIEILGIEYMGELTQFTESQLQSHFGEKNG
SWLYAMCRGIEHDPVKPRQLPKTIGCSKNFPGKTALATREQVQWWLLQLAQELEERLT
KDRNDNDRVATQLVVSIRVQGDKRLSSLRCCALTRYDAHKMSHDAFTVIKNCNTSGI
QTEWSPPLTMLFLCATKFSASAPSSSTDITSFLSSDPSSLPKVPVTSSEAKTQGS GPAVTA
TKKATTSLLESFFQKAAERQKVKEASLSSLTAPTQAPMSN (SEQ ID NO:1169); and has a
length of 511 amino acids.

(275) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase kappa (κ) having the following amino acid sequence:

(276) TABLE-US-00005 (SEQ ID NO: 1170)

MDSTKEKCD SYKDDLLRMGLNDNKAGMEGLDKEKINKII
MEATKGS RFYGNELKKEKQVNQRIENMMQQKAQITSQQLR
KAQLQVDRFAMELEQSRNLSNTIVHIDMDAFYAAVEMRDN
PELKDKPIAVGSM SMLSTS NYHARRFGVRAAMPGFIAKRL
CPQLIIVPPNFDKYRAVSKEVKEILADYDPNFMAMSLDEA
YLNITKHLEERQNW PEDKRRYFIKMGSSVENDNPGKEV NK
LSEHERSISPLLFEESPSDVQPPGDPFQVNFEEQNNPQIL
QNSVVFGTSAQEVVKEIRFRIEQKTTLTASAGIAPNTMLA
KVCSDKNKPNGQYQILPNRQAVMDFIKDLPIRKVSGIGKV
TEKMLKALGIITCTELYQQRALLSLLFSETSWHYFLHISL
GLGSTHLTRDGERKSMSVERTFSEINKAEEQYSLCQELCS
ELAQDLQKERLKGRTVTIKLKNVNFVVKTRASTVSSVVST
AEEIFAIKELLKTEIDADFP HPLRLRLMGVRISFPNEE
DRKHQQR SIIGFLQAGNQALSATECTLEKTDKDKFVKPLE
MSHKKSFFDKKR SERKWSHQDTFKCEAVNKQSFQTSQPFQ
VLKKKM NENLEISENSDDCQILT CPVCFRAQGCISLEALN
KHVDECLDGP SISENFKMFSCSHVSATKVNKKENVPASSL
CEKQDYEAHPKIKEISSVDCIALVD TIDNSSKAESIDALS
NKHSKEECSSLP SKSFNIEHCHQNSSSTVSLENEDVGSFR
QEYRQPYLCEVKTGQALVCPVCNVEQKTS DLT LFNVHVDV
CLNKSFIQELRKDKFNPVNQPKESSRSTGSSSGVQKAVTR
TKRPGLMTKYSTSKKIKPNNPKHTLDIFFK.

(277) In some cases, a suitable error-prone DNA polymerase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid

sequence identity to a DNA polymerase kappa (κ) having the following amino acid sequence: MDSTKEKCD SYKDDLLLRMGLNDNKAGMEGLDKEKINKIIMEATKGS RFYGNELKKE KQVNQRIENMMQQAQITSQQLRKAQLQVDRFAMELEQSRNLSNTIVHIDMDAFYAAV EMRDNP ELKDKPIAVGSM SMLSTSNYHARRFGVRAAMPGFIAKRLCPQLIIVPPNFDKY RAVSKEVKEILADYDPNFMAMSLDEAYLNITKHLEERQNW PEDKRRYFIKMGSSVEND NPGKEVNKLSEHERSISPLLFEESPSDVQPPGDPFQVNFEEQNNPQILQNSVVFGTSAQEV VKEIRFRIEQKTTLTASAGIAPNTMLAKVCSDKNKPNGQYQILPNRQAVMDFIKDLPIRK VSGIGKVTEKMLKALGIITCTELYQQRALLSLLFSETSWHYFLHISLGLGSTHLTRDGER KSMSVERTFSEINKAEEQYSLCQELCSELAQDLQKERLKGRTVTIKLKNVNFEVKTRAST VSSVVSTAEEIFAIAKELLKTEIDADFP HPLRLRLMGVRISFPNEEDRKHQQR SIIGFLQA GNQALSATECTLEKTDKDKFVKPLE (SEQ ID NO:1171); and has a length of 560 amino acids.

(278) Linkers

(279) In some cases, a fusion polypeptide of the present disclosure comprises a linker positioned between the DNA polymerase and the RNA-guided endonuclease.

(280) In some embodiments, a subject fusion polypeptide can be fused to a fusion partner via a linker polypeptide (e.g., one or more linker polypeptides). The linker polypeptide may have any of a variety of amino acid sequences. Proteins can be joined by a spacer peptide, generally of a flexible nature, although other chemical linkages are not excluded. Suitable linkers include polypeptides of between 4 amino acids and 40 amino acids in length, or between 4 amino acids and 25 amino acids in length. These linkers can be produced by using synthetic, linker-encoding oligonucleotides to couple the proteins, or can be encoded by a nucleic acid sequence encoding the fusion protein. Peptide linkers with a degree of flexibility can be used. The linking peptides may have virtually any amino acid sequence, bearing in mind that the preferred linkers will have a sequence that results in a generally flexible peptide. The use of small amino acids, such as glycine and alanine, are of use in creating a flexible peptide. The creation of such sequences is routine to those of skill in the art. A variety of different linkers are commercially available and are considered suitable for use.

(281) Examples of linker polypeptides include glycine polymers (G)_n, glycine-serine polymers (including, for example, (GS)_n, (GSGGS)_n (SEQ ID NO:1154), (GGSGGS)_n (SEQ ID NO:1155), and (GGGS)_n (SEQ ID NO:1156), where n is an integer of at least one), glycine-alanine polymers, alanine-serine polymers. Exemplary linkers can comprise amino acid sequences including, but not limited to, GSGG (SEQ ID NO:1157), GSGGG (SEQ ID NO:1158), GSGSG (SEQ ID NO:1159), GSGGG (SEQ ID NO:1160), GGGSG (SEQ ID NO:1161), GSSSG (SEQ ID NO:1162), and the like. The ordinarily skilled artisan will recognize that design of a peptide conjugated to any desired element can include linkers that are all or partially flexible, such that the linker can include a flexible linker as well as one or more portions that confer less flexible structure.

(282) Additional Polypeptides

(283) A fusion polypeptide of the present disclosure can comprise one or more additional polypeptides. Suitable additional polypeptides include, but are not limited to, a nuclear localization signal (NLS); a DNA-binding polypeptide that increases the processivity of the DNA polymerase; a tag for ease of tracking and/or purification (e.g., a fluorescent protein, e.g., green fluorescent protein (GFP), yellow fluorescent protein (YFP), red fluorescent protein (RFP), cyan fluorescent protein (CFP), mCherry, tdTomato, and the like; a histidine tag, e.g., a 6×His tag; a hemagglutinin (HA) tag; a FLAG tag; a Myc tag; a flap endonuclease; a DNA ligase; etc.

(284) In some cases, e.g., in the context of a base editor, the additional polypeptide promotes or provides for efficient ligation of the nick generated by the RNA-guided endonuclease after the polymerase makes a mis-incorporation. For example, in some cases, a fusion polypeptide of the present disclosure comprises a flap endonuclease and/or a DNA ligase.

(285) A suitable flap endonuclease comprises an amino acid sequence having at least 85%, at least

90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the FEN1 amino acid sequence depicted in FIG. 20.

(286) A suitable DNA ligase comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the T4 DNA ligase amino acid sequence depicted in FIG. 21.

(287) In some cases, a fusion polypeptide of the present disclosure comprises, in order from N-terminus to C-terminus: a) an error-prone DNA polymerase; b) a flap endonuclease; and c) an enzymatically active RNA-guided endonuclease that introduces a single-stranded break in a target DNA. In some cases, a fusion polypeptide of the present disclosure comprises, in order from N-terminus to C-terminus: a) an error-prone DNA polymerase; b) an enzymatically active RNA-guided endonuclease that introduces a single-stranded break in a target DNA; and c) a DNA ligase.

(288) Localization Signals

(289) In some cases, a fusion polypeptide of the present disclosure comprises one or more localization signal peptides. Suitable localization signals (“subcellular localization signals”) include, e.g., a nuclear localization signal (NLS) for targeting to the nucleus; a sequence to keep the fusion protein out of the nucleus, e.g., a nuclear export sequence (NES); a sequence to keep the fusion protein retained in the cytoplasm; a mitochondrial localization signal for targeting to the mitochondria; a chloroplast localization signal for targeting to a chloroplast; an endoplasmic reticulum (ER) retention signal; and ER export signal; and the like. In some cases, a fusion polypeptide does not include a NLS so that the protein is not targeted to the nucleus (which can be advantageous, e.g., when the target nucleic acid is an RNA that is present in the cytosol).

(290) In some cases, a fusion polypeptide includes (is fused to) a nuclear localization signal (NLS) (e.g., in some cases 2 or more, 3 or more, 4 or more, or 5 or more NLSs). Thus, in some cases, a fusion polypeptide includes one or more NLSs (e.g., 2 or more, 3 or more, 4 or more, or 5 or more NLSs). In some cases, one or more NLSs (2 or more, 3 or more, 4 or more, or 5 or more NLSs) are positioned at or near (e.g., within 50 amino acids of) the N-terminus and/or the C-terminus. In some cases, one or more NLSs (2 or more, 3 or more, 4 or more, or 5 or more NLSs) are positioned at or near (e.g., within 50 amino acids of) the N-terminus. In some cases, one or more NLSs (2 or more, 3 or more, 4 or more, or 5 or more NLSs) are positioned at or near (e.g., within 50 amino acids of) the C-terminus. In some cases, one or more NLSs (3 or more, 4 or more, or 5 or more NLSs) are positioned at or near (e.g., within 50 amino acids of) both the N-terminus and the C-terminus. In some cases, an NLS is positioned at the N-terminus and an NLS is positioned at the C-terminus.

(291) In some cases, a fusion polypeptide includes (is fused to) between 1 and 10 NLSs (e.g., 1-9, 1-8, 1-7, 1-6, 1-5, 2-10, 2-9, 2-8, 2-7, 2-6, or 2-5 NLSs). In some cases, a fusion polypeptide includes (is fused to) between 2 and 5 NLSs (e.g., 2-4, or 2-3 NLSs).

(292) Non-limiting examples of NLSs include an NLS sequence derived from: the NLS of the SV40 virus large T-antigen, having the amino acid sequence PKKKRKV (SEQ ID NO:1172); the NLS from nucleoplasmin (e.g., the nucleoplasmin bipartite NLS with the sequence KRPAATKKAGQAKKKK (SEQ ID NO:1173)); the c-myc NLS having the amino acid sequence PAAKRVKLD (SEQ ID NO:1174) or RQRRNELKRSP (SEQ ID NO:1175); the hRNPA1 M9 NLS having the sequence NQSSNFGPMKGGNFGGRSSGPYGGGGQYFAKPRNQGGY (SEQ ID NO:1176); the sequence RMRIZFKNKGKDTAELRRRRVEVSVELRKAKKDEQILKRRNV (SEQ ID NO:1177) of the IBB domain from importin-alpha; the sequences VSRKRPRP (SEQ ID NO:1178) and PPKKARED (SEQ ID NO:1179) of the myoma T protein; the sequence PQPKKKPL (SEQ ID NO:1180) of human p53; the sequence SALIKKKKKMAP (SEQ ID NO:1181) of mouse c-abl IV; the sequences DRLRR (SEQ ID NO:1182) and PKQKKRK (SEQ ID NO:1183) of the influenza virus NS1; the sequence RKLKKKIKKL (SEQ ID NO:1184) of the Hepatitis virus delta antigen; the sequence REKKKFLKRR (SEQ ID NO:1185) of the mouse Mx1 protein; the sequence KRKGDEV DGVDEVAKKKSKK (SEQ ID NO:1186) of the human

poly(ADP-ribose) polymerase; and the sequence RKCLQAGMNLEARKTKK (SEQ ID NO:1187) of the steroid hormone receptors (human) glucocorticoid. In some cases, an NLS comprises the amino acid sequence MDSLLMNRKFLYQFKNVRWAKGRRETYLC (SEQ ID NO:1188). In general, NLS (or multiple NLSs) are of sufficient strength to drive accumulation of the fusion polypeptide in a detectable amount in the nucleus of a eukaryotic cell. Detection of accumulation in the nucleus may be performed by any suitable technique. For example, a detectable marker may be fused to the fusion polypeptide such that location within a cell may be visualized. Cell nuclei may also be isolated from cells, the contents of which may then be analyzed by any suitable process for detecting protein, such as immunohistochemistry, Western blot, or enzyme activity assay.

Accumulation in the nucleus may also be determined indirectly.

(293) In some cases, the NLS is located N-terminal of the RNA-guided endonuclease present in a fusion polypeptide of the present disclosure. In some cases, the NLS is located between the DNA polymerase and the RNA-guided endonuclease present in a fusion polypeptide of the present disclosure. In some cases, the NLS is located N-terminal of the DNA polymerase present in a fusion polypeptide of the present disclosure. In some cases, the NLS is located C-terminal of the DNA polymerase present in a fusion polypeptide of the present disclosure.

(294) In some cases, a fusion polypeptide includes a “Protein Transduction Domain” or PTD (also known as a CPP—cell penetrating peptide), which refers to a polypeptide, polynucleotide, carbohydrate, or organic or inorganic compound that facilitates traversing a lipid bilayer, micelle, cell membrane, organelle membrane, or vesicle membrane. A PTD attached to another molecule, which can range from a small polar molecule to a large macromolecule and/or a nanoparticle, facilitates the molecule traversing a membrane, for example going from extracellular space to intracellular space, or cytosol to within an organelle. In some embodiments, a PTD is covalently linked to the amino terminus of a polypeptide. In some embodiments, a PTD is covalently linked to the carboxyl terminus of a polypeptide. In some cases, the PTD is inserted internally in the fusion polypeptide (i.e., is not at the N- or C-terminus of the fusion polypeptide) at a suitable insertion site. In some cases, a subject fusion polypeptide includes (is conjugated to, is fused to) one or more PTDs (e.g., two or more, three or more, four or more PTDs). In some cases, a PTD includes a nuclear localization signal (NLS) (e.g., in some cases 2 or more, 3 or more, 4 or more, or 5 or more NLSs). Thus, in some cases, a fusion polypeptide includes one or more NLSs (e.g., 2 or more, 3 or more, 4 or more, or 5 or more NLSs). In some embodiments, a PTD is covalently linked to a nucleic acid (e.g., a guide nucleic acid, a polynucleotide encoding a guide nucleic acid, a polynucleotide encoding a fusion polypeptide, a donor polynucleotide, etc.). Examples of PTDs include but are not limited to a minimal undecapeptide protein transduction domain (corresponding to residues 47-57 of HIV-1 TAT comprising YGRKKRRQRRR; SEQ ID NO:1189); a polyarginine sequence comprising a number of arginines sufficient to direct entry into a cell (e.g., 3, 4, 5, 6, 7, 8, 9, 10, or 10-50 arginines); a VP22 domain (Zender et al. (2002) *Cancer Gene Ther.* 9(6):489-96); an *Drosophila* Antennapedia protein transduction domain (Noguchi et al. (2003) *Diabetes* 52(7):1732-1737); a truncated human calcitonin peptide (Trehin et al. (2004) *Pharm. Research* 21:1248-1256); polylysine (Wender et al. (2000) *Proc. Natl. Acad. Sci. USA* 97:13003-13008); RRQRRTSKLMKR (SEQ ID NO:1190); Transportan GWTLNSAGYLLGKINLKALAALAKKIL (SEQ ID NO:1191); KALAWKAKLAKALAKALAKHLAKALAKALKCEA (SEQ ID NO:1192); and RQIKIWFQNRRMKWKK (SEQ ID NO:1193). Exemplary PTDs include but are not limited to, YGRKKRRQRRR (SEQ ID NO:1189), RKKRRQRRR (SEQ ID NO:1194); an arginine homopolymer of from 3 arginine residues to 50 arginine residues; Exemplary PTD domain amino acid sequences include, but are not limited to, any of the following: YGRKKRRQRRR (SEQ ID NO:1189); RKKRRQRR (SEQ ID NO:1195); YARAAARQARA (SEQ ID NO:1196); THRLPRRRRRR (SEQ ID NO:1197); and GGRRARRRRRRR (SEQ ID NO:1198). In some embodiments, the PTD is an activatable CPP (ACPP) (Aguilera et al. (2009) *Integr Biol (Camb)* June; 1(5-6): 371-381). ACPPs comprise a polycationic CPP (e.g., Arg9 or “R9”) connected via a

cleavable linker to a matching polyanion (e.g., Glu9 or "E9"), which reduces the net charge to nearly zero and thereby inhibits adhesion and uptake into cells. Upon cleavage of the linker, the polyanion is released, locally unmasking the polyarginine and its inherent adhesiveness, thus "activating" the ACP to traverse the membrane.

(295) DNA-Binding Polypeptides that Increase the Processivity of the DNA Polymerase

(296) In some cases, a fusion polypeptide of the present disclosure comprises a DNA-binding polypeptide that increases the processivity of the DNA polymerase. Suitable DNA-binding polypeptides that increase the processivity of the DNA polymerase include, but are not limited to, an Sso7d polypeptide, a helix-hairpin-helix domain of topoisomerase I, a thioredoxin binding domain of a T7 DNA polymerase, or a thioredoxin binding domain of a T3 polymerase.

(297) Suitable Sso7d polypeptides comprise an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the Sso7d amino acid sequence depicted in FIG. 12.

(298) Suitable thioredoxin binding domains comprise an amino acid sequence having at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 90%, or 100%, amino acid sequence identity to the following amino acid sequence:

(299) TABLE-US-00006 (SEQ ID NO: 1199)

TETFGSWYQPKGGTEMFCHPRTGKPLPKYPRIKTPKVGGI
FKKPKNKAQREGREPCELDTREYVAGAPYTPVEHV.

Systems

(300) The present disclosure provides a system comprising: a) a fusion polypeptide of the present disclosure; and b) a guide RNA that comprises: i) a protein-binding segment comprising a nucleotide sequence that binds to the RNA-guided endonuclease; and ii) a target-binding segment comprising a nucleotide sequence that is complementary to a target nucleotide sequence in a target nucleic acid. Fusion polypeptides of the present disclosure are described above. Suitable guide RNAs are described above. A system of the present disclosure (e.g., a system comprising: a) a fusion protein comprising: i) an enzymatically active RNA-guided endonuclease that introduces a single-stranded break in a target DNA; and ii) an error-prone DNA polymerase; and b) a guide RNA) is also referred to herein as "EvolvR."

(301) In some cases, the guide RNA is a single-molecule guide RNA. In some cases, the guide RNA is a dual-molecule guide RNA. In some cases, the guide RNA comprises one or more of: a) a modified base; b) a modified backbone; c) a modified sugar moiety; and d) a non-natural internucleoside linkage. Such modifications are described above.

(302) Nucleic Acids; Recombinant Expression Vectors

(303) The present disclosure provides a nucleic acid comprising a nucleotide sequence encoding a fusion polypeptide of the present disclosure.

(304) In some cases, a nucleic acid comprising a nucleotide sequence encoding a fusion polypeptide of the present disclosure is contained within an expression vector. Thus, the present disclosure provides a recombinant expression vector comprising a nucleic acid comprising a nucleotide sequence encoding a fusion polypeptide of the present disclosure. In some cases, the nucleotide sequence encoding a fusion polypeptide of the present disclosure is operably linked to a transcriptional control element (e.g., a promoter; an enhancer; etc.). In some cases, the transcriptional control element is inducible. In some cases, the transcriptional control element is constitutive. In some cases, the promoters are functional in eukaryotic cells. In some cases, the promoters are cell type-specific promoters. In some cases, the promoters are tissue-specific promoters.

(305) Depending on the host/vector system utilized, any of a number of suitable transcription and translation control elements, including constitutive and inducible promoters, transcription enhancer

elements, transcription terminators, etc. may be used in the expression vector (see e.g., Bitter et al. (1987) *Methods in Enzymology*, 153:516-544).

(306) A promoter can be a constitutively active promoter (i.e., a promoter that is constitutively in an active/"ON" state), it may be an inducible promoter (i.e., a promoter whose state, active/"ON" or inactive/"OFF", is controlled by an external stimulus, e.g., the presence of a particular temperature, compound, or protein.), it may be a spatially restricted promoter (i.e., transcriptional control element, enhancer, etc.)(e.g., tissue specific promoter, cell type specific promoter, etc.), and it may be a temporally restricted promoter (i.e., the promoter is in the "ON" state or "OFF" state during specific stages of embryonic development or during specific stages of a biological process).

(307) Suitable promoter and enhancer elements are known in the art. For expression in a bacterial cell, suitable promoters include, but are not limited to, lacd, lacZ, T3, T7, gpt, lambda P and trc. For expression in a eukaryotic cell, suitable promoters include, but are not limited to, light and/or heavy chain immunoglobulin gene promoter and enhancer elements; cytomegalovirus immediate early promoter; herpes simplex virus thymidine kinase promoter; early and late SV40 promoters; promoter present in long terminal repeats from a retrovirus; mouse metallothionein-I promoter; and various art-known tissue specific promoters.

(308) Suitable reversible promoters, including reversible inducible promoters are known in the art. Such reversible promoters may be isolated and derived from many organisms, e.g., eukaryotes and prokaryotes. Such reversible promoters, and systems based on such reversible promoters but also comprising additional control proteins, include, but are not limited to, alcohol regulated promoters (e.g., alcohol dehydrogenase I (alcA) gene promoter, promoters responsive to alcohol transactivator proteins (AlcR), etc.), tetracycline regulated promoters, (e.g., promoter systems including TetActivators, TetON, TetOFF, etc.), steroid regulated promoters (e.g., rat glucocorticoid receptor promoter systems, human estrogen receptor promoter systems, retinoid promoter systems, thyroid promoter systems, ecdysone promoter systems, mifepristone promoter systems, etc.), metal regulated promoters (e.g., metallothionein promoter systems, etc.), pathogenesis-related regulated promoters (e.g., salicylic acid regulated promoters, ethylene regulated promoters, benzothiadiazole regulated promoters, etc.), temperature regulated promoters (e.g., heat shock inducible promoters (e.g., HSP-70, HSP-90, soybean heat shock promoter, etc.), light regulated promoters, synthetic inducible promoters, and the like.

(309) Inducible promoters suitable for use include any inducible promoter described herein or known to one of ordinary skill in the art. Examples of inducible promoters include, without limitation, chemically/biochemically-regulated and physically-regulated promoters such as alcohol-regulated promoters, tetracycline-regulated promoters (e.g., anhydrotetracycline (aTc)-responsive promoters and other tetracycline-responsive promoter systems, which include a tetracycline repressor protein (tetR), a tetracycline operator sequence (tetO) and a tetracycline transactivator fusion protein (tTA)), steroid-regulated promoters (e.g., promoters based on the rat glucocorticoid receptor, human estrogen receptor, moth ecdysone receptors, and promoters from the steroid/retinoid/thyroid receptor superfamily), metal-regulated promoters (e.g., promoters derived from metallothionein (proteins that bind and sequester metal ions) genes from yeast, mouse and human), pathogenesis-regulated promoters (e.g., induced by salicylic acid, ethylene or benzothiadiazole (BTH)), temperature/heat-inducible promoters (e.g., heat shock promoters), and light-regulated promoters (e.g., light responsive promoters from plant cells).

(310) Examples of constitutive plant promoters include the cauliflower mosaic virus (CaMV) 35S promoter, which confers constitutive, high-level expression in most plant tissues (see, e.g., Odell et al. (1985) *Nature* 313: 810-812); the nopaline synthase promoter (An et al. (1988) *Plant Physiol.* 88: 547-552); and the octopine synthase promoter (Fromm et al. (1989) *Plant Cell* 1: 977-984).

(311) A variety of plant gene promoters that regulate gene expression in response to environmental, hormonal, chemical, developmental signals, and in a tissue-active manner can be used for expression of a nucleotide sequence (e.g., a nucleotide sequence encoding a fusion polypeptide of

the present disclosure) in plants. The choice of a promoter can be determined by such factors as tissue (e.g., seed, fruit, root, pollen, vascular tissue, flower, carpel, etc.), inducibility (e.g., in response to wounding, heat, cold, drought, light, pathogens, etc.), timing, developmental stage, and the like. Numerous known promoters have been characterized and can be employed to promote expression of a polynucleotide of the invention in a transgenic plant or cell of interest. For example, tissue specific promoters include: seed-specific promoters (such as the napin, phaseolin or DC3 promoter described in U.S. Pat. No. 5,773,697), fruit-specific promoters that are active during fruit ripening (such as the *dru 1* promoter (U.S. Pat. No. 5,783,393), or the 2A11 promoter (U.S. Pat. No. 4,943,674) and the tomato polygalacturonase promoter (Bird et al. (1988) *Plant Mol. Biol.* 11: 651-662), root-specific promoters, such as those disclosed in U.S. Pat. Nos. 5,618,988, 5,837,848 and 5,905,186, pollen-active promoters such as PTA29, PTA26 and PTA13 (U.S. Pat. No. 5,792,929), promoters active in vascular tissue (Ringli and Keller (1998) *Plant Mol. Biol.* 37: 977-988), flower-specific (Kaiser et al. (1995) *Plant Mol. Biol.* 28: 231-243), pollen (Baerson et al. (1994) *Plant Mol. Biol.* 26: 1947-1959), carpels (Ohl et al. (1990) *Plant Cell* 2: 837-848), pollen and ovules (Baerson et al. (1993) *Plant Mol. Biol.* 22: 255-267), auxin-inducible promoters (such as that described in van der Kop et al. (1999) *Plant Mol. Biol.* 39: 979-990 or Baumann et al. (1999) *Plant Cell* 11: 323-334), cytokinin-inducible promoter (Guevara-Garcia (1998) *Plant Mol. Biol.* 38: 743-753), promoters responsive to gibberellin (Shi et al. (1998) *Plant Mol. Biol.* 38: 1053-1060, Willmott et al. (1998) 38: 817-825) and the like. Additional promoters are those that elicit expression in response to heat (Ainley et al. (1993) *Plant Mol. Biol.* 22: 13-23), light (e.g., the pea *rbcS-3A* promoter, Kuhlemeier et al. (1989) *Plant Cell* 1: 471-478, and the maize *rbcS* promoter, Schaffner and Sheen (1991) *Plant Cell* 3: 997-1012); wounding (e.g., *wun1*, Siebertz et al. (1989) *Plant Cell* 1: 961-968); pathogens (such as the PR-1 promoter described in Buchel et al. (1999) *Plant Mol. Biol.* 40: 387-396, and the PDF1.2 promoter described in Manners et al. (1998) *Plant Mol. Biol.* 38: 1071-1080), and chemicals such as methyl jasmonate or salicylic acid (Gatz (1997) *Annu. Rev. Plant Physiol. Plant Mol. Biol.* 48: 89-108). In addition, the timing of the expression can be controlled by using promoters such as those acting at senescence (Gan and Amasino (1995) *Science* 270: 1986-1988); or late seed development (Odell et al. (1994) *Plant Physiol.* 106: 447-458).

(312) In some cases, a nucleic acid comprising a nucleotide sequence encoding a fusion polypeptide of the present disclosure is a recombinant expression vector. In some embodiments, the recombinant expression vector is a viral construct, e.g., a recombinant adeno-associated virus (AAV) construct, a recombinant adenoviral construct, a recombinant lentiviral construct, a recombinant retroviral construct, etc. In some cases, a nucleic acid comprising a nucleotide sequence encoding a fusion polypeptide of the present disclosure is a recombinant lentivirus vector. In some cases, a nucleic acid comprising a nucleotide sequence encoding a fusion polypeptide of the present disclosure is a recombinant AAV vector.

(313) Suitable expression vectors include, but are not limited to, viral vectors (e.g. viral vectors based on vaccinia virus; poliovirus; adenovirus (see, e.g., Li et al., *Invest Ophthalmol Vis Sci* 35:2543-2549, 1994; Borrás et al., *Gene Ther* 6:515-524, 1999; Li and Davidson, *PNAS* 92:7700-7704, 1995; Sakamoto et al., *Hum Gene Ther* 5:1088-1097, 1999; WO 94/12649, WO 93/03769; WO 93/19191; WO 94/28938; WO 95/11984 and WO 95/00655); adeno-associated virus (see, e.g., Ali et al., *Hum Gene Ther* 9:81-86, 1998, Flannery et al., *PNAS* 94:6916-6921, 1997; Bennett et al., *Invest Ophthalmol Vis Sci* 38:2857-2863, 1997; Jomary et al., *Gene Ther* 4:683-690, 1997, Rolling et al., *Hum Gene Ther* 10:641-648, 1999; Ali et al., *Hum Mol Genet* 5:591-594, 1996; Srivastava in WO 93/09239, Samulski et al., *J. Vir.* (1989) 63:3822-3828; Mendelson et al., *Viol.* (1988) 166:154-165; and Flotte et al., *PNAS* (1993) 90:10613-10617); SV40; herpes simplex virus; human immunodeficiency virus (see, e.g., Miyoshi et al., *PNAS* 94:10319-23, 1997; Takahashi et al., *J Virol* 73:7812-7816, 1999); a retroviral vector (e.g., Murine Leukemia Virus, spleen necrosis virus, and vectors derived from retroviruses such as Rous Sarcoma Virus, Harvey Sarcoma Virus, avian

leukosis virus, a lentivirus, human immunodeficiency virus, myeloproliferative sarcoma virus, and mammary tumor virus); and the like. In some cases, the vector is a lentivirus vector. Also suitable are transposon-mediated vectors, such as piggyback and sleeping beauty vectors.

(314) A number of expression vectors suitable for stable transformation of plant cells or for the establishment of transgenic plants have been described including those described in Weissbach and Weissbach (1989) *Methods for Plant Molecular Biology*, Academic Press, and Gelvin et al. (1990) *Plant Molecular Biology Manual*, Kluwer Academic Publishers. Specific examples include those derived from a Ti plasmid of *Agrobacterium tumefaciens*, as well as those disclosed by Herrera-Estrella et al. (1983) *Nature* 303: 209, Bevan (1984) *Nucleic Acids Res.* 12: 8711-8721, Klee (1985) *Bio/Technology* 3: 637-642, for dicotyledonous plants.

(315) Alternatively, non-Ti vectors can be used to transfer a nucleic acid into monocotyledonous plants and cells by using free DNA delivery techniques. Such methods can involve, for example, the use of liposomes, electroporation, microprojectile bombardment, silicon carbide whiskers, and viruses. By using these methods transgenic plants such as wheat, rice (Christou (1991) *Bio/Technology* 9: 957-962) and corn (Gordon-Kamm (1990) *Plant Cell* 2: 603-618) can be produced. An immature embryo can also be a good target tissue for monocots for direct DNA delivery techniques by using the particle gun (Weeks et al. (1993) *Plant Physiol.* 102: 1077-1084; Vasil (1993) *Bio/Technology* 10: 667-674; Wan and Lemeaux (1994) *Plant Physiol.* 104: 37-48, and for *Agrobacterium*-mediated DNA transfer (Ishida et al. (1996) *Nature Biotechnol.* 14: 745-750).

(316) Cells

(317) The present disclosure provides a cell comprising a fusion polypeptide of the present disclosure. The present disclosure provides a cell comprising a system of the present disclosure. The present disclosure provides a cell comprising a nucleic acid (e.g., a recombinant expression vector) of the present disclosure.

(318) Suitable host cells include, e.g. a bacterial cell; an archaeal cell; a cell of a single-cell eukaryotic organism; a plant cell; an algal cell, e.g., *Botryococcus braunii*, *Chlamydomonas reinhardtii*, *Nannochloropsis gaditana*, *Chlorella pyrenoidosa*, *Sargassum patens*, *C. Agardh*, and the like; a fungal cell (e.g., a yeast cell); an animal cell; a cell from an invertebrate animal (e.g. fruit fly, cnidarian, echinoderm, nematode, etc.); a cell from a vertebrate animal (e.g., fish, amphibian, reptile, bird, mammal); a cell from a mammal (e.g., a cell from a rodent, a cell from a human, etc.); and the like.

(319) A suitable host cell can be a stem cell (e.g. an embryonic stem (ES) cell, an induced pluripotent stem (iPS) cell); a germ cell; a somatic cell, e.g. a fibroblast, a hematopoietic cell, a neuron, a muscle cell, a bone cell, a hepatocyte, a pancreatic cell; an in vitro or in vivo embryonic cell of an embryo at any stage, e.g., a 1-cell, 2-cell, 4-cell, 8-cell, etc. stage zebrafish embryo; etc.). Cells may be from established cell lines or they may be primary cells, where “primary cells”, “primary cell lines”, and “primary cultures” are used interchangeably herein to refer to cells and cells cultures that have been derived from a subject and allowed to grow in vitro for a limited number of passages of the culture. For example, primary cultures include cultures that may have been passaged 0 times, 1 time, 2 times, 4 times, 5 times, 10 times, or 15 times, but not enough times go through the crisis stage. Primary cell lines can be maintained for fewer than 10 passages in vitro. Host cells are in some cases unicellular organisms, or are grown in culture.

(320) If the cells are primary cells, they may be harvest from an organism (e.g., an individual) by any convenient method. For example, leukocytes may be conveniently harvested by apheresis, leukocytapheresis, density gradient separation, etc., while cells from tissues such as skin, muscle, bone marrow, spleen, liver, pancreas, lung, intestine, stomach, etc. are most conveniently harvested by biopsy. An appropriate solution may be used for dispersion or suspension of the harvested cells. Such solution will generally be a balanced salt solution, e.g. normal saline, phosphate-buffered saline (PBS), Hank's balanced salt solution, etc., conveniently supplemented with fetal calf serum

or other naturally occurring factors, in conjunction with an acceptable buffer at low concentration, e.g., from 5-25 mM. Convenient buffers include HEPES, phosphate buffers, lactate buffers, etc. The cells may be used immediately, or they may be stored, frozen, for long periods of time, being thawed and capable of being reused. In such cases, the cells can be frozen in 10% dimethyl sulfoxide (DMSO), 50% serum, 40% buffered medium, or some other such solution as is commonly used in the art to preserve cells at such freezing temperatures, and thawed in a manner as commonly known in the art for thawing frozen cultured cells.

(321) In some cases, a subject genetically modified host cell is in vitro. In some embodiments, a subject genetically modified host cell is in vivo. In some embodiments, a subject genetically modified host cell is a prokaryotic cell or is derived from a prokaryotic cell. In some embodiments, a subject genetically modified host cell is a bacterial cell or is derived from a bacterial cell. In some cases, a subject genetically modified host cell is an archaeal cell or is derived from an archaeal cell. In some embodiments, a subject genetically modified host cell is a eukaryotic cell or is derived from a eukaryotic cell. In some cases, a subject genetically modified host cell is a plant cell or is derived from a plant cell. In some cases, a subject genetically modified host cell is an animal cell or is derived from an animal cell. In some embodiments, a subject genetically modified host cell is an invertebrate cell or is derived from an invertebrate cell. In some cases, a subject genetically modified host cell is a vertebrate cell or is derived from a vertebrate cell. In some cases, a subject genetically modified host cell is a mammalian cell or is derived from a mammalian cell. In some cases, a subject genetically modified host cell is a rodent cell or is derived from a rodent cell. In cases embodiments, a subject genetically modified host cell is a human cell or is derived from a human cell.

(322) The present disclosure provides a genetically modified plant cell, where the genetically modified plant cell is genetically modified with a nucleic acid (e.g., a recombinant expression vector) comprising a nucleotide sequence encoding a fusion polypeptide of the present disclosure. In some cases, the plant cell is a cell of a monocot (a monocotyledon). In some cases, the plant cell is a cell of a dicot (a dicotyledon). A plant cell can be a cell of the xylem, the phloem, the cambium layer, a leaf, a root, etc.

(323) Suitable plants include, e.g., soybean, wheat, corn, potato, cotton, rice, oilseed rape, sunflower, alfalfa, clover, sugarcane, turf, banana, blackberry, blueberry, strawberry, raspberry, cantaloupe, carrot, cauliflower, coffee, cucumber, eggplant, grapes, honeydew, lettuce, mango, melon, onion, papaya, peas, peppers, pineapple, pumpkin, spinach, squash, sweet corn, tobacco, tomato, watermelon, mint and other labiates, rosaceous fruits, and vegetable brassicas.

(324) Plant protoplasts are also suitable for some applications. For example, a nucleic acid is introduced into plant tissues, cultured plant cells or plant protoplasts by standard methods including electroporation (Fromm et al. (1985) Proc. Natl. Acad. Sci. 82: 5824-5828), infection by viral vectors such as cauliflower mosaic virus (CaMV) (Hohn et al. (1982) Molecular Biology of Plant Tumors Academic Press, New York, N.Y., pp. 549-560; U.S. Pat. No. 4,407,956), high velocity ballistic penetration by small particles with the nucleic acid either within the matrix of small beads or particles, or on the surface (Klein et al. (1987) Nature 327: 70-73), use of pollen as vector (WO 85/01856), or use of *Agrobacterium tumefaciens* or *A. rhizogenes* carrying a T-DNA plasmid in which a nucleic acid encoding a fusion polypeptide of the present disclosure is cloned. The T-DNA plasmid is transmitted to plant cells upon infection by *Agrobacterium tumefaciens*, and a portion is stably integrated into the plant genome (Horsch et al. (1984) Science 233: 496-498; Fraley et al. (1983) Proc. Natl. Acad. Sci. 80: 4803-4807).

(325) The present disclosure further provides progeny of a subject genetically modified cell, where the progeny can comprise the same exogenous nucleic acid or polypeptide as the subject genetically modified cell from which it was derived. The present disclosure further provides a composition comprising a subject genetically modified host cell.

(326) Methods

(327) A fusion polypeptide of the present disclosure is useful for introducing mutations into a target region of a target nucleic acid; i.e., in some cases, a fusion polypeptide of the present disclosure functions as a mutator. Thus, the present disclosure provides methods of introducing mutations into a target region of a target nucleic acid. A fusion polypeptide of the present disclosure is useful for correcting a mutation in a target nucleic acid; i.e., in some cases, a fusion polypeptide of the present disclosure functions as a base editor. Thus, the present disclosure provides methods correcting a mutation in a target nucleic acid.

(328) Method of Introducing Mutations into a Target Region of a Target Nucleic Acid

(329) The present disclosure provides a method of mutagenizing a target nucleic, e.g., introducing mutations into a target nucleic acid. The method comprises contacting the target nucleic acid with a system of the present disclosure (e.g., EvolvR); i.e., the method comprises contacting the target nucleic acid with a complex comprising: a) a fusion polypeptide of the present disclosure; and b) a guide RNA. In some cases, the target nucleic acid is present in a cell; and the method comprises introducing a system of the present disclosure into the cell. In some cases, the cell is a prokaryotic cell. In some cases, the cell is a eukaryotic cell. In some cases, the cell is a plant cell. In some cases, the plant cell is a cell of a dicotyledonous plant. In some cases, the plant cell is a cell of a monocotyledonous plant. In some cases, the cell is in vitro. In some cases, the cell is in vivo.

(330) A method of the present disclosure finds use in a variety of applications. Non-limiting examples of applications involving use of an EvolvR system of the present disclosure include: a) diversifying antibody-encoding genes; b) diversifying protein-coding nucleotide sequences; c) diversifying regulatory elements; d) optimizing antibody affinity; e) diversifying T cells; f) engineering T-cell activity; g) discovering disease-causing genotypes; h) engineering desirable traits into plants.

(331) In some cases, contacting a target nucleic acid with a complex comprising: a) a fusion polypeptide of the present disclosure; and b) a guide RNA results in introduction of from 1 mutation to 103 mutations within a target region of a target nucleic acid. For example, in some cases, contacting a target nucleic acid with a complex comprising: a) a fusion polypeptide of the present disclosure; and b) a guide RNA results in introduction of from 1 mutation to 5 mutations, from 5 mutations to 10 mutations, from 10 mutations to 50 mutations, from 50 mutations to 10^{sup.2} mutations, from 10^{sup.2} mutations to 5×10^{sup.2} mutations, or from to 5×10^{sup.2} mutations to 103 mutations within a target region of a target nucleic acid. As noted above, in some cases, the target region of a target nucleic acid is from 1 nucleotide (nt) to 10 nucleotides (nt), from 10 nt to 50 nt, from 50 nt to 100 nt, from 100 nt to 500 nt, from 500 nt to 10^{sup.3} nt, from 10^{sup.3} nt to 5×10^{sup.3} nt, or from 5×10^{sup.3} nt to 10^{sup.4} nt from a nick in a target DNA introduced by the RNA-guided endonuclease.

(332) Introducing mutations into a target region of a target nucleic acid provides for generation of a plurality of mutants, which can then be selected for a particular desired trait. Alternatively, an undesirable trait can be selected against. A desired trait can be selected for simultaneously with selecting against an undesired trait.

(333) In some cases, a method of the present disclosure comprises: a) mutagenizing a target nucleic acid, generating a plurality of mutated nucleic acids; and b) applying a selection to the mutated nucleic acids. Applying a selection to the mutated nucleic acids can comprise: i) selecting a mutated nucleic acid(s) that confers a desirable trait (phenotype) on a genetically modified host cell comprising the mutated nucleic acid; or ii) selecting a mutated nucleic acid(s) that confers a desirable trait (phenotype) on a transgenic non-human organism that is genetically modified to comprise the mutated nucleic acid. Selection methods are well known in the art, and any known method can be applied.

(334) For example, in some cases, a mutated nucleic acid may confer increased drought resistance to a plant; and the mutated nucleic acid is identified by subjecting a plurality of plants (or a plurality of plant seeds), each of which is genetically modified with a single member of the

plurality of mutated nucleic acids, to drought conditions, and selecting plants that exhibit increased resistance to the drought conditions. Drought assays can be applied to identify a mutated nucleic acid that confers better plant survival after short-term, severe water deprivation. Ion leakage can be measured in the context of a drought assay.

(335) As another example, in some cases, a mutated nucleic acid may confer increased resistance of a plant to a pathogen (e.g., a fungus; an insect; etc.); and the mutated nucleic acid is identified by subjecting a plurality of plants (or a plurality of plant seeds), each of which is genetically modified with a single member of the plurality of mutated nucleic acids, to the pathogen, and selecting plants that exhibit increased resistance to the pathogen.

(336) As another example, in some cases, a mutated nucleic acid may confer increased resistance of a plant to salt stress (e.g., high salt conditions; e.g., high NaCl concentrations); and the mutated nucleic acid is identified by subjecting a plurality of plants (or a plurality of plant seeds), each of which is genetically modified with a single member of the plurality of mutated nucleic acids, to high salt conditions, and selecting plants that exhibit increased resistance to the high salt conditions. Plants differ in their tolerance to NaCl depending on their stage of development; therefore seed germination, seedling vigor, and plant growth responses can be evaluated to determine resistance to high salt conditions.

(337) As another example, in some cases, a mutated nucleic acid may confer increased resistance of a plant to freezing; and the mutated nucleic acid is identified by subjecting a plurality of plants (or a plurality of plant seeds), each of which is genetically modified with a single member of the plurality of mutated nucleic acids, to low temperature (e.g., freezing) conditions, and selecting plants that exhibit increased resistance to the low temperature conditions.

(338) As another example, in some cases, a mutated nucleic acid may confer increased ability to germinate under high temperature conditions; and the mutated nucleic acid is identified by subjecting a plurality of plants (or a plurality of plant seeds), each of which is genetically modified with a single member of the plurality of mutated nucleic acids, to high temperature conditions, and selecting plants that exhibit increased resistance to the high temperature conditions. Parameters that can be tested include seed germination, seedling vigor, and plant growth.

(339) As another example, in some cases, a mutated nucleic acid may confer increased resistance to hyperosmotic stress; and the mutated nucleic acid is identified by subjecting a plurality of plants (or a plurality of plant seeds), each of which is genetically modified with a single member of the plurality of mutated nucleic acids, to hyperosmotic stress conditions, and selecting plants that exhibit increased resistance to the hyperosmotic stress conditions. Plants that are resistant to hyperosmotic stress may be more tolerant to drought or freezing.

(340) Sugar sensing assays can be conducted to identify mutated nucleic acids that provide for sugar sensing by germinating seeds on high concentrations of sucrose and glucose and looking for degrees of hypocotyl elongation. The germination assay on mannitol controls for responses related to hyperosmotic stress. Sugars are key regulatory molecules that affect diverse processes in higher plants including germination, growth, flowering, senescence, sugar metabolism and photosynthesis. Sucrose is the major transport form of photosynthate and its flux through cells has been shown to affect gene expression and alter storage compound accumulation in seeds (source-sink relationships). Glucose-specific hexose-sensing has also been described in plants and is implicated in cell division and repression of “famine” genes (photosynthetic or glyoxylate cycles).

(341) Crop productivity is in part limited by its rate of CO₂ fixation by the RuBisCo enzyme. In some cases, all RuBisCo catalytic subunits are simultaneously targeted within cyanobacteria using a system of the present disclosure; growing this microbe under high temperature and/or low CO₂ conditions will enrich for RuBisCo variants with improved catalytic efficiency and specificity.

(342) Base Editor

(343) In some cases, a system of the present disclosure (a fusion polypeptide of the present

disclosure and a guide RNA) is a base editor system that comprises a non-processive DNA polymerase that is biased in which mismatches it generates. Such as system provides for targeted base substitutions.

(344) In some cases, a base editor system of the present disclosure provides for generation of a G.fwdarw.T substitution. In some cases, a base editor system of the present disclosure provides for generation of a G.fwdarw.C substitution. In some cases, a base editor system of the present disclosure provides for generation of a C.fwdarw.A substitution. In some cases, a base editor system of the present disclosure provides for generation of a C.fwdarw.G substitution. In some cases, a base editor system of the present disclosure provides for generation of an A.fwdarw.T substitution. In some cases, a base editor system of the present disclosure provides for generation of an A.fwdarw.G substitution. In some cases, a base editor system of the present disclosure provides for generation of an A.fwdarw.C substitution. In some cases, a base editor system of the present disclosure provides for generation of a T.fwdarw.A substitution. In some cases, a base editor system of the present disclosure provides for generation of a T.fwdarw.G substitution. In some cases, a base editor system of the present disclosure provides for generation of a T.fwdarw.C substitution. In some cases, a base editor system of the present disclosure provides for generation of a G.fwdarw.A substitution. In some cases, a base editor system of the present disclosure provides for generation of a C.fwdarw.T substitution.

(345) A base-editor system of the present disclosure provides for generation of targeted substitutions that can reverse pathogenic single nucleotide polymorphisms. A base-editor system of the present disclosure provides for creation of crop plants with new alleles.

(346) In some cases, the substitution rate is 1 mutation per nucleotide per generation.

(347) A base editor system of the present disclosure comprises a DNA polymerase such as DNA polymerase beta, DNA polymerase iota, DNA polymerase nu, DNA polymerase eta, or DNA polymerase kappa. Variants of these polymerases that place the wrong nucleotide three times more often across from the template base than the correct complementary base can be used.

(348) In some cases, a fusion polypeptide of the present disclosure comprises: a) an enzymatically active RNA-guided endonuclease; and b) a DNA polymerase selected from a DNA polymerase-beta, a DNA polymerase-iota, a DNA polymerase nu, a DNA polymerase eta, and a DNA polymerase kappa.

(349) In some cases, a DNA polymerase beta suitable for inclusion in a fusion polypeptide of the present disclosure comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase R having the following amino acid sequence:

(350) TABLE-US-00007 (SEQ ID NO: 1163)

MSKRKAPQETLNGGITDMLTELANFEKNVSQAIHKYNAYR
KAASVIAKYPHKIKSGAEAKKLPGVGTKIAEKIDEFLÅTG
KLRKLEKIRQDDTSSSINFLTRVSGIGPSAARKFVDEGIK
TLEDLRKNEDKLNHHQRIGLKYFGDFEKRIPREEMLQMQR
IVLNEVKKVDSEYIATVCGSFRRGAESSGDMDVLLTHPSF
TSESTKQPKLLHQVVEQLQKVHFITDTLSKGETKFMGVQC
LPSKNDEKEYPHRRIDIRLIPKDQYYCGVLYFTGSDIFNK
NMRAHALEKGF TINEYTIRPLGV TGVAGEPLPVDSEKDIF DYIQWKYREPKDRSE.

(351) In some cases, a DNA polymerase iota suitable for inclusion in a fusion polypeptide of the present disclosure comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase iota having the following amino acid sequence:

MEKLGVEPEEEGGDDDEEDAEAWAMELADVGA AASSQGVHDQVLPTPNASSRVIVH
VDLDCFYAQVEMISNP ELKDKPLGVQQKYLVT CNYEARKLG VKKLMNVRDAKEKCP
QLVLVNGEDLTRYREMSYKVTELLEEFSPVVERLGF DENFVDLTEMVEKRLQQQLQSDE

LSAVTVSGHVYNNQNSINLLDVLHIRLLVGSQIAAEMREAMYNQLGLTGCAGVASNKLL
AKLVSGVFKPNQQT VLLPESCQHSLNHIKEIPGIGYKTAKCLEALGINSVRDLQTFSP
KILEKELGISVAQRIQKLSFGEDNSPVILSGPPQSFSEEDSFKKCSSEVEAKNKIEELLASLL
NRVCQDGRKPHTVRLIIRRYSSSEKHYGRESRQCPIPSHVIQKLTGTGNYDVMTPMVDILM
KLFRNMVNVKMPFHLTLLSVCFCNLKALNTAKKGLIDYYLMPSLSTTSRSGKHSFKMK
DTHMEDFPKDKETNRDFLPSGRIESTRTRESPLDTTNFSKEKDINEFPLCSLPEGVDQEVF
KQLPVDIQEEILSGKSREKFQKGSGVSCPLHASRGVLSFFSKKQMQDIPINPRDHLSSSKQ
VSSVSPCEPGTSGFNSSSSSYMSSQKDYSYYLDNRLKDERISQGPKEPQGFHFTNSNPAV
SAFHSFPNLQSEQLFSRNHTTDSHKQTVATDSHEGLTENREPDSVDEKITFPSPDIDPQVFI
ELPEAVQKELLA EWKRAGSDFHGHK (SEQ ID NO:1164). In some cases, such a DNA
polymerase generates T.fwdarw.G substitutions.

(352) In some cases, a DNA polymerase iota suitable for inclusion in a fusion polypeptide of the present disclosure comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase iota having the following amino acid sequence (amino acids 1-445 of DNA polymerase iota):

MEKLGVEPEEEGGDDDEEDAEAWAMELADVGAASSQGVHDQVLPTPNASSRVIVH
VDLDCFYAQVEMISNP ELKDKPLGVQKYLVTCTNYEAR KLGVKKLMNVRDAKEKCP
QLVLVNGEDLTRYREMSYKVTELLEEFSPVVERLGF DENFVDLTEMVEKRLQQLQSDE
LSAVTVSGHVYNNQNSINLLDVLHIRLLVGSQIAAEMREAMYNQLGLTGCAGVASNKLL
AKLVSGVFKPNQQT VLLPESCQHSLNHIKEIPGIGYKTAKCLEALGINSVRDLQTFSP
KILEKELGISVAQRIQKLSFGEDNSPVILSGPPQSFSEEDSFKKCSSEVEAKNKIEELLASLL
NRVCQDGRKPHTVRLIIRRYSSSEKHYGRESRQCPIPSHVIQKLTGTGNYDVMTPMVDILM
KLFRNMVNVKMPFHLTLLSVCFCNLKALNTAK (SEQ ID NO:1165); and having a length of
445 amino acids. In some cases, such a DNA polymerase generates T.fwdarw.G substitutions. In
some cases, such a DNA polymerase has a T.fwdarw.G error rate approaching 1.

(353) In some cases, a DNA polymerase iota suitable for inclusion in a fusion polypeptide of the present disclosure comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase iota having the following amino acid sequence (amino acids 26-445 of DNA polymerase iota):

ELADVGAASSQGVHDQVLPTPNASSRVIVHVDLDCFYAQVEMISNP ELKDKPLGVQK
KYLVTCTNYEAR KLGVKKLMNVRDAKEKCPQLVLVNGEDLTRYREMSYKVTELLEEF
SPVVERLGF DENFVDLTEMVEKRLQQLQSDELSAVTVSGHVYNNQNSINLLDVLHIRLLV
GSQIAAEMREAMYNQLGLTGCAGVASNKLLAKLVSGVFKPNQQT VLLPESCQHSLN
NHIKEIPGIGYKTAKCLEALGINSVRDLQTFSPKILEKELGISVAQRIQKLSFGEDNSPVILS
GPPQSFSEEDSFKKCSSEVEAKNKIEELLASLLNRVCQDGRKPHTVRLIIRRYSSSEKHYG
RESRQCPIPSHVIQKLTGTGNYDVMTPMVDILM KLFRNMVNVKMPFHLTLLSVCFCNLKA
LNTAK (SEQ ID NO:1166); and having a length of 419 amino acids. In some cases, such a DNA
polymerase generates T.fwdarw.G substitutions. In some cases, such a DNA polymerase has a
T.fwdarw.G error rate approaching 1.

(354) In some cases, a DNA polymerase iota suitable for inclusion in a fusion polypeptide of the present disclosure comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase nu (v) having the following amino acid sequence:

ENYEALVGFDLCNTPLSSVAQKIMSAMHSGDLVDSKTWKGSTETMEVINKSSVKYSVQ
LED RKTQSPEKKDLKSLRSQTSRGS AKLSPQSFSVRLTDQLSADQKQKSISSLTSSCLIP
QYNQEASVLQKKGHKRKHFLMENINNENKGSINLKRKHITYNNLSEKTSKQMALEEDT
DDAEGYLNSGNSGALKKHFC DIRHLDDWAKS QLIEMLKQAAALVITVMYTDGSTQLG
ADQTPVSSVRGIVVLVKRQAEGGHGCPDAPACGPVLEGFVSDDP CIYIQIEHSAIWDQE
EAHQQFARNVLFQTMKCKCPVICFNAKDFVRIVLQFFGNDG SWKHVADFIGLDPRIAA
WLIDPSDATPSFEDLVEKYCEKSITVKVNSTYGNSSRNIVNQNVRENKTL YRLTMDLCS

KLKDYGLFWQLFRLPIILAVMESHAIQVNKEEMKTSALLGARLKELEQEAHFVA
GERFLITSNNQLREILFGKLKLHLLSQRNSLPRTGLQKYPSTSEAVLNALRDLHPLPKIILE
YRQVHKIKSTFVDGLLACMKKGSISSTWNQTGTVTGRLSAKHPNIQGISKHPIQITTPKN
FKGKEDKILTISPRAMFVSSKGHTFLAADFSQIELRILTHLSGDPPELLKLFQESERDDVFST
LTSQWKDVPVEQVTHADREQTKKVYAVVYGAGKERLAACLGVP IQEAAQFLESFLQ
KYKKIKDFARAAIAQCHQTGCVVSIMGRRRPLPRIHAHDQQLRAQAERQAVNFVVQGS
AADLCKLAMIHVFTAVAASHTLTARLVAQIHDELLFEVEDPQIPECAALVRRTMESLEQ
VQALELQLQVPLKVSLSAGRSWGHVPLQEAWGPPPGPCRTESSNSLAAPGSPASTQP
PPLHFSPSFCL (SEQ ID NO:1167). In some cases, such a DNA polymerase generates G.fwdarw.T
substitutions.

(355) In some cases, a DNA polymerase eta suitable for inclusion in a fusion polypeptide of the
present disclosure comprises an amino acid sequence having at least 85%, at least 90%, at least
95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase eta
(η) having the following amino acid sequence:

(356) TABLE-US-00008 (SEQ ID NO: 1168)

MATGQDRVVALVDMDCFFVQVEQRQNPHLRNKPCAVVQYK
SWKGGGIIAVSYEARAFGVTRSMWADDAKKLCPDLLLAQV
RESRGKANLTKYREASVEVMEIMSRFAVIERASIDEAYVD
LTSAVQERLQKLQGQPISADLLPSTYIEGLPQGPTTAEET
VQKEGMRKQGLFQWLDLQIDNLTSPDLQLTVGAVIVEEM
RAAIERETGFQCSAGISHNKVLAKLACGLNKP NRQTLVSH
GSVPQLFSQMPIRKIRSLGGKLGASVIEILGIEYMGELTQ
FTESQLQSHFGEKNGSWLYAMCRGIEHDPVKPRQLPKTIG
CSKNFPGKTALATREQVQWWLLQLAQELEERLT KDRNDND
RVATQLVVSIRVQGD KRLSSLRRCALTRYDAH KMSHDAF
TVIKNCNTSGIQTEWSPPLTMLFLCATKFSASAPSSSTDI
TSFLSSDPSSLPKVPVTSSEAKTQGS GPAVTATKKATTSL
ESFFQKAAERQKVKEASLSSLTAPTQAPMSNSPSKPSLPF
QTSQSTGTEPFFKQKSLLLKQKQLNNSVSSPQQNPWSNC
KALPNSLPTEYPGCVPVCEGVSKLEESSKATPAEMDLAHN
SQSMHASSASKSVLEVTQKATPNPSLLAAEDQVPCEKCGS
LVPVWDMPEHMDYHFALELQKSFLQPHSSNPQVVSAVSHQ
GKRNPKSPLACTNKRPRPEGMQTLESFFKPLTH.

(357) In some cases, a DNA polymerase eta suitable for inclusion in a fusion polypeptide of the
present disclosure comprises an amino acid sequence having at least 85%, at least 90%, at least
95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase eta
(η) having the following amino acid sequence:

MATGQDRVVALVDMDCFFVQVEQRQNPHLRNKPCAVVQYKSWKGGGIIAVSYEARAF
GVTRSMWADDAKKLCPDLLLAQVRESRGKANLTKYREASVEVMEIMSRFAVIERASID
EAYVDLTSAVQERLQKLQGQPISADLLPSTYIEGLPQGPTTAEETVQKEGMRKQGLFQW
LDLQIDNLTSPDLQLTVGAVIVEEMRAAIERETGFQCSAGISHNKVLAKLACGLNKP NR
QTLVSHGSVPQLFSQMPIRKIRSLGGKLGASVIEILGIEYMGELTQFTESQLQSHFGEKNG
SWLYAMCRGIEHDPVKPRQLPKTIGCSKNFPGKTALATREQVQWWLLQLAQELEERLT
KDRNDNDRVATQLVVSIRVQGD KRLSSLRRCALTRYDAH KMSHDAFTVIKNCNTSGI
QTEWSPPLTMLFLCATKFSASAPSSSTDITSFLSSDPSSLPKVPVTSSEAKTQGS GPAVTA
TKKATTSL ESFFQKAAERQKVKEASLSSLTAPTQAPMSN (SEQ ID NO:1169); and has a
length of 511 amino acids.

(358) In some cases, a DNA polymerase kappa suitable for inclusion in a fusion polypeptide of the
present disclosure comprises an amino acid sequence having at least 85%, at least 90%, at least
95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase

kappa (κ) having the following amino acid sequence:

(359) TABLE-US-00009 (SEQ ID NO: 1170)

MDSTKEKCD SYKDDLLRLMGLNDNKAGMEGLDKEKINKII
MEATKGS RFYGNELKKEKQVNQRIENMMQQKAQITSQQLR
KAQLQVDRFAMELEQSRNLSNTIVHIDMDAFYAAVEMRDN
PELKDKPIAVGSM SMLSTS NYHARRFGVRAAMPGFIAKRL
CPQLIIVPPNFDKYRAVSKEVKEILADYDPNFMAMSLDEA
YLNITKHLEERQNW PEDKRRYFIKMGSSVENDNPGKEV NK
LSEHERSISPLLFEESPSDVQPPGDPFQVNFEEQNNPQIL
QNSVVFGTSAQEVVKEIRFRIEQKTTLTASAGIAPNTMLA
KVCSDKNKPNGQYQILPNRQAVMDFIKDLPIRKVSGIGKV
TEKMLKALGIITCTELYQQRALLSLLFSETSWHYFLHISL
GLGSTHLTRDGERKSMSVERTFSEINKAEEQYSLCQELCS
ELAQDLQKERLKGRTVTIKLKNVNFEVKTRASTVSSVVST
AEEIFAIKELLKTEIDADFP HPLRLRLMGVRIS SFPNEE
DRKHQQR SIIGFLQAGNQALSATECTLEKTDKDKFVKPLE
MSHKKSFFDKKR SERKWSHQDTFKCEAVNKQSFQTSQPFQ
VLKKKM NENLEISENSDDCQILTCPVCFRAQGCISLEALN
KHVDECLDGPSISENF KMFSCSHVSATKVNKKENVPASSL
CEKQDYEAHPKIKEISSVDCIALVD TIDNSSKAESIDALS
NKHSKEECSSLP SKSFNIEHCHQNSSSTVSLENEDVGSFR
QEYRQPYLCEVKTGQALVCPVCNVEQKTS DLT LFN VHVDV
CLNKSFIQELRKDKFNPVNQPKESSRSTGSSSGVQKAVTR
TKRPGLMTKYSTSKKIKPNNPKHTLDIFFK.

(360) In some cases, a DNA polymerase kappa suitable for inclusion in a fusion polypeptide of the present disclosure comprises an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to a DNA polymerase kappa (κ) having the following amino acid sequence:

MDSTKEKCD SYKDDLLRLMGLNDNKAGMEGLDKEKINKIIMEATKGS RFYGNELKKE
KQVNQRIENMMQQKAQITSQQLRKAQLQVDRFAMELEQSRNLSNTIVHIDMDAFYAAV
EMRDNPELKDKPIAVGSM SMLSTS NYHARRFGVRAAMPGFIAKRLCPQLIIVPPNFDKY
RAVSKEVKEILADYDPNFMAMSLDEAYLNITKHLEERQNW PEDKRRYFIKMGSSVEND
NPGKEV NK LSEHERSISPLLFEESPSDVQPPGDPFQVNFEEQNNPQILQNSVVFGTSAQEV
VKEIRFRIEQKTTLTASAGIAPNTMLAKVCSDKNKPNGQYQILPNRQAVMDFIKDLPIRK
VSGIGKVTEKMLKALGIITCTELYQQRALLSLLFSETSWHYFLHISLGLGSTHLTRDGER
KSMSVERTFSEINKAEEQYSLCQELCSELAQDLQKERLKGRTVTIKLKNVNFEVKTRAST
VSSVVSTAEEIFAIKELLKTEIDADFP HPLRLRLMGVRIS SFPNEEDRKHQQR SIIGFLQA
GNQALSATECTLEKTDKDKFVKPLE (SEQ ID NO:1171); and has a length of 560 amino acids.

(361) In some cases, a base editor system of the present disclosure comprises: a) a fusion polypeptide of the present disclosure (e.g., a fusion polypeptide comprising: i) an enzymatically active RNA-guided endonuclease; and ii) a DNA polymerase selected from a DNA polymerase-beta, a DNA polymerase-iota, a DNA polymerase nu, a DNA polymerase eta, and a DNA polymerase kappa); and b) a guide RNA that comprises a nucleotide sequence that comprises: i) a protein-binding segment comprising a nucleotide sequence that binds to the RNA-guided endonuclease; and ii) a target-binding segment comprising a nucleotide sequence that is complementary to a target nucleotide sequence in a target nucleic acid.

(362) In some cases, a base editor system of the present disclosure comprises: a) a nucleic acid (e.g., a recombinant expression vector) comprising a nucleotide sequence encoding a fusion polypeptide of the present disclosure (e.g., a fusion polypeptide comprising: i) an enzymatically active RNA-guided endonuclease; and ii) a DNA polymerase selected from a DNA polymerase-

beta, a DNA polymerase-iota, a DNA polymerase nu, a DNA polymerase eta, and a DNA polymerase kappa); and b) a guide RNA that comprises a nucleotide sequence that comprises: i) a protein-binding segment comprising a nucleotide sequence that binds to the RNA-guided endonuclease; and ii) a target-binding segment comprising a nucleotide sequence that is complementary to a target nucleotide sequence in a target nucleic acid.

(363) A fusion polypeptide of the present disclosure is useful for correcting a mutation in a target nucleic acid; i.e., in some cases, a fusion polypeptide of the present disclosure functions as a base editor. Thus, the present disclosure provides methods correcting a mutation in a target nucleic acid.

(364) In some cases, the target nucleic acid comprises a target nucleotide sequence associated with a disease or disorder. In some cases, the target nucleotide sequence comprises a point mutation associated with a disease or disorder. In some cases, a base editor system of the present disclosure introduces a single nucleotide mutation; i.e., changes a single nucleotide in a target nucleotide sequence.

(365) In some cases, the disease or disorder is cystic fibrosis; phenylketonuria; epidermolytic hyperkeratosis (EHK); Charcot-Marie-Toot disease type 4J; neuroblastoma (NB); von Willebrand disease (vWD); myotonia congenital; hereditary renal amyloidosis; dilated cardiomyopathy (DCM); hereditary lymphedema; familial Alzheimer's disease; Prion disease; chronic infantile neurologic cutaneous particular syndrome (CINCA); desmin-related myopathy (DRM); or a neoplastic disease associated with a mutant PI3KCA protein, a mutant CTNNB1 protein, a mutant HRAS protein, or a mutant p53 protein.

(366) A base editor system of the present disclosure is introduced into a cell (in vivo or in vitro), where the cell comprises a target nucleic acid comprising a target nucleotide sequence having a single nucleotide mutation that gives rise to a disease or disorder. For example, a eukaryotic cell comprising a mutation to be corrected, e.g., a cell carrying a point mutation, is contacted with a base editor system of the present disclosure. Suitable cells include mammalian cells, including, e.g., non-human primate cells, human cells, canine cells, feline cells, ungulate cells, etc. Suitable cells include plant cells. Suitable cells include insect cells. Suitable cells include arachnid cells.

(367) A base editor system of the present disclosure is capable of modifying a specific nucleotide base without generating a significant proportion of indels. An “indel”, as used herein, refers to the insertion or deletion of a nucleotide base within a nucleic acid. Such insertions or deletions can lead to frame shift mutations within a coding region of a gene. In some cases, it is desirable to generate base editors that efficiently substitute a specific nucleotide within a nucleic acid, without generating a large number of insertions or deletions (i.e., indels) in the nucleic acid. In cases, a base editor system of the present disclosure is capable of generating a greater proportion of intended modifications (e.g., nucleotide substitutions) versus indels. In some cases, a base editor system of the present disclosure is capable of generating a ratio of intended point mutations to indels that is greater than 1:1. In some cases, a base editor system of the present disclosure is capable of generating a ratio of intended point mutations to indels that is at least 1.5:1, at least 2:1, at least 2.5:1, at least 3:1, at least 3.5:1, at least 4:1, at least 4.5:1, at least 5:1, at least 5.5:1, at least 6:1, at least 6.5:1, at least 7:1, at least 7.5:1, at least 8:1, at least 10:1, at least 12:1, at least 15:1, at least 20:1, at least 25:1, at least 30:1, at least 40:1, at least 50:1, at least 100:1, at least 200:1, at least 300:1, at least 400:1, at least 500:1, at least 600:1, at least 700:1, at least 800:1, at least 900:1, or at least 1000:1, or more than 1000:1.

(368) Fusion Polypeptide Comprising a Modifying Enzyme, an RNA-Guided Nickase, and a DNA Polymerase

(369) The present disclosure provides a fusion polypeptide comprising: a) an RNA-guided enzyme that exhibits nickase activity; b) a modifying enzyme (e.g., a DNA-modifying enzyme or a protein-modifying enzyme); and c) a DNA polymerase. The RNA-guided enzyme that exhibits nickase activity, together with a guide RNA, guides the fusion polypeptide to a target nucleotide sequence in a target nucleic acid (e.g., a genome).

(370) In some cases, a fusion polypeptide of the present disclosure (e.g., a fusion polypeptide comprising: a) an RNA-guided enzyme that exhibits nickase activity; b) a DNA- or protein-modifying enzyme; and c) a DNA polymerase) is a base editor. Thus, in some cases, a base editor of the present disclosure is a fusion polypeptide comprising: a) an RNA-guided enzyme that exhibits nickase activity; b) a DNA- or protein-modifying enzyme; and c) a DNA polymerase. In some cases, a base editor fusion polypeptide of the present disclosure comprises, in order from N-terminus to C-terminus: a) an RNA-guided enzyme that exhibits nickase activity; b) a DNA- or protein-modifying enzyme; and c) a DNA polymerase. In some cases, a base editor fusion polypeptide of the present disclosure comprises, in order from N-terminus to C-terminus: a) a DNA polymerase; b) an RNA-guided enzyme that exhibits nickase activity; and c) a DNA-modifying enzyme. In some cases, a base editor fusion polypeptide of the present disclosure comprises, in order from N-terminus to C-terminus: a) a DNA- or protein-modifying enzyme; b) an RNA-guided enzyme that exhibits nickase activity; and c) a DNA polymerase. In these embodiments, the DNA polymerase can be a high-fidelity DNA polymerase, or a DNA polymerase that lacks substantial proofreading activity. In some cases, the DNA polymerase comprises a D424A substitution.

(371) Suitable DNA- and protein-modifying enzymes are known in the art. In some cases, the modifying enzyme is a nucleic acid-modifying enzyme. Examples of suitable nucleic acid-modifying enzymes include DNA editing enzymes; deaminases, such as activation induced deaminase and APOBEC proteins; nucleases; recombinases; glycosylases; methyltransferases; and the like. In other cases, the modifying enzyme is a protein-modifying enzyme. For example, the modifying enzyme may be one that modifies a protein associated with a nucleic acid. Examples of suitable protein-modifying enzymes include histone-modifying enzymes, acetylases, kinases, methyltransferases, ubiquitin ligases, SUMO ligases, demethylases, deacetylases, phosphatases, and the like.

(372) In some cases, a base editor fusion polypeptide of the present disclosure comprises, in order from N-terminus to C-terminus: a) a deaminase; b) a DNA polymerase; and c) an RNA-guided enzyme that exhibits nickase activity. Suitable deaminases include a cytidine deaminase and an adenosine deaminase. In some cases, a base editor fusion polypeptide of the present disclosure comprises, in order from N-terminus to C-terminus: a) a cytidine deaminase; b) an RNA-guided enzyme that exhibits nickase activity; and c) a DNA polymerase. In some cases, a base editor fusion polypeptide of the present disclosure comprises, in order from N-terminus to C-terminus: a) an adenine deaminase; b) an RNA-guided enzyme that exhibits nickase activity; and c) a DNA polymerase. In these embodiments, the DNA polymerase can be a high-fidelity DNA polymerase, or a DNA polymerase that lacks substantial proofreading activity. In some cases, the DNA polymerase comprises a D424A substitution. In some cases, the fusion polypeptide comprises a linker between the deaminase and the DNA polymerase, between the deaminase and the gene-editing enzyme, or between the gene-editing enzyme and the DNA polymerase.

(373) In some cases, a deaminase-containing base editor system of the present disclosure provides for generation of a G.fwdarw.T substitution. In some cases, a deaminase-containing base editor system of the present disclosure provides for generation of a G.fwdarw.C substitution. In some cases, a deaminase-containing base editor system of the present disclosure provides for generation of a C.fwdarw.A substitution. In some cases, a deaminase-containing base editor system of the present disclosure provides for generation of a C.fwdarw.G substitution. In some cases, a deaminase-containing base editor system of the present disclosure provides for generation of an A.fwdarw.T substitution. In some cases, a deaminase-containing base editor system of the present disclosure provides for generation of an A.fwdarw.G substitution. In some cases, a deaminase-containing base editor system of the present disclosure provides for generation of an A.fwdarw.C substitution. In some cases, a deaminase-containing base editor system of the present disclosure provides for generation of a T.fwdarw.A substitution. In some cases, a deaminase-containing base editor system of the present disclosure provides for generation of a T.fwdarw.G substitution. In

some cases, a deaminase-containing base editor system of the present disclosure provides for generation of a T.fwdarw.C substitution. In some cases, a deaminase-containing base editor system of the present disclosure provides for generation of a G.fwdarw.A substitution. In some cases, a deaminase-containing base editor system of the present disclosure provides for generation of a C.fwdarw.T substitution.

(374) DNA Polymerase

(375) A DNA- or protein-modifying enzyme-containing fusion polypeptide of the present disclosure (e.g., a deaminase-containing base editor of the present disclosure) comprises a DNA polymerase that is a high-fidelity DNA polymerase. In some cases, a DNA- or protein-modifying enzyme-containing fusion polypeptide of the present disclosure comprises a DNA polymerase lacks proof-reading activity. In some cases, a DNA- or protein-modifying enzyme-containing fusion polypeptide of the present disclosure comprises a DNA polymerase comprises a D424A substitution that exhibits increase nick translation activity. In some cases, a DNA- or protein-modifying enzyme-containing fusion polypeptide of the present disclosure comprises a DNA polymerase is not an error-prone DNA polymerase.

(376) Previous base-editors fused a DNA-editing enzyme to a programmable nuclease; such a fusion protein will chemically modify a target nucleotide in such a way that DNA polymerases will incorporate across from the chemically modified nucleotide a nucleotide that is not the same nucleotide type originally at this position. However, these previous base-editors rely on replication or repair machinery to perform this DNA polymerase-mediated new nucleotide insertion. A base editor fusion polypeptide of the present disclosure fuses a nick translating DNA polymerase to a deaminase and an RNA-guided enzyme that provides nickase activity, such that the chemically altered nucleotide will be used as a template for the fused DNA polymerase-mediated nick translation. This removes the dependence on replication and repair to achieve base-editing.

(377) In some cases, a deaminase-containing base editor of the present disclosure comprises a DNA polymerase lacks proof-reading activity. In some cases, a deaminase-containing base editor of the present disclosure comprises a DNA polymerase comprises a D424A substitution that exhibits increase nick translation activity. In some cases, a deaminase-containing base editor of the present disclosure comprises a DNA polymerase is not an error-prone DNA polymerase.

(378) In some cases, DNA- or protein-modifying enzyme-containing fusion polypeptide of the present disclosure (e.g., a deaminase-containing base editor of the present disclosure) comprises a DNA polymerase that comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following *E. coli* DNA polymerase I amino acid sequence:

(379) TABLE-US-00010 (SEQ ID NO: 1246)

VQIPQNPLILVDGSSYLYRAYHAFPPLTNSAGEPTGAMYG
VLNMLRSLIMQYKPTHAAVVFDAKGKTFRDELFEHYKSH
RPPMPDDLRAQIEPLHAMVKAMGLPLLAVSGVEADDVIGT
LAREAEKAGRPVLSTGDKDMAQLVTPNITLINTMTNTIL
GPEEVVNKYGVPELIIDFLALMGDSSDNIPGVPGVGEKT
AQALLQGLGLDLYAEPEKIAGLSFRGAKTMAAKLEQNK
EVAYLSYQLATIKTDVELELTCEQLEVQQPAAEELLGLFK
KYEFKRWTADVEAGKWLQAKGAKPAKPKQETSVADEAPEV
TATVISYDNYVTILDEETLKAWIAKLEKAPVFAFDTETDS
LDNISANLVGLSFAIEPGVAAYIPVAHDYLDAPDQISRER
ALELLKPLLEDEKALKVGQNLKYDRGILANYGIELRGIAF
DTMLESYILNSVAGRHDMDSLAERWLKHKTITFEEIAGKG
KNQLTFNQIALEEAGRYAAEDADVTLQLHLKMWPDQLQHK
GPLNVFENIEMPLVPVLSRIERNGVKIDPKVLHNSSELT
LRLAELEKKAHEIAGEEFNLSSTKQLQTILFEKQGIKPLK

KTPGGSEAPTEELALDYPLPKVILEYRGLAKLKSTY
TDKLPLMINPKTGRVHTSYHQAVTATGRLSSTDPNLQNIP
VRNEEGRRIRQAFIAPEDYVIVSADYSQIELRIMAHLSRD
KGLLTAFAGKDIHRATAAEVFGFLPLETVTSEQRRSAKAI
NFGLIYGMSAFGLARQLNIPRKEAQKYMDLYFERYPGVLE
YMERTRAQAKEQGYVETLDGRRRLYLPDIKSSNGARRAAAE
RAAINAPMQGTAADIHKRAMIAVDWLQAEQPRVRMIMQV
HDELVFEVHKDDVDAVAKQIHQLMENCTRLDVPLLVEVGS GENWDQAH.

(380) In some cases, a DNA- or protein-modifying enzyme-containing fusion polypeptide of the present disclosure (e.g., a deaminase-containing base editor of the present disclosure) comprises a DNA polymerase that comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following exonuclease-deficient *E. coli* DNA polymerase I amino acid sequence:

(381) TABLE-US-00011 (SEQ ID NO: 1144)

VQIPQNPLILVDGSSYLYRAYHAFPLTNSAGEPTGAMYGVNLMLRSLI
MQYKPTHAAVVFDAKKGKTRDELFEHYKSHRPPMPDDLRAQIEPLHAMV
KAMGLPLLAVSGVEADDVIGTLAREAEKAGRPVLISTGDKDMAQLVTPN
ITLINTMTNTILGPEEVVNKYGVPPELIIDFLALMGDSSDNIPGVPGVG
EKTAQALLQGLGGLDTLYAEPEKIAGLSFRGAKTMAAKLEQNKEVAYLS
YQLATIKTDVELELTCEQLEVQQPAAEELLGLFKKYEFKRWTADVEAGK
WLQAKGAKPAAKPQETSVADEAPEVTATVISYDNYVTILDEETLKAWIA
KLEKAPVFAFDTTETDSDLNISANLVGLSFAIEPGVAAYIPVAHDYLDAP
DQISRERALELLKPLLEDEKALKVGQNLKYARGILANYGIELRGIAFDT
MLESYILNSVAGRHDMDSLAERWLKHKTTITFEEIAGKGKNQLTFNQIAL
EEAGRYAAEDADVTLQLHLKMWPDQLQKHKGPLNVFENIEMPLVPVLSRI
ERNGVKIDPKVLHNHSEELTLRLAELEKKAHEIAGEEFNLSSTKQLQTI
LFEKQGKIKPLKKTPGGAPSTSEEVLLEALDYPLPKVILEYRGLAKLKS
TYTDKLPLMINPKTGRVHTSYHQAVTATGRLSSTDPNLQNIPVRNEEGR
RIRQAFIAPEDYVIVSADYSQIELRIMAHLSRDKGLLTAFAGKDIHRA
TAAEVFGFLPLETVTSEQRRSAKAINFGLIYGMSAFGLARQLNIPRKEAQ
KYMDLYFERYPGVLEYMERTRAQAKEQGYVETLDGRRRLYLPDIKSSNGA
RRAAAERAAINAPMQGTAADIHKRAMIAVDWLQAEQPRVRMIMQVHDE
LVFEVHKDDVDAVAKQIHQLMENCTRLDVPLLVEVGS GENWDQAH

Modifying Enzymes

(382) As noted above, suitable DNA- and protein-modifying enzymes are known in the art. In some cases, the modifying enzyme is a nucleic acid-modifying enzyme. Examples of suitable nucleic acid-modifying enzymes include DNA editing enzymes; deaminases, such as activation induced deaminase and APOBEC proteins; nucleases; recombinases; glycosylases; methyltransferases; and the like. In other cases, the modifying enzyme is a protein-modifying enzyme. For example, the modifying enzyme may be one that modifies a protein associated with a nucleic acid. Examples of suitable protein-modifying enzymes include histone-modifying enzymes, acetylases, kinases, methyltransferases, ubiquitin ligases, SUMO ligases, demethylases, deacetylases, phosphatases, and the like.

(383) Adenosine Deaminases

(384) Adenosine deaminases suitable for inclusion in a deaminase-containing base editor of the present disclosure include any enzyme that is capable of deaminating adenosine in DNA. In some cases, the deaminase is a TadA deaminase.

(385) In some cases, a suitable adenosine deaminase comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following amino acid sequence:

(386) TABLE-US-00012 (SEQ ID NO: 1247)
MSEVEFSHEYWMRHALTLAKRAWDEREVPVGAVLVHNNRVIGEGWNRPI
GRHDPTAHAEIMALRQGGLVMQNYRLIDATLYVTLEPCVMCAGAMIHSR
IGRVVFGARDAKTGAAGSLMDVLHHPGMNHRVEITEGILADECAALLSD
FFRMRRQEIKAKKAQSSTD

(387) In some cases, a suitable adenosine deaminase comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following amino acid sequence:

(388) TABLE-US-00013 (SEQ ID NO: 1248)
MRRAFITGVFFLSEVEFSHEYWMRHALTLAKRAWDEREVPVGAVLVHNN
RVIGEGWNRPIGRHDPTAHAEIMALRQGGLVMQNYRLIDATLYVTLEPC
VMCAGAMIHSRIGRVVFGARDAKTGAAGSLMDVLHHPGMNHRVEITEGI
LADECAALLSDDFFRMRRQEIKAKKAQSSTD.

(389) In some cases, a suitable adenosine deaminase comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following *Staphylococcus aureus* TadA amino acid sequence:

(390) TABLE-US-00014 (SEQ ID NO: 1200)
MGSHMTNDIYFMTLAIEEAKKAAQLGEVPIGAITKDDEVIARAHNLRE
TLQQPTAHAEHIAIERAAKVLGSRLEGCTLYVTLEPCVMCAGTIVMSR
IPRVVYGADDPKGGCSGLMNLLQQSNFNHRAIVDKGVLKEACSTLLTT FFK
NLRANKKSTN:

(391) In some cases, a suitable adenosine deaminase comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following *Bacillus subtilis* TadA amino acid sequence:

(392) TABLE-US-00015 (SEQ ID NO: 1201)
MTQDELYMKEAIKEAKKAEKGEVPIGAVLVINGEIIARAHNLRETEQR
SIAHAEMLVIDEACKALGTWRLEGATLYVTLEPCPMCAGAVVLSRVEKV
VFGAFDPKGGCSGTLMNLLQEERFNHQAEEVSGVLEEECGGMLSAFFRE
LRKKKKAARKNLSE

(393) In some cases, a suitable adenosine deaminase comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following *Salmonella typhimurium* TadA:

(394) TABLE-US-00016 (SEQ ID NO: 1202)
MPPAFITGVTSLSDVELDHEYWMRHALTLAKRAWDEREVPVGAVLVHNNH
RVIGEGWNRPIGRHDPTAHAEIMALRQGGLVLQNYRLDITLYVTLEPC
VMCAGAMVHSRIGRVVFGARDAKTGAAGSLIDVLHHPGMNHRVEIIEGV
LRDECATLLSDDFFRMRRQEIKALKKADRAEGAGPAV

(395) In some cases, a suitable adenosine deaminase comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following *Shewanella putrefaciens* TadA amino acid sequence: NO:1203)

(396) TABLE-US-00017 (SEQ ID NO: 1203)
MDEYWMQVAMQMAEKAEAAAGEVPVGAVLVKDGQQIATGYNLSISQHDPT
AHAEILCLRSAGKKLENYRLLDATLYITLPCAMCAGAMVHSRIARVVY
GARDEKTGAAGTVVNLLQHAPFNHQVEVTSGVLAEACSAQLSRFFKRRR
DEKKALKLAQRAQQGIE

(397) In some cases, a suitable adenosine deaminase comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following *Haemophilus influenzae* F3031 TadA amino acid sequence:

(398) TABLE-US-00018 (SEQ ID NO: 1204)
MDAAKVRSEFDEKMMRYALELADKAEALGEIPVGAVLVDDARNIIGEGW

NLSIVQSDPTAHAIALRNGAKNIQNYRLLNSTLYVTLEPCTMCAGAI
LHSRIKRLVFGASDYKTGAIGSRFHFFDDYKMNHTLEITSGVLAE ECSQ KLS
TFFQKRREEKKIEKALLKSLSDK

(399) In some cases, a suitable adenosine deaminase comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following *Caulobacter crescentus* TadA amino acid sequence:

(400) TABLE-US-00019 (SEQ ID NO: 1205)

MRTDESEDQDHRMMRLALDAARAAAEAGETPVGAVILDPSTGEVIATAG
NGPIAAHDPTAHAEIAAMRAAAAKLGNRYRLTDLTLVVTLEPCAMCAGAI
SHARIGRVVFGADDPKGGAVVHGPKEFAQPTCHWRPEVTGGVLADESAD
LLRGFFRARRKAKI

(401) In some cases, a suitable adenosine deaminase comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following *Geobacter sulfurreducens* TadA amino acid sequence:

(402) TABLE-US-00020 (SEQ ID NO: 1206)

MSSLKKTPIRDDAYWMGKAIREAAKAAARDEVPIGAVIVRDGAVIGRGH
NLREGSNDPSAHAEMIAIRQAARRSANWRLTGATLYVTLEPCLMCMGAI
ILARLERVVFGCYDPKGGAGSLYDLSADPRLNHQVRLSPGVCQEECGT
MLSDFFRDLRRRKAKATPALFIDERKVPPEP

Cytidine Deaminases

(403) Cytidine deaminases suitable for inclusion in a deaminase-containing base editor of the present disclosure include any enzyme that is capable of deaminating cytidine in DNA.

(404) In some cases, the cytidine deaminase is a deaminase from the apolipoprotein B mRNA-editing complex (APOBEC) family of deaminases. In some cases, the APOBEC family deaminase is selected from the group consisting of APOBEC1 deaminase, APOBEC2 deaminase, APOBEC3A deaminase, APOBEC3B deaminase, APOBEC3C deaminase, APOBEC3D deaminase, APOBEC3F deaminase, APOBEC3G deaminase, and APOBEC3H deaminase. In some cases, the cytidine deaminase is an activation induced deaminase (AID).

(405) In some cases, a suitable cytidine deaminase comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following amino acid sequence:

(406) TABLE-US-00021 (SEQ ID NO: 1207)

MDSLLMNRRKFLYQFKNVRWAKGRRETYLCYVVKRRDSATSFSLDFGYL
RNKNGCHVELLFLRYISDWDLDPGRCYRVTWFTSWSPCYDCARHVADFL
RGNPNLSLRIFTARLYFCEDRKAEPGLRRLHRAGVQIAIMTFKDYFYC
WNTFVENHERTFKAWEGLHENSRLSRQLRRILLPLYEVDDLRFRTLGL

(407) In some cases, a suitable cytidine deaminase is an AID and comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following amino acid sequence: MDSLLMNRRKFLYQFKNVRWAKGRRETYLCYVVKRRDSATSFSLDFGYLRNKNGCHVELLFLRYISDWDLDPGRCYRVTWFTSWSPCYDCARHVADFLRGNPNLSLRIFTARLYFCEDRKAEPGLRRLHRAGVQIAIMTFKENHERTFKAWEGLHENSRLSRQLRRILLPLYEVDDLRFRTLGL (SEQ ID NO:1208).

(408) In some cases, a suitable cytidine deaminase is an AID and comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following amino acid sequence: MDSLLMNRRKFLYQFKNVRWAKGRRETYLCYVVKRRDSATSFSLDFGYLRNKNGCHVELLFLRYISDWDLDPGRCYRVTWFTSWSPCYDCARHVADFLRGNPNLSLRIFTARLYFCEDRKAEPGLRRLHRAGVQIAIMTFKDYFYCWNTFVENHERTFKAWEGLHENSRLSRQLRRILLPLYEVDDLRFRTLGL (SEQ ID NO:1209).

(409) RNA-Guided Enzyme Exhibiting Nickase Activity

(410) As noted above, a DNA- or protein-modifying enzyme-containing fusion polypeptide of the present disclosure (e.g., a deaminase-containing base editor of the present disclosure) comprises an RNA-guided enzyme that exhibits nickase activity. Suitable nickases are described elsewhere herein.

(411) In some cases, a suitable RNA-guided enzyme that exhibits nickase activity comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following “nicking high fidelity” Cas9 amino acid sequence:

(412) TABLE-US-00022 (SEQ ID NO: 1210)

DKKYSIGLAIGTNSVGWAVITDEYKVPSSKKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDSSFFH
RLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDK
ADLRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFE
ENPINASGVDAKAILSARLSKSRLENLIAQLPGEKKNGLFGNLIALLSL
GLTPNFKSNFDLAEDAKLQLSKDITYDDDLNLLAQIGDQYADLFLAAKN
LSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQPL
EKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKL
NREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYFPLKDNREKIEK
ILTRIPYVVGPLARGNSRFAWMTRKSEETITPWNFEVVDKGASAQSF
IERMTAFDKNLPNEKVLPKHSLLYEYFTVYNELTKVKYVTEGMRKPAFL
SGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNA
SLGTYHDLKIIKDKDFLDNEENEDILEDIVLTTLTFEDREMIEERLKT
YAHLFDDKVMKQLKRRRYTGWGALSRLKINGIRDKQSGKTILDFLKSDG
FANRNFMALIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIIKG
ILQTVKVVDLKVVMGRHKPENIVIMARENQTTQKGQKNSRERMKRIE
EGIKELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVDQELDINRLS
DYDVDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMKMKNYW
RQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKRQLVETRAITKHVA
QILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKVVREINNY
HHAHDAYLNAVVGTAIIKKYPKLESEFVYGDYKVYDVRKMIKSEQEIG
KATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRD
FATVRKVLSPQVNIVKKTEVQTGGFSKESILPKRNSDKLIARKKDWDP
KKYGGFDSPTVAYSVLVVAKEKKGSKKLKSVKELLGITIMERSSSFEN
PIDFLEAKGYKEVKKDLIIKLPKYSLEFLENGRKRMLASAGELQKGNEL
ALPSKYVNFYLYASHYEKLKGSPEDNEQKQLFVEQHKHYLDEIIEQISE
FSKRVLADANLDKVL SAYNKH RDKPIREQAENIIHLFTLTNLGAPAAF
KYFDTTIDRKRYTSTKEVL DATLIHQ SITGLYETRIDLSQLGGD.

(413) In some cases, a suitable RNA-guided enzyme that exhibits nickase activity comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following “nicking enhanced” Cas9 amino acid sequence:

(414) TABLE-US-00023 (SEQ ID NO: 1211)

DKKYSIGLAIGTNSVGWAVITDEYKVPSSKKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDSSFFH
RLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDK
ADLRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFE
ENPINASGVDAKAILSARLSKSRLENLIAQLPGEKKNGLFGNLIALLSL
GLTPNFKSNFDLAEDAKLQLSKDITYDDDLNLLAQIGDQYADLFLAAKN
LSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQPL

EKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKL
NREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYFPLKDNREKIEK
ILTRIPYYYVGPLARGNSRFAWMTRKSEETITPWNFEVVDKGASAQSF
IERMTNFDKNLPNEKVLPHKSLLEYFTVYNELTKVKYVTEGMRKPAFL
SGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNA
SLGTYHDLLKIIKDKDFLDNEENEDILEDIVLTLTLFEDREMIEERLKT
YAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDG
FANRNFQMQLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIIKG
ILQTVKVVDDELVKVMGRHKPENIVIAMARENQTTQKGQKNSRERMKRIE
EGIKELGSQILKEHPVENTQLQNEKLYLYYLQNGRDMYVDQELDINRLS
DYDVDHIVPQSFLADDSIDNKVLTRSDKNRGKSDNVPSEEVVKMKMKNYW
RQLLNAKLITQRKFDNLTAKAERGGLSELDKAGFIKRQLVETRQITKHVA
QILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKVVREINNY
HHAHDAYLNAVVGTAIIKKYPALESEFVYGDYKVYDVRKMIKSEQEIG
KATAKYFFYSNIMNFFKTEITLANGEIRKAPLIETNGETGEIVWDKGRD
FATVRKVLSPQVNIVKKTEVQTGGFSKESILPKRNSDKLIARKKDWDP
KKYGGFDSPTVAYSVLVVAKEKKGSKKLKSVKELLGITIMERSSEFEKN
PIDFLEAKGYKEVKKDLIIKLPKYSLFELNGRKRMLASAGELQKGNEL
ALPSKYVNFYLYLASHYEKLKGSPEDEQKQLFVEQHKHYLDEIIEQISE
FSKRVLADANLDKVL SAYNKH RD KPIREQAENIIHLFTLTNLGAPAAF
KYFDTTIDRKRYTSTKEVL DATLIHQ SITGLYETRIDLSQLGGD.

(415) In some cases, a suitable RNA-guided enzyme that exhibits nickase activity comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following “nicking” Cas9 amino acid sequence:

(416) TABLE-US-00024 (SEQ ID NO: 1212)

DKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGA
LLFDSGETAEATRLKRTARRRYTRRKNRICYLQEIFS NEMAKVDDSFH
RLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDK
ADRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLF
ENPINASGVDAKAILSARLSKSRLENLIAQLPGEKKNGLFGNLIALSL
GLTPNFKSNFDLAEDAKLQLSKD TYDDDLDNLLAQIGDQYADLFLAAKN
LSDAILSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLP
EKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKL
NREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYFPLKDNREKIEK
ILTRIPYYYVGPLARGNSRFAWMTRKSEETITPWNFEVVDKGASAQSF
IERMTNFDKNLPNEKVLPHKSLLEYFTVYNELTKVKYVTEGMRKPAFL
SGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNA
SLGTYHDLLKIIKDKDFLDNEENEDILEDIVLTLTLFEDREMIEERLKT
YAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDG
FANRNFQMQLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIIKG
ILQTVKVVDDELVKVMGRHKPENIVIAMARENQTTQKGQKNSRERMKRIE
EGIKELGSQILKEHPVENTQLQNEKLYLYYLQNGRDMYVDQELDINRLS
DYDVDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMKMKNYW
RQLLNAKLITQRKFDNLTAKAERGGLSELDKAGFIKRQLVETRQITKHVA
QILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKVVREINNY
HHAHDAYLNAVVGTAIIKKYPKLESEFVYGDYKVYDVRKMIKSEQEIG
KATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRD
FATVRKVLSPQVNIVKKTEVQTGGFSKESILPKRNSDKLIARKKDWDP
KKYGGFDSPTVAYSVLVVAKEKKGSKKLKSVKELLGITIMERSSEFEKN

PIDFLEAKGYEVKLIKLPKYSLFLENGRKRM LASAGELQKGNEL
ALPSKYVNFYLYASHYEKLKGSPEDNEQKQLFVEQHKHYLDEIIEQISE
FSKRVLADANLDKVL SAYNKH RD KPIREQAENIIHLFTLTNLGAPAAF
KYFDTTIDRKRYTSTKEVL DATLIHQ SITGLYETRIDLSQLGGD.

UGI

(417) In some cases, a DNA- or protein-modifying enzyme-containing fusion polypeptide of the present disclosure (e.g., a deaminase-containing base editor of the present disclosure) further includes a uracil glycosylase inhibitor (UGI) polypeptide. The UGI polypeptide can be positioned at the N-terminus, at the C-terminus, or internally within the fusion polypeptide.

(418) In some cases, a suitable UGI polypeptide comprises an amino acid sequence having at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the following amino acid sequence:

(419) TABLE-US-00025 (SEQ ID NO: 1213)
TNLSDIIEKETGKQLVIQESILMLPEEVEEVIGNKPESDILVHTAYDES
TDENVMLLTSDAPEYKPWALVIQDSNGENKIKML.

NLS

(420) In some cases, a DNA- or protein-modifying enzyme-containing fusion polypeptide of the present disclosure (e.g., a deaminase-containing base editor of the present disclosure) includes a nuclear localization signal (NLS). In some cases, a DNA- or protein-modifying enzyme-containing fusion polypeptide of the present disclosure (e.g., a deaminase-containing base editor of the present disclosure) comprises a single NLS at the N-terminus of the fusion polypeptide. In some cases, a DNA- or protein-modifying enzyme-containing fusion polypeptide of the present disclosure (e.g., a deaminase-containing base editor of the present disclosure) comprises 2, 3, or 4 NLSs at the N-terminus of the fusion polypeptide. In other instances, a DNA- or protein-modifying enzyme-containing fusion polypeptide of the present disclosure (e.g., a deaminase-containing base editor of the present disclosure) comprises a single NLS at the C-terminus of the fusion polypeptide. In some cases, a DNA- or protein-modifying enzyme-containing fusion polypeptide of the present disclosure (e.g., a deaminase-containing base editor of the present disclosure) comprises 2, 3, or 4 NLSs at the C-terminus of the fusion polypeptide.

(421) Non-limiting examples of suitable NLSs include an NLS sequence derived from: the NLS of the SV40 virus large T-antigen, having the amino acid sequence PKKKRKV (SEQ ID NO:1172); the NLS from nucleoplasmin (e.g., the nucleoplasmin bipartite NLS with the sequence KRPAATKKAGQAKKKK (SEQ ID NO:1173)); the c-myc NLS having the amino acid sequence PAAKRVKLD (SEQ ID NO:1174) or RQRRNELKRSP (SEQ ID NO:1175); the hRNPA1 M9 NLS having the sequence NQSSNFGPMKGGNFGGRSSGPYGGGGQYFAKPRNQGGY (SEQ ID NO:1176); the sequence RMRIZFKNKGKDTAELRRRRVEVSVELRKAKKDEQILKRRNV (SEQ ID NO:1177) of the IBB domain from importin-alpha; the sequences VSRKRPRP (SEQ ID NO:1178) and PPKKARED (SEQ ID NO:1179) of the myoma T protein; the sequence PQPKKKPL (SEQ ID NO:1180) of human p53; the sequence SALIKKKKKMAP (SEQ ID NO:1181) of mouse c-abl IV; the sequences DRLRR (SEQ ID NO:1182) and PKQKKRK (SEQ ID NO:1183) of the influenza virus NS1; the sequence RKLKKKIKKL (SEQ ID NO:1185) of the Hepatitis virus delta antigen; the sequence REKKKFLKRR (SEQ ID NO:1185) of the mouse Mx1 protein; the sequence KRKGDEV DGVDEVAKKKSKK (SEQ ID NO:1186) of the human poly(ADP-ribose) polymerase; and the sequence RKCLQAGMNLEARKTKK (SEQ ID NO:1187) of the steroid hormone receptors (human) glucocorticoid. In some cases, an NLS comprises the amino acid sequence MDSL LMNR RKFLYQFKNVRWAKGRRETYLC (SEQ ID NO:1188). In general, NLS (or multiple NLSs) are of sufficient strength to drive accumulation of the fusion polypeptide in a detectable amount in the nucleus of a eukaryotic cell. Detection of accumulation in the nucleus may be performed by any suitable technique. For example, a detectable marker may be fused to the fusion polypeptide such that location within a cell may be visualized. Cell nuclei may

also be isolated from cells, the contents of which may then be analyzed by any suitable process for detecting protein, such as immunohistochemistry, Western blot, or enzyme activity assay. Accumulation in the nucleus may also be determined indirectly.

(422) System Comprising a Modifying Enzyme-Containing Fusion Protein

(423) The present disclosure provides a system comprising a DNA- or protein-modifying enzyme-containing fusion polypeptide of the present disclosure (e.g., a deaminase-containing base editor of the present disclosure). The system comprises: a) the DNA- or protein-modifying enzyme-containing fusion polypeptide; and b) a guide RNA.

(424) The present disclosure provides a base-editing system comprising a deaminase-containing base editor of the present disclosure. The system comprises: a) a base-editing system comprising a deaminase-containing base editor of the present disclosure; and b) a guide RNA. In some cases, the guide RNA is a single-molecule guide RNA. In some cases, the guide RNA is a dual-molecule guide RNA. In some cases, the guide RNA comprises a modified base and/or a modified sugar and/or a non-naturally occurring internucleoside linkage, as described above.

EXAMPLES OF NON-LIMITING ASPECTS OF THE DISCLOSURE

(425) Aspects, including embodiments, of the present subject matter described above may be beneficial alone or in combination, with one or more other aspects or embodiments. Without limiting the foregoing description, certain non-limiting aspects of the disclosure numbered 1-84 are provided below. As will be apparent to those of skill in the art upon reading this disclosure, each of the individually numbered aspects may be used or combined with any of the preceding or following individually numbered aspects. This is intended to provide support for all such combinations of aspects and is not limited to combinations of aspects explicitly provided below:

(426) Aspect 1. A fusion polypeptide comprising: a) an enzymatically active RNA-guided endonuclease that introduces a single-stranded break in a target DNA; and b) an error-prone DNA polymerase.

(427) Aspect 2. The fusion polypeptide of aspect 1, wherein the RNA-guided endonuclease is a class 2 CRISPR/Cas endonuclease.

(428) Aspect 3. The fusion polypeptide of aspect 2, wherein the class 2 CRISPR/Cas endonuclease is a type V or type VI CRISPR/Cas endonuclease.

(429) Aspect 4. The fusion polypeptide of aspect 2, wherein the class 2 CRISPR/Cas endonuclease is a Cas9 polypeptide.

(430) Aspect 5. The fusion polypeptide of aspect 4, wherein the Cas9 polypeptide comprises a mutation that reduces off-target binding.

(431) Aspect 6. The fusion polypeptide of aspect 3, wherein the RNA-guided endonuclease is a Cpf1 polypeptide.

(432) Aspect 7. The fusion polypeptide of any one of aspects 1-6, further comprising a nuclear localization signal.

(433) Aspect 8. The fusion polypeptide of any one of aspects 1-7, comprising a linker interposed between the RNA-guided endonuclease and the error-prone DNA polymerase.

(434) Aspect 9. The fusion polypeptide of any one of aspects 1-8, further comprising a DNA-binding polypeptide that increases the processivity of the DNA polymerase.

(435) Aspect 10. The fusion polypeptide of aspect 9, wherein the DNA-binding polypeptide that increases the processivity of the DNA polymerase is an Sso7d polypeptide.

(436) Aspect 11. The fusion polypeptide of aspect 9, wherein the DNA-binding polypeptide that increases the processivity of the DNA polymerase is a helix-hairpin-helix domain of topoisomerase I.

(437) Aspect 12. The fusion polypeptide of aspect 9, wherein the DNA-binding polypeptide that increases the processivity of the DNA polymerase is a thioredoxin binding domain of a T7 DNA polymerase or T3 polymerase.

(438) Aspect 13. The fusion polypeptide of any one of aspects 1-12, wherein the fusion

polypeptide, when complexed with a guide RNA, exhibits a target mutation rate of from 10.sup.-8 to 10.sup.-2 mutations per nucleotide per genome replication event.

(439) Aspect 14. The fusion polypeptide of any one of aspects 1-12, wherein the fusion polypeptide, when complexed with a guide RNA, exhibits a target mutation rate of from 10.sup.-6 to 10.sup.-5 mutations per nucleotide per genome replication event.

(440) Aspect 15. The fusion polypeptide of any one of aspects 1-12, wherein the fusion polypeptide, when complexed with a guide RNA, exhibits a target mutation rate of from 10.sup.-5 to 10.sup.-3 mutations per nucleotide per genome replication event.

(441) Aspect 16. The fusion polypeptide of any one of aspects 1-12, wherein the fusion polypeptide, when complexed with a guide RNA, exhibits a target mutation rate of from 10.sup.-3 to 10.sup.-2 mutations per nucleotide per genome replication event.

(442) Aspect 17. The fusion polypeptide of any one of aspects 1-12, wherein the fusion polypeptide, when complexed with a guide RNA, exhibits a target mutation rate that is at least 5-fold higher than the global mutation rate exhibited by the error-prone DNA polymerase not fused to the RNA-guided endonuclease.

(443) Aspect 18. The fusion polypeptide of any one of aspects 1-12, wherein the fusion polypeptide, when complexed with a guide RNA, exhibits a target mutation rate that is at least 10-fold higher than the global mutation rate exhibited by the error-prone DNA polymerase not fused to the RNA-guided endonuclease.

(444) Aspect 19. The fusion polypeptide of any one of aspects 1-12, wherein the fusion polypeptide, when complexed with a guide RNA, exhibits a target mutation rate that is at least 102-fold higher than the global mutation rate exhibited by the error-prone DNA polymerase not fused to the RNA-guided endonuclease.

(445) Aspect 20. The fusion polypeptide of any one of aspects 1-19, wherein the fusion polypeptide, when complexed with a guide RNA, introduces mutations at a distance of from 1 nucleotide to 104 nucleotides from a nick in a target DNA introduced by the RNA-guided endonuclease.

(446) Aspect 21. The fusion polypeptide of any one of aspects 1-19, wherein the fusion polypeptide, when complexed with a guide RNA, introduces mutations at a distance of from 1 nucleotide to 100 nucleotides from a nick in a target DNA introduced by the RNA-guided endonuclease.

(447) Aspect 22. The fusion polypeptide of any one of aspects 1-19, wherein the fusion polypeptide, when complexed with a guide RNA, introduces mutations at a distance of from 1 nucleotide to 50 nucleotides from a nick in a target DNA introduced by the RNA-guided endonuclease.

(448) Aspect 23. The fusion polypeptide of any one of aspects 1-22, wherein the fusion polypeptide comprises, in order from N-terminus to C-terminus: a) the enzymatically active RNA-guided endonuclease; and b) the error-prone DNA polymerase.

(449) Aspect 24. The fusion polypeptide of any one of aspects 1-22, wherein the fusion polypeptide comprises, in order from N-terminus to C-terminus: a) the enzymatically active RNA-guided endonuclease; b) a linker; and c) the error-prone DNA polymerase.

(450) Aspect 25. The fusion polypeptide of any one of aspects 1-22, wherein the fusion polypeptide comprises, in order from N-terminus to C-terminus: a) the error-prone DNA polymerase; and b) the enzymatically active RNA-guided endonuclease.

(451) Aspect 26. The fusion polypeptide of any one of aspects 1-22, wherein the fusion polypeptide comprises, in order from N-terminus to C-terminus: a) the error-prone DNA polymerase; b) a linker; and c) the enzymatically active RNA-guided endonuclease.

(452) Aspect 27. The fusion polypeptide of any one of aspects 1-26, wherein the fusion polypeptide comprises, in order from N-terminus to C-terminus a) a nuclear localization signal; b) an enzymatically inactive RNA-guided endonuclease; and c) an error-prone DNA polymerase.

(453) Aspect 28. The fusion polypeptide of any one of aspects 1-26, wherein the fusion polypeptide comprises, in order from N-terminus to C-terminus: a) a nuclear localization signal; b) an error-prone DNA polymerase; and c) an enzymatically inactive RNA-guided endonuclease.

(454) Aspect 29. The fusion polypeptide of any one of aspects 28, wherein the DNA polymerase comprises an amino acid sequence having at least 85% amino acid sequence to the DNA polymerase I amino acid sequence depicted in FIG. 8, wherein the DNA polymerase has one or more of the following: an Ala at amino acid position 242, an Asn at amino acid position 709, an Arg at amino acid position 759, a Tyr at amino acid position 742, and a His at amino acid position 796.

(455) Aspect 30. The fusion polypeptide of any one of aspects 28, wherein the DNA polymerase is a DNA polymerase beta, a DNA polymerase iota, a DNA polymerase nu, a DNA polymerase eta, or a DNA polymerase kappa.

(456) Aspect 31. The fusion polypeptide of aspect 30, wherein the fusion polypeptide, when complexed with a guide RNA, exhibits a target mutation rate of 1 mutation per nucleotide per genome replication event.

(457) Aspect 32. A system comprising: a) the fusion polypeptide of any one of aspects 1-31; and b) a guide RNA that comprises: i) a protein-binding segment comprising a nucleotide sequence that binds to the RNA-guided endonuclease; and ii) a target-binding segment comprising a nucleotide sequence that is complementary to a target nucleotide sequence in a target nucleic acid.

(458) Aspect 33. The system of aspect 32, wherein the guide RNA is a single-molecule guide RNA.

(459) Aspect 34. The system of aspect 32, wherein the guide RNA is a dual-molecule guide RNA.

(460) Aspect 35. The system of any one of aspects 32-34, wherein the guide RNA comprises one or more of: a) a modified base; b) a modified backbone; c) a modified sugar moiety; and d) a non-natural internucleoside linkage.

(461) Aspect 36. A cell comprising the fusion polypeptide of any one of aspects 1-31.

(462) Aspect 37. The cell of aspect 36, wherein the cell is a prokaryotic cell.

(463) Aspect 38. The cell of aspect 36, wherein the cell is a eukaryotic cell.

(464) Aspect 39. The cell of any one of aspects 36-38, wherein the cell is in vitro.

(465) Aspect 40. A cell comprising the system of any one of aspects 36-39.

(466) Aspect 41. The cell of aspect 40, wherein the cell is a prokaryotic cell.

(467) Aspect 42. The cell of aspect 40, wherein the cell is a eukaryotic cell.

(468) Aspect 43. The cell of any one of aspects 40-42, wherein the cell is in vitro.

(469) Aspect 44. A method of mutagenizing a target DNA, the method comprising contacting the target DNA with the system of any one of aspects 32-35.

(470) Aspect 45. The method of aspect 44, wherein the target DNA is present in a cell.

(471) Aspect 46. The method of aspect 44, wherein the cell is a prokaryotic cell.

(472) Aspect 47. The method of aspect 45, wherein the cell is a eukaryotic cell.

(473) Aspect 48. The method of aspect 47, wherein the cell is an animal cell.

(474) Aspect 49. The method of aspect 47, wherein the cell is a plant cell.

(475) Aspect 50. The method of aspect 49, wherein the plant cell is a cell of a dicotyledon.

(476) Aspect 51. The method of aspect 49, wherein the plant cell is a cell of a monocotyledon.

(477) Aspect 52. The method of any one of aspects 44-51, wherein the cell is in vitro.

(478) Aspect 53. The method of any one of aspects 44-51, wherein the cell is in vivo.

(479) Aspect 54. The method of any one of aspects 44-53, wherein mutations are introduced into the target DNA at a mutation rate of from 10^8 to 10^5 mutations per nucleotide per genome replication event.

(480) Aspect 55. The method of any one of aspects 44-53, wherein mutations are introduced into the target DNA at a mutation rate of from 10^6 to 10^5 mutations per nucleotide per genome replication event.

(481) Aspect 56. The method of any one of aspects 44-53, wherein mutations are introduced into

the target DNA at a mutation rate of from 10^5 to $10^{\text{sup.}-3}$ mutations per nucleotide per genome replication event.

(482) Aspect 57. The method of any one of aspects 44-53, wherein mutations are introduced into the target DNA at a mutation rate of from 10^3 to $10^{\text{sup.}-2}$ mutations per nucleotide per genome replication event.

(483) Aspect 58. A base editor system comprising: a) the fusion polypeptide of any one of aspects 29-31; and b) a guide RNA that comprises: i) a protein-binding segment comprising a nucleotide sequence that binds to the RNA-guided endonuclease; and ii) a target-binding segment comprising a nucleotide sequence that is complementary to a target nucleotide sequence in a target nucleic acid.

(484) Aspect 59. The system of aspect 58, wherein the guide RNA is a single-molecule guide RNA.

(485) Aspect 60. The system of aspect 58, wherein the guide RNA is a dual-molecule guide RNA.

(486) Aspect 61. The system of any one of aspects 58-60, wherein the guide RNA comprises one or more of: a) a modified base; b) a modified backbone; c) a modified sugar moiety; and d) a non-natural internucleoside linkage.

(487) Aspect 62. A method of editing a target nucleotide sequence in a target nucleic acid, the method comprising contacting the target nucleotide sequence with the base editor system of any one of aspects 58-61.

(488) Aspect 63. The method of aspect 62, wherein the target nucleotide sequence comprises a single nucleotide mutation that results in a disease or disorder in an organism.

(489) Aspect 64. A fusion polypeptide comprising: a) an enzymatically active RNA-guided endonuclease that introduces a single-stranded break in a target DNA; b) a DNA polymerase; and c) modifying enzyme.

(490) Aspect 65. The fusion polypeptide of aspect 64, wherein the modifying enzyme is a protein-modifying enzyme.

(491) Aspect 66. The fusion polypeptide of aspect 64, wherein the modifying enzyme is a DNA-modifying enzyme.

(492) Aspect 67. The fusion polypeptide of aspect 66, wherein the DNA-modifying enzyme is a cytidine deaminase or an adenosine deaminase.

(493) Aspect 68. The fusion polypeptide of aspect 64, wherein the RNA-guided endonuclease is a class 2 CRISPR/Cas endonuclease.

(494) Aspect 69. The fusion polypeptide of aspect 68, wherein the class 2 CRISPR/Cas endonuclease is a type V or type VI CRISPR/Cas endonuclease.

(495) Aspect 70. The fusion polypeptide of aspect 68, wherein the class 2 CRISPR/Cas endonuclease is a Cas9 polypeptide.

(496) Aspect 71. The fusion polypeptide of aspect 70, wherein the Cas9 polypeptide comprises a mutation that reduces off-target binding.

(497) Aspect 72. The fusion polypeptide of aspect 64, wherein the RNA-guided endonuclease is a Cpf1 polypeptide.

(498) Aspect 73. The fusion polypeptide of any one of aspects 64-72, further comprising a nuclear localization signal.

(499) Aspect 74. The fusion polypeptide of any one of aspects 64-73, comprising a linker between the RNA-guided endonuclease and the DNA polymerase.

(500) Aspect 75. The fusion polypeptide of any one of aspects 64-74, wherein the DNA polymerase is a high-fidelity DNA polymerase.

(501) Aspect 76. The fusion polypeptide of any one of aspects 64-75, wherein the fusion polypeptide further comprises a uracil glycosylase inhibitor (UGI).

(502) Aspect 77. A system comprising: a) the fusion polypeptide of any one of aspects 64-76; and b) a guide RNA that comprises: i) a protein-binding segment comprising a nucleotide sequence that binds to the RNA-guided endonuclease; and ii) a target-binding segment comprising a nucleotide

sequence that is complementary to a target nucleotide sequence in a target nucleic acid.

(503) Aspect 78. The system of aspect 77, wherein the guide RNA is a single-molecule guide RNA.

(504) Aspect 79. The system of aspect 77, wherein the guide RNA is a dual-molecule guide RNA.

(505) Aspect 80. The system of any one of aspects 77-79, wherein the guide RNA comprises one or more of: a) a modified base; b) a modified backbone; c) a modified sugar moiety; and d) a non-natural internucleoside linkage.

(506) Aspect 81. A method of editing a target nucleotide sequence in a target nucleic acid, the method comprising contacting the target nucleotide sequence with the system of any one of aspects 77-80.

(507) Aspect 82. A nucleic acid comprising a nucleotide sequence encoding the fusion polypeptide of any one of aspects 64-76.

(508) Aspect 83. A recombinant expression vector comprising the nucleic acid of aspect 82.

(509) Aspect 84. A cell comprising the nucleic acid of aspect 82 or the recombinant expression vector of aspect 83.

EXAMPLES

(510) The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure and description of how to make and use the present invention, and are not intended to limit the scope of what the inventors regard as their invention nor are they intended to represent that the experiments below are all or the only experiments performed. Efforts have been made to ensure accuracy with respect to numbers used (e.g. amounts, temperature, etc.) but some experimental errors and deviations should be accounted for. Unless indicated otherwise, parts are parts by weight, molecular weight is weight average molecular weight, temperature is in degrees Celsius, and pressure is at or near atmospheric. Standard abbreviations may be used, e.g., bp, base pair(s); kb, kilobase(s); pl, picoliter(s); s or sec, second(s); min, minute(s); h or hr, hour(s); aa, amino acid(s); kb, kilobase(s); bp, base pair(s); nt, nucleotide(s); i.m., intramuscular (ly); i.p., intraperitoneal (ly); s.c., subcutaneous (ly); and the like.

Example 1

(511) A schematic depiction of a mutagenesis tool of the present disclosure is provided in FIG. 1. A fusion polypeptide comprising an error-prone nick-translating DNAPol and a variant of Cas9 that nicks its target sequence (nCas9) is depicted in FIG. 1. The specificity of the polymerase initiation site created by the nCas9 offers single-base resolution for the start site of the editing window while the mutagenesis window length, mutation rate, and substitution bias are controlled by the polymerase processivity, fidelity, and misincorporation bias, characteristics chosen through polymerase variant selection. Because this genetic device will not rely on host machinery for the mutagenesis, it is readily adaptable to any transformation-competent organism. This mutagenesis tool is termed “EvolvR”, reflecting how evolution often begins with replicative errors.

(512) FIG. 1. Fusion of DNAPol to nCas9 for targeted mutagenesis. A nicking Cas9 (nCas9) fused to an error prone DNA polymerase is targeted to the region to be mutated with an sgRNA. After the nCas9 makes a single stranded break, the polymerase binds to the nick, and synthesizes a new strand with low fidelity while displacing and degrading the original strand.

(513) Mutator designs are tested for their ability to mutagenize target loci through fluctuation analysis, which determines mutation rates from the rate of reversing a nonsense mutation in an antibiotic resistance gene. In this assay, depicted schematically in FIG. 2A, a plasmid expressing the EvolvR components (pMut) is cotransformed into *E. coli* with a plasmid containing a nonfunctional antibiotic resistance gene (pNARG) containing a disabling nonsense mutation. After sixteen hours of growth without the antibiotic, the cultures are plated on the antibiotic and the mutation rates are determined by applying the Ma-Sandri-Sarkar statistical analysis to the colony counts.

(514) Preliminary fluctuation analysis estimated the mutation rate of wild-type *E. coli* to be approximately 10⁻¹⁰ mutations per nucleotide per generation, which is similar to a previously

reported 5.4×10^{-10} . The first mutator architecture tested was a translational fusion of nCas9 and the most well characterized nick-translating DNA polymerase, *E. coli* DNA polymerase I, with three fidelity-reducing mutations (PolI3M). These mutations include D424A, which inactivates proofreading activity, 1709N, which is thought to enlarge the substrate-binding pocket by disrupting the hydrophobic pocket, and A759R, which is thought to stabilize the polymerase's closed conformation promoting translocation after a misincorporation. Combining these mutations results in a polymerase that has a reported mutation rate of approximately 8.1×10^{-4} mutations per nucleotide.

(515) To control for changes in global mutation rate that occur as a result of expressing an error-prone polymerase in the cell, the rate of non-specific mutation accumulation will be determined by measuring the antibiotic-resistance reversion rate of cells carrying a pMut plasmid targeting the (fitness-neutral) *dbpA* gene in the *E. coli* genome. The targeted mutation rate will be determined with an sgRNA nicking 11 bases 5' of the nonsense mutation. FIG. 2B shows that nCas9 alone has no difference in global and targeted mutation rate. Expressing nCas9 and PolI3M as separate proteins elevated the global mutation rate ~100 fold and the targeted mutation rate ~1,000 fold. An N-terminally fused PolI3M maintains the ~100 fold increase in global mutation rate but provides a greater than 10,000 fold increase of mutation rate over wild-type at the targeted locus.

(516) To promote the unbinding of the nuclease after cleavage, it was hypothesized that DNA residence time could be reduced by introducing a set of mutations into the fused nCas9 that are suggested to lower Cas9's non-specific DNA affinity. Using this mutant nCas9 (enCas9) increased the global mutation rate 1.9 fold while the targeted mutation rate increased 8.6 fold compared to nCas9.

(517) FIG. 2C shows that the fidelity of the DNAPol confers the targeted and global mutation rate. PolI with a D424A mutation (PolI1M) was less mutagenic than PolI with both D424A and I709N mutations (PolI3M). PolI3M was the most mutagenic.

(518) FIG. 2A-2C. A fluctuation analysis using different mutator architectures will determine targetability of mutagenesis, a) A fluctuation analysis will determine mutation rates by cotransforming *E. coli* with a plasmid coding the mutator architecture (pMut) and a plasmid containing an antibiotic resistance gene with a nonsense mutation (pNARG) and counting resistant colonies after 16 hours of growth. b) A fluctuation analysis showed that nCas9 and PolI3M-nCas9 do not increase the global or targeted mutation rate, as determined by an off target sgRNA and sgRNA nicking 11 nucleotides 5' of the nonsense mutation, respectively. nCas9-PolI3M increased the global mutation rate 100 fold and increased the targeted mutation rate 10,000 fold. Replacing nCas9 with a variant proposed to have lower affinity for DNA (enCas9) increased the targeted mutation rate 8.6 fold while the global mutation rate increased 1.9 fold. c) enCas9 fused to PolI with a D424A mutation (PolI1M) was less mutagenic than enCas9 fused to PolI with D424A and 1709 mutations (PolI2M). enCas9 fused to PolI3M was that most mutagenic.

(519) The mutagenesis window length can be determined with a fluctuation analysis using sgRNAs targeting different distances from the antibiotic resistance gene's nonsense mutation. To test whether the mutagenesis window length is determined by the processivity of the polymerase, PolI with T7 DNA polymerase's thioredoxin binding domain (PolI-TBD) was tested. This chimera has previously been shown to have higher processivity than PolI. Indeed, FIG. 3 shows that while PolI does not show any increase in mutagenesis over background when targeted 56 nucleotides from the nonsense mutation, PolI-TBD does. This suggests that the mutagenesis window length can be tuned by modifying the polymerase's processivity. It was found that a mutation that abolishes the flap endonuclease activity only slightly decreases the mutation rate. This suggests that DNA polymerases without flap endonuclease activity can be used, likely relying on endogenous single stranded exonucleases for degrading the displaced strand.

(520) FIG. 3. Mutagenesis window length can be tuned by changing polymerase processivity, enCas9-PolI3M has a mutagenesis window length between 11 and 56 nucleotides. enCas9-PolI3M-

TBD has a mutagenesis window length between 56 and 197 nucleotides. Inactivating flap endonuclease activity slightly decreases targeted mutation rate.

(521) As shown in FIG. 4A, cells are transformed with EvolvR constructs targeted to a neutral genomic locus. The number of cell generations is estimated by quantifying the viable cell count before and after the growth period. The genomic target is amplified from purified genomes using polymerase chain reaction (PCR) primers that simultaneously add Illumina adapters. A second round of PCR add indexes to de-multiplex pooled sequence runs. The samples are then analyzed with 300-base pair paired-end sequencing on an Illumina MiSeq.

(522) FIG. 4A. Targeted amplification of genomic loci targeted with a mutator for a certain number of generations is sequenced with 300-bp completely overlapping paired-end sequencing with an Illumina MiSeq. Unmatching paired reads and low frequency variants are discarded. The variants are used to determine the distribution of types of substitution as well as the mutagenesis window length.

(523) High throughput sequencing is used to characterize the mutation rate, frequency of each type of nucleotide substitution and mutagenesis window length of the optimal architecture and polymerase combinations. The target region is amplified from purified genomes using PCR primers that simultaneously add Illumina adapters. A second round of PCR adds indexes to de-multiplex pooled sequencing runs. The samples are then analyzed with 300 base pair paired-end sequencing on an Illumina MiSeq. To prevent sequencing errors contributing false positive mutational events to the analysis, the paired-end reads is compared to each other and any unidentical pairs are discarded. After aligning to a reference, VarScan2 is used to call the variants. Any variant represented by more than 0.4% of the total reads is considered a true mutational event. The mutations identified are used to quantify the relative frequency of the 12 types of substitutions and will determine a 5 bp window rolling average mutation rate. FIG. 4B shows that variants can be captured at a frequency greater than 0.4% within a window of twenty nucleotides 3' of the nick using the enCas9-PolI3M construct and that all four nucleotides were substituted.

(524) FIG. 4B. (left panel) Frequency of single nucleotide variants after 24 hours of growth (“on”=sgRNA causing nicking at position 0; “off”=sgRNA targeting dbpA, a safespot in the genome; “vector”=empty plasmid without enCas9-PolITM construct). (right panel) Distribution of point mutants' original nucleotide identity.

Example 2

(525) EvolvR, a system that can continuously diversify all nucleotides within a tunable window length at user-defined loci was developed. This was achieved by directly generating mutations using engineered DNA polymerases targeted to desired loci via CRISPR-guided nickases. Nickase and polymerase variants that offer a range of targeted mutation rates up to 7,770,000-fold greater than wild type cells, and editing window lengths up to 350 nucleotides were identified. EvolvR was then used to identify novel ribosomal mutations that confer resistance to the antibiotic spectinomycin. The results demonstrate that CRISPR-guided DNA polymerases enable multiplexed and continuous diversification of user-defined genomic loci that will be useful for a broad range of basic and biotechnological applications.

(526) It was hypothesized-that recruiting an error-prone, nick-translating DNA polymerase with a nicking variant of Cas9 (nCas9) could offer an ideal HDR-independent targeted mutagenesis tool, termed EvolvR (FIG. 22A). The specificity of the polymerase initiation site created by the nCas9 specifies the start site of the editing window, while the mutagenesis window length, mutation rate, and substitution bias are controlled by the polymerase variant's processivity, fidelity, and misincorporation bias, respectively.

(527) FIG. 22A-22E| EvolvR (error-prone DNA polymerase fused to nicking Cas9) enables targeted mutagenesis. a, The EvolvR system consists of a CRISPR-guided nickase that nicks the target locus and a fused DNA polymerase which performs error-prone nick translation. b, High-throughput sequencing shows that fusing nicking Cas9 (nCas9) to *E. coli* DNA polymerase I with

the fidelity-reducing mutations D424A, I709N, and A759R (Poli3M) resulted in substitutions in a ~17 nucleotide window 3' from the nick. Only data points above 0.04% frequency were considered true substitutions (dotted line; see Methods). Expressing nCas9 fused to Poli3M with an off-target guide yielded only one substitution and at low-frequency, while an unfused nCas9 and Poli3M did not show substitutions at a frequency above 0.04%. c, All four nucleotides were substituted by nCas9-Poli3M. d, Description of fluctuation analysis workflow used to sensitively quantify targeted and global mutation rates. e, The global and targeted mutation rates of wild-type *E. coli*, nCas9 fused to Poli3M, unfused nCas9 and Poli3M, Poli3M alone, and nCas9 alone were determined by fluctuation analysis. On-target gRNA yields nicks 11 nucleotides 5' of the nonsense mutation. Fluctuation analysis was performed with ten replicates of each group, and error bars represent the 95% confidence intervals.

(528) In the initial design, nCas9 (*Streptococcus pyogenes* Cas9 harboring a D10A mutation) was fused to the N-terminus of a fidelity-reduced variant of *E. coli* DNA polymerase I (Poli) harboring the mutations D424A, I709N, and A759R (Poli3M).^{sup.21} A plasmid (pEvolvR) expressing the nCas9-Poli3M and a gRNA in *E. coli* was tested for its ability to mutate a second plasmid targeted by the gRNA over 24 hours of propagation. High-throughput targeted amplicon sequencing revealed that the target plasmid accrued substitutions in a ~17 nucleotide window 3' of the nick site (FIG. 22B), consistent with the established 15-20 nucleotide processivity of Poli.^{sup.22} The presence of low-frequency substitutions 5' of the nick site may be due to endogenous 3' to 5' exonucleases removing a few nucleotides 5' of the nick before the polymerase initiates synthesis. Critically, substitutions of all four nucleotide types were observed (FIG. 22C). In addition to substitutions, Poli is known to create deletions, which accounted for 25% of the observed variants in the 17 nucleotide window.^{sup.23} No insertions were detected. As controls, expressing nCas9-Poli3M with an off-target guide only showed one low-frequency substitution, while targeting pTarget with an unfused nCas9 and Poli3M, as well as nCas9 alone, did not show substitutions at a frequency greater than 0.04%.

(529) To sensitively quantify the mutation rate and mutagenesis window length of EvolvR variants, a fluctuation analysis was designed.^{sup.24} For this assay, the pEvolvR plasmid was cotransformed into *E. coli* with a plasmid (pTarget) containing the *aadA* spectinomycin resistance gene disabled by a nonsense mutation (FIG. 22D). After 16 hours of growth, the cultures were plated on spectinomycin and the mutation rates were determined from the number of resistant colony forming units (CFU). As shown in FIG. 22E, fluctuation analysis estimated the mutation rate of wild-type *E. coli* to be approximately 10.^{sup.}–10 mutations per nucleotide per generation, similar to previously reported values.^{sup.25} The global mutation rate (the mutation rate of the untargeted genome in cells expressing EvolvR) was determined by measuring the spectinomycin-resistance reversion rate of cells carrying a gRNA targeting *dbpA*, a fitness-neutral RNA helicase gene in the *E. coli* genome.^{sup.26} The targeted mutation rate was determined with a gRNA nicking 11 nucleotides 5' of the nonsense mutation in pTarget. Expressing nCas9 fused to the N-terminus of Poli3M dramatically increased the mutation rate at the targeted locus 24,500-fold over wild type while increasing the global mutation rate 120-fold over wild type (FIG. 22E), a global mutation rate comparable to that of previous *E. coli* targeted mutagenesis techniques.^{sup.1,27,28} By comparison, expressing nCas9 and Poli3M as separate proteins, Poli3M alone, nCas9 alone, or catalytically inactive Cas9 (dCas9) fused to Poli3M, showed significantly lower targeted mutation rates ($p < 0.05$ with a simple two-sided student's T test), suggesting that both Poli3M and the nick created by nCas9 are essential for EvolvR-mediated mutagenesis. Finally, by replacing the D10A nCas9, which nicks the strand complementary to the gRNA, with the H840A nickase, which nicks the strand non-complementary to the gRNA, it was found that the direction of EvolvR-mediated mutagenesis relative to the gRNA is dependent on which strand is nicked (FIG. 25).

(530) FIG. 23A-23H| EvolvR provides tunable mutation rates and mutagenesis window lengths, combinatorial mutations, multiplexed targeting, and continuous diversification of genomic loci. a,

Introducing mutations found to lower non-specific DNA affinity into the fused nCas9 (enCas9).sup.29 increased the global mutation rate 223-fold compared to the wild type mutation rate (1.9-fold greater than nCas9-PolI3M), while increasing the targeted mutation rate 212,038-fold above wild type (8.7-fold greater than nCas9-PolI3M). b, Mutagenesis rates were dependent on the fidelity of the polymerase. PolI with a D424A mutation (PolI1M) was less mutagenic than PolI with both D424A and I709N mutations (PolI2M), while PolI3M (D424A, I709N, A759R) was the most mutagenic. c, Screening mutations in PolI3M previously shown to decrease wild type PolI fidelity revealed that PolI3M with additional mutations F742Y and P796H (PolI5M) had a mutation rate 7,770,000-fold greater than wild type cells one nucleotide from the nick. Introduction of these mutations had no effect on the global mutation rate compared to PolI3M. Off-target mutation rates for the other variants were not measured. d, The editing window length was increased by incorporating the thioredoxin binding domain of bacteriophage T7 DNA polymerase into PolI3M (PolI3M-TBD). enCas9-PolI3M-TBD provided a targeted mutation rate 56 nucleotides from the nick that was 555-fold above the global mutation rate, while enCas9-PolI3M showed no targeted mutagenesis 56 nucleotides from the nick. e, enCas9-PolI3M-TBD targeted to a plasmid containing two nonsense mutations in the spectinomycin resistance gene (pTarget2) showed that EvolvR is able to generate combinations of multiple mutations. f, enCas9-PolI3M targeted to *E. coli*'s endogenous genomic ribosomal protein subunit 5 gene, rpsE, generated 16,000-fold more spectinomycin resistant colony forming units (SpecR CFU) than when targeted to the dbpA locus. g, enCas9-PolI3M-TBD targeted to rpsL increased the rate of acquiring streptomycin resistance without increasing the rate of acquiring spectinomycin resistance. Coexpression of both rpsL and rpsE gRNA's increased both spectinomycin and streptomycin resistant CFUs. h, Cultures expressing enCas9-PolI3M-TBD and either the rpsL gRNA or both rpsE and rpsL gRNAs grew in streptomycin-supplemented media while cultures expressing an off-target gRNA or the rpsE gRNA did not. After back-dilution into spectinomycin- and streptomycin-supplemented media, only cultures expressing both rpsE and rpsL gRNAs grew. (For all figures, on-target gRNA nicked 11 nucleotides 5' of the nonsense mutation unless labeled otherwise. Ten replicates of each group were performed for bar and scatter plots; error bars indicate 95% confidence intervals for mutation rates and one standard deviation from the mean for SpecR CFU/viable CFU and OD 600 nm. “*” denotes $p < 0.05$ with a simple two-sided student's T test).

(531) It was hypothesized that the targeted mutation rate could be further increased by promoting the dissociation of nCas9 from DNA after nicking the target locus. Therefore, a set of mutations (K848A, K1003A, R1060A) previously suggested to lower Cas9's non-specific DNA affinity.sup.30 were introduced into the fused nCas9. The resulting enhanced nicking Cas9 (enCas9) fused to PolI3M increased the global mutation rate 223-fold compared to wild type cells (1.9-fold greater than nCas9-PolI3M), yet elevated the mutation rate at the targeted locus by 212,000-fold (8.7-fold greater than nCas9-PolI3M) (FIG. 23A).

(532) PolI3M was initially chosen because it was the most error-prone variant of PolI previously characterized. However, it was hypothesized that the modularity of EvolvR would enable tuning of the mutation rate by using polymerases with different fidelities. First, it was confirmed that the fidelity of the polymerase determines mutation rates by comparing PolI variants, in decreasing order of fidelity: PolI1M (D424A), PolI2M (D424A, I709N), and PolI3M (D424A, I709N, A759R) (FIG. 23B). Next, to further increase EvolvR's targeted mutation rate, several additional mutations previously shown to individually decrease wild type PolI fidelity were screened.sup.21,31,32 (FIG. 24C). PolI3M with the additional mutations F742Y and P796H (PolI5M) displayed a mutation rate one nucleotide from the nick 7,770,000-fold greater than wild type cells, and 33-fold higher than PolI3M, making it the most error-prone PolI mutant ever reported. Surprisingly, PolI5M did not exhibit a higher global rate of mutagenesis than PolI3M (FIG. 24B) and did not show higher mutation rates than PolI3M 11 nucleotides from the nick (FIG. 26).

(533) It was hypothesized that using more processive DNA polymerases could increase the editing

window length, so PolI5M was exchanged for the more processive bacteriophage Phi29 DNA polymerase (Phi29). Coexpression of gRNAs targeting different distances from the nonsense mutation, either wild type Phi29 or variants with previously reported fidelity-reducing and thermostabilizing mutations, and the Phi29 single-stranded binding protein led to targeted mutagenesis both 56 and 347 nucleotides from the nick site (FIG. 27). However, the mutation rate at these distances was not as high as what was achieved with PolI3M at shorter distances.

(534) It was hypothesized that an alternative method to increase the editing window length while retaining high mutation rates would be to increase the processivity of PolI. Previous work has shown that fusing the thioredoxin binding domain (TBD) of bacteriophage T7 DNA polymerase increases PolI's processivity.^{sup.34} FIG. 23D shows that while the original enCas9-PolI3M did not show a difference between global and targeted mutation rates 56 nucleotides from the nick, incorporation of the TBD into the PolI3M EvolvR gene (enCas9-PolI3M-TBD) produced a 555-fold increase over the global mutation rate at this range. To leverage this increased editing window length, enCas9-PolI3M-TBD was targeted to a plasmid (pTarget2) containing two nonsense mutations (11 and 37 nucleotides from the nick) in the antibiotic resistance gene, and EvolvR's ability to generate combinations of multiple mutations with a single gRNA was thereby shown (FIG. 23E).

(535) It was hypothesized that unintended translation products consisting of functional DNA polymerase not fused to a functional CRISPR-guided protein contributed to undesirable off-target mutagenesis.^{sup.35} Therefore, the EvolvR coding sequence (enCas9-PolI3M-TBD-CO) was codon-optimized to remove three strong internal RBSs identified with the RBS Calculator.^{sup.36} The off-target mutation rate decreased 4.14-fold when expressing enCas9-PolI3M-TBD-CO compared to enCas9-PolI3M-TBD while the on-target mutation rate only decreased 1.23-fold (FIG. 28).

(536) The ability to couple EvolvR-mediated mutagenesis to a non-selectable genetic screen would considerably broaden EvolvR's utility. It was found that after targeting EvolvR to a plasmid containing a GFP cassette with an early termination codon, 0.06% and 0.07% of the population was expressing GFP while no cells were expressing GFP using an off-target gRNA (FIG. 29). Importantly, EvolvR also showed the capacity to diversify chromosomal loci by increasing the fraction of the population resistant to spectinomycin 16,000-fold after targeting enCas9-PolI3M to *E. coli*'s endogenous ribosomal protein subunit 5 gene, *rpsE*, which has mutations known to confer resistance to spectinomycin.^{sup.37} (FIG. 23F).

(537) Next, it was hypothesized that EvolvR avoids the cell-viability and evolutionary escape issues associated with non-targeted mutagenesis systems.^{sup.38} It was found that, unlike two previously developed non-targeted continuous mutagenesis systems, EvolvR does not impede cell viability or growth rate (FIGS. 30A and 30B). Additionally, targeting EvolvR to the *rpsE* gene evolved more spectinomycin resistant CFU's per mL compared to these previous *E. coli* non-targeted mutagenesis systems, even when normalized by optical density (FIG. 30).

(538) It was hypothesized that EvolvR could enable simultaneous diversification of distant genomic loci through coexpression of multiple gRNAs. First, expression of a gRNA targeting enCas9-PolI3M-TBD to *rpsL*, a ribosomal protein subunit gene with mutations known to confer streptomycin resistance.^{sup.39} increased the rate of acquiring streptomycin resistance compared to wildtype cells, without increasing spectinomycin resistant CFUs (FIG. 23G). In comparison, coexpression of the gRNAs targeting *rpsE* and *rpsL* generated approximately the same number of spectinomycin and streptomycin resistant CFUs as expression of the *rpsE* gRNA or *rpsL* gRNA alone, respectively. This capacity to simultaneously diversify multiple loci will be useful for identifying epistatic interactions. Expression of two gRNA's that nick separate strands at genomic loci separated by 100 base-pairs was lethal, while nicking the same strand at the same distance was not lethal. Therefore, if multiple gRNAs were used to increase the length of the target region, targeting the same strand is suggested.

(539) To evolve resistance to both spectinomycin and streptomycin, it was hypothesized that EvolvR's continuous diversity generation could be utilized for continuous directed evolution, in which mutagenesis, selection, and amplification occur simultaneously, to allow adaptation to modulated selection pressures with minimal researcher intervention. First, cultures expressing enCas9-PolI3M-TBD and either the rpsL gRNA or both rpsE and rpsL gRNAs grew in liquid media supplemented with streptomycin, whereas cultures expressing an off-target gRNA or the rpsE gRNA did not (FIG. 23H). After the cultures were diluted 1000-fold into liquid media supplemented with both spectinomycin and streptomycin, only cultures expressing both rpsE and rpsL gRNAs grew.

(540) FIG. 24A-24D| EvolvR identified novel mutations to *E. coli*'s Ribosomal Subunit 5 gene, rpsE, that confer spectinomycin resistance. a, Spectinomycin inhibits protein synthesis through interactions with the 30S ribosome. b, enCas9-PolI3M-TBD targeted to different parts of the endogenous rpsE gene with five gRNAs showed higher rates of spectinomycin resistance than targeting dbpA (off-target). c, After selection, high-throughput sequencing of the resistant cells containing gRNA's A, B, and C revealed that all twelve types of substitutions as well as deletions (del) were generated. Error bars indicate one standard deviation from the mean. d, Five mutations not previously described as conferring spectinomycin resistance were regenerated in a new strain of *E. coli* (RE1000), and growth curves in varying concentrations of spectinomycin confirmed the mutations provide spectinomycin resistance. Shaded area represents $\pm$ one standard deviation from the mean of three biological replicates.

(541) Spectinomycin's clinical utility as a broad-spectrum antibiotic has motivated previous efforts to characterize genomic mutations conferring spectinomycin resistance.^{sup.40,41} EvolvR's capacity was used to diversify the genomic rpsE gene to identify novel mutations that confer spectinomycin resistance by disrupting the spectinomycin binding pocket of the 30S ribosome (FIG. 24A). First, enCas9-PolI3M-TBD was targeted to five dispersed loci in the endogenous rpsE gene using gRNA's that nick after the 119.^{sup.th}, 187.^{sup.th}, 320.^{sup.th}, 403.^{sup.rd}, or 492.^{sup.nd} base pair within the 504 base pair rpsE coding sequence (FIG. 31). Then, the cell populations were challenged for growth on agar plates supplemented with varying concentrations of spectinomycin and observed that resistance was highest with the gRNAs targeted to the domain of the ribosomal subunit protein proposed to interact with spectinomycin (FIG. 24). After selection, high-throughput sequencing of the resistant cells containing gRNA's A, B, and C revealed that all twelve types of substitutions, as well as deletions, were generated (FIG. 24C). For functional analysis, five of the candidate mutations not previously described as providing spectinomycin resistance were introduced into a different strain of *E. coli* (RE1000) using oligonucleotide-mediated recombination. Growth curves in varying concentrations of spectinomycin confirmed that each of the five mutations (Δ 17-19; K23N, Δ 24; Δ 24; Δ 26; G27D) provided varying levels of spectinomycin resistance, but reduced fitness in the absence of selection (FIG. 24D). Using these mutations, it was hypothesized that mutations that move Lys26 relative to the spectinomycin binding pocket confer resistance to spectinomycin by removing a hydrogen bond that stabilizes spectinomycin's interaction with the ribosome, which led to the discovery of more novel spectinomycin resistance-conferring mutations (FIG. 32). These mutations were accordingly submitted to the Antibiotic Resistance Genes Database.^{sup.42} This rapid method to determine genotypes conferring antibiotic resistance is generally useful for improving the effective use of antibiotics.

(542) FIG. 25| The direction of EvolvR-mediated mutagenesis relative to the gRNA is dependent on which strand is nicked. The previous fluctuation analysis in FIG. 22E demonstrated that nCas9-PolI3M mutates a window 3' of the nick site. Here the study directly tested whether mutations are being generated 5' of the nick site using a different gRNA. Since DNA polymerases synthesize in the 5'-3' direction, it was hypothesized that nCas9-PolI3M would not provide an elevated mutation rate 5' of the nick site. The study indeed found that expressing a guide RNA which targeted nCas9

(D10A)-PolI3M to nick 16 nucleotides 3' from the nonsense mutation (red "x") did not show targeted mutagenesis. It was hypothesized that targeted mutagenesis could be induced using the same gRNA by using a Cas9 variant harboring the H840A mutation, which nicks the DNA strand non-complementary to the gRNA, rather than the D10A mutation, which nicks the strand complementary to the gRNA. It was found that nCas9 (H840A)-PolI3M increased the mutation rate 16 nucleotides 3' from the nick by 52 fold compared to the global mutation rate of cells expressing an off-target gRNA. The D10A nCas9 variant was used for all subsequent experiments. "*" denotes $p < 0.05$ with a simple two-sided student's T test).

(543) FIG. 26| PolI5M elevates mutation rate 1 but not 11 nucleotides from the nick compared to PolI3M. PolI3M with F742Y and P796H mutations (PolI5M) elevates the mutation rate 33 fold 1 nucleotide from the nick compared to PolI3M. PolI5M did not have a higher mutation rate than PolI3M 11 nucleotides from the nick. "*" denotes $p < 0.05$ with a simple two-sided student's T test).

(544) FIG. 27| Fusing a more processive DNA polymerase to enCas9 increases the target window length. PolI was exchanged for a more processive and higher fidelity bacteriophage Phi29 DNA polymerase (Phi29). Due to Phi29 not having a flap endonuclease, residues 1-325 of PolI were inserted between enCas9 and Phi29. Using gRNA's targeting different distances from the nonsense mutation, it was found that Phi29 with two previously reported fidelity-reducing mutations (N62D and L384R) elevated the mutation rate 56 nucleotides from the nick compared to the global mutation rate.^{sup.33,48}. When Phi29's single-stranded binding protein (ssb), known to improve the activity of Phi29, was expressed, an elevation in targeted mutation rate was seen.^{sup.49}. Finally, since the activity of Phi29 is known to decrease at temperatures above 30° C. and the fluctuation analysis was performed at 37° C., mutations previously reported to improve the thermostability of Phi29 (iPhi29) were added and a targeted mutation rate 347 nucleotides from the nick site that was significantly greater than the global mutation rate was observed.^{sup.50}. Unfortunately, mutations decreasing Phi29's fidelity are known to decrease its processivity explaining the inability to identify Phi29 variants that retain high processivity while offering as high of a mutation rate as PolI3M.^{sup.33}. (Ten replicates of each group; error bars indicate 95% confidence intervals for mutation rates; "*" denotes $p < 0.05$ with a simple two-sided student's T test).

(545) FIG. 28| Removing internal ribosome binding sequences decreases EvolvR-mediated off-target mutagenesis. enCas9-PolI3M-TBD-CO is enCas9-PolI3M-TBD that was codon optimized to remove strong ribosome binding sites in the EvolvR coding sequence predicted to produce an untethered DNA polymerase. The off-target mutation rate decreased 4.14-fold when expressing enCas9-PolI3M-TBD-CO compared to enCas9-PolI3M-TBD while the on-target mutation rate only decreased 1.23-fold.

(546) FIG. 29A-29B| EvolvR-mediated mutagenesis can be coupled with a non-selectable genetic screen. a, To test the capability for EvolvR-mediated mutagenesis to be coupled with a non-selectable genetic screen, a target plasmid containing a GFP cassette with an early termination codon in the GFP coding sequence (pTarget-GFP*) was designed. After cotransforming pEvolvR with pTarget-GFP* and growing for 24 hours, the GFP positive fraction was analyzed and sorted. In the two replicates expressing an off-target gRNA, no GFP cells were detected or sorted. In contrast, for the two replicates expressing a gRNA nicking four nucleotides away from the chain-terminating mutation in GFP's coding sequence, it was found that 0.06% and 0.07% of the total cells were GFP positive. These results agree with sequencing outcomes from FIG. 22B which showed that expressing nCas9-PolI3M for 24 hours produces substitutions in the target region at frequencies between 0.5% to 1%. b, After culturing the sorted populations, both replicates expressing an off-target gRNA did not show growth, while both replicates expressing the on-target gRNA grew bright green.

(547) FIGS. 30A-30C| EvolvR enables targeted genome diversification without affecting viability or growth rate. a, The viability of TG1 *E. coli* expressing EvolvR targeted to the essential rpsE gene was significantly higher than TG1 *E. coli* transformed with the MP6 plasmid and induced

with 25 mM arabinose and 25 mM glucose (as previously described) as well as XL1-Red *E. coli*. Viability was measured relative to TG1 *E. coli* transformed with an empty control plasmid. b, TG1 *E. coli* transformed with an empty control plasmid and TG1 *E. coli* transformed with pEvolvR targeting the *rpsE* gene showed similar growth curves while XL1-Red *E. coli* and TG1 *E. coli* transformed with MP6 plasmid and induced with 25 mM arabinose and 25 mM glucose grew much slower and saturated at lower final optical densities. c, The spectinomycin resistant CFU per mL saturated culture normalized by optical density of TG1 targeting EvolvR to the *rpsE* gene was significantly higher than XL1-Red *E. coli* and TG1 *E. coli* transformed with MP6 plasmid and induced with 25 mM arabinose and 25 mM glucose. Asterisks (*) denote $p < 0.05$ in two-tailed t-test.

(548) FIG. 31| Locations of gRNA targets and novel mutations relative to the *rpsE* gene. enCas9-PolI3M-TBD was targeted to five dispersed loci in the endogenous *rpsE* gene using gRNA's that nick after the 119.sup.th, 187.sup.th, 320.sup.th, 403.sup.rd, or 492.sup.nd base pair of the 504 base pair *rpsE* coding sequence. The locations of the previously identified *rpsE* mutations that provide spectinomycin resistance are colored orange and the region where new spectinomycin resistance mutations were identified is highlighted in red.

(549) FIGS. 32A-32B| Deletions in ribosomal protein S5 confer spectinomycin resistance. a, The mutations that were discovered confer spectinomycin resistance would be expected to move Lys26, which is predicted to hydrogen bond with spectinomycin, relative to the spectinomycin binding pocket. It was hypothesized that mutations that move Lys26 relative to the spectinomycin binding pocket removes that hydrogen bond and destabilizes spectinomycin's interaction with the ribosome, thereby conferring spectinomycin resistance. b, Therefore, the study tested whether deleting any single amino acid between residue 16 and 35 confers spectinomycin resistance. It was found that deleting residue 23, 24, 25, 26, 27, or 28 all provide spectinomycin resistance, while deleting any of the residues between 16 to 22 or 29 to 35 do not. This supports the hypothesis that one mechanism of resistance to spectinomycin is disruption of the interaction between Lys26 and spectinomycin.

(550) Methods

(551) Plasmid Construction: All plasmids were constructed using a modular Golden Gate strategy.sup.43. pEvolvR consisted of EvolvR and gRNA expression cassettes, a pBR322 origin of replication, and a kanamycin resistance cassette. pTarget consisted of a p15a origin of replication carrying both a functional trimethoprim resistance cassette for selection and a disabled spectinomycin resistance gene (*aadA*) harboring a Leu106Ter non-sense mutation. pTarget2 is identical to pTarget with the exception that the *aadA* gene now carried both Glu98Ter and Leu106Ter mutations. The full plasmid sequences are provided in FIG. 36.

(552) High-throughput sequencing of pTarget sample preparation: A pTarget and pEvolvR plasmid were cotransformed into 50 μ L of chemically competent TG1 *E. coli* prepared by a TSS/KCM method. Cells were allowed to recover in the TSS/LB solution for 1 hour, before 4 μ L of the transformation mix was inoculated into 2 mL of LB containing 25 μ g/mL kanamycin and 15 μ g/mL trimethoprim. The cultures were grown for 24 hours at 37° C. while shaking at 750 rpm. 1.5 mLs of each culture was miniprepmed using a Zippy Plasmid Prep kit (Zymo Research).

(553) The oligos pTarget-F and pTarget-R were used to amplify the target region in a 20 cycle PCR reaction using 100 ng of miniprepmed DNA as the template. A second PCR reaction added Illumina sequencing adapters and indices to the previous PCR product over 10 thermocycles. A Qubit fluorimeter was used to quantify the DNA prior to pooling samples. The sample pool was submitted to the UC Berkeley Vincent J. Coates Genomics Sequencing Laboratory for quality control and sequencing. Quality control consisted of fragment analysis (Advanced Analytical) and concentration measurement of the sequenceable fraction by quantitative PCR (Kapa Biosystems). The pooled sample was mixed with Illumina PhiX sequencing control library at 10% molarity, diluted to 14 pM, denatured, and run on an Illumina MiSeq using a 150 bp paired-end read MiSeq

Reagent Kit v2. Resulting basecall files were converted into demultiplexed fastq format using Illumina's version 2.17 bcl2fastq.

(554) High-throughput sequencing data analysis: Perfectly complementary paired reads were filtered and the 5 randomized nucleotides, amplification primer sequences, and first and last three nucleotides were trimmed using a custom python script. Bwa and samtools were used to generate alignment files using the wild type aadA gene sequence as a reference. VarScan2 was used for variant calling with the parameters: min-coverage 1; min-reads2 1; variants 1; min-var-freq 0.0005; p-value 0.99.sup.44. The limit of detection was determined by sequencing a culture transformed with an empty vector as a control. The highest frequency variant was 0.04% so all substitutions with a frequency under 0.05% were discarded.

(555) Fluctuation analysis assay: 50 μ L of chemically competent TG1 *E. coli* were contrtransformed with pEvolvR and pTarget or pTarget2. After 1 hour of recovery at 37° C., 4 μ L was inoculated into a 1.996 mL LB containing 25 μ g/mL Kanamycin and 15 μ g/mL Trimethoprim. After shaking at 37° C. for 16 hours, 1 mL and 1 μ L of culture were plated on separate LB agar plates containing 50 μ g/mL spectinomycin. Additionally, for viable CFU counting, 300 μ L of 1:50,000,000 diluted culture was plated on LB agar plates. After 24 hours of incubation at 37° C., spectinomycin resistant CFU and viable CFU were counted. Ten replicates were used for each condition.

(556) Calculation of mutation rate and statistics: The Ma-Sandri-Sarkar Maximum Likelihood Estimator was used to determine mutation rates as it is the most accurate and valid for all mutation rates.sup.24. Falcor was used to calculate the mutation rates by inputting the viable and resistant CFU counts for the ten replicates.sup.45. A simple two-tailed students t-test was carried out to determine p values as previously described.sup.46.

(557) Fluorescence-activated cell sorting of EvolvR libraries: Sony Cell Sorter SH800: pEvolvR expressing either an on-or off-target gRNA was contrtransformed with pTarget-GFP* and shaken at 37° C. for 24 hours. For each sample, the GFP positive fraction of a million events was sorted with a Cell Sorter SH800 (Sony) using a 488 nm laser and a 525/50 nm emission filter.

(558) Continuous evolution of *E. coli* resistant to both spectinomycin and streptomycin: pEvolvR expressing enCas9-PolI3M-TBD and either the off-target gRNA (targeting dbpA), rpsL gRNA, rpsE gRNA, or both rpsL and rpsE gRNAs was transformed into TG1 *E. coli* as previously described. After recovering for one hour, 4 μ L of transformation mix was inoculated into 2 mL of LB supplemented with 25 μ g/mL kanamycin and cultures were propagated over 16 hours at 37° C. 2 μ L of culture was inoculated into 198 μ L of LB supplemented with 50 μ g/mL of streptomycin. A Tecan M1000 Pro spectrophotometer was used to measure each well's optical density over 12 hours of growth at 37° C. Each well was then diluted 1000-fold in LB supplemented with 50 μ g/mL of streptomycin and 25 μ g/mL of spectinomycin and the optical density of 200 μ L of culture was again measured with a Tecan M1000 Pro spectrophotometer over 24 hours of growth at 37° C. Three biological replicates of each gRNA was characterized.

(559) High-throughput sequencing of spectinomycin resistant *E. coli*: A pEvolvR plasmid expressing enCas9-PolI3M-TBD with rpsE-gRNA-A, -B, -C, -D, or -E were transformed into chemically competent TG1 *E. coli* and recovered for 1 hour before innoculating 4 μ L of the transformation mix into 1.996 mL of LB supplemented with 25 μ g/mL kanamycin. The cultures were grown for 16 hours at 37° C. while shaking. 1 mL and 1 μ L of each culture were plated on separate LB agar plates containing 10, 100, or 1000 μ g/mL spectinomycin. Resistant CFUs were counted in the same manner as the fluctuation assays. The colonies of each plate were then scraped into separate cultures containing 2 mL of LB supplemented with 50 μ g/mL spectinomycin and grown for 16 hours at 37° C. Genomic DNA was purified using the Wizard Genomic DNA Purification Kit (Promega). 100 ng of purified genome was then processed and sequenced in the same manner as already described for the sequencing analysis of pTarget, with the one exception that oligos rpsE-F and rpsE-R were used in the first round of PCR.

(560) Oligonucleotide recombination: Re-introduction of rpsE mutations was performed using

RE1000 *E. coli* (EcNR1ilvG+dam@λTerm dualTetO-pTet< > {kil, λ cI} ΔampR pConst::araE pConst::araC lacIQ1 recJ_off xonA_off dnaG.Q576A cymR< >SS7) developed for recombineering. Electro-competent cells were prepared fresh from overnight cultures of bacteria. The saturated culture was back-diluted 1:70 into 5 mL LB with 100 ng/μl anhydrous tetracycline and shaken at 37° C. until the OD reached 0.5. Cultures were then transferred to an ice-water bath and swirled for approximately 30 seconds before being chilled on ice for 10 minutes. Chilled cultures were centrifuged at 8000 RPM for 1 minute. The supernatant was aspirated and the pellet was resuspended in 1 mL ice-chilled 10% glycerol. The aspiration and resuspension was repeated twice. The final pellet was resuspended in 70 μL chilled 10% glycerol for each transformation. 1 μg of oligonucleotide was electroporated into the cells. The cells were recovered for 1 hr at 37° C. in 1 mL LB and streaked out on LB agar plates containing 50 μg/mL spectinomycin. Successful recombination was verified by Sanger sequencing a PCR-amplification of the genomic *rpsE* gene. (561) Spectinomycin resistance characterization: Single colonies of sequence-verified *rpsE* mutants were grown overnight in LB media and then back-diluted 1:200 into LB containing 0, 100, or 1000 μg/mL spectinomycin. A Tecan M1000 Pro spectrophotometer was used to measure each well's optical density over 8 hours of growth at 37° C. Three biological replicates of each mutant at each spectinomycin concentration were characterized.

Example 3

(562) The EvolvR system was adapted for use in eukaryotic cells. EvolvR provides for diversification of all nucleotides at user-defined positions without generating double stranded breaks or relying on HDR.

(563) A HEK293T cell line with a genomically-integrated, constitutively-expressed blue fluorescent protein (BFP) gene (HEK293T-BFP) was used. As depicted in FIG. 39A, a BFP gene (SEQ ID NO:1215) can become a green fluorescent protein (GFP) gene (SEQ ID NO:1217) by undergoing a particular single-nucleotide substitution (SEQ ID NO:1218) causing an H67Y missense mutation. The frequency of GFP positive cells in an EvolvR-expressing population was used as a relative proxy for the cell's targeted mutation rate.

(564) An EvolvR expression plasmid, designated pEvolvR-HT, was constructed. The pEvolvR-HT expression plasmid can be transiently transfected into human cells. As depicted in FIG. 39B, the pEvolvR-HT expression plasmid consists of a gRNA expression cassette driven by the human U6 promoter as well as a CMV promoter-driven enCas9-PolI5M gene tagged with two SV40 nuclear localization sequences (NLSs) and an mCherry fluorescent reporter. Two days after transiently transfecting pEvolvR-HT into HEK293T-BFP cells, transfectants were enriched by sorting mCherry positive cells using FACS. As depicted in FIG. 39C, the sorted population was allowed to expand for five days; the frequency of GFP-expressing cells in the expanded population was analyzed using flow cytometry. As shown in FIG. 39D, while a population not expressing EvolvR and a population expressing EvolvR with an off-target gRNA targeting the *aavsl* locus did not produce any GFP-expressing cells, a population expressing EvolvR with a gRNA that nicks 15 base-pairs away from the H67Y mutation showed 0.05% GFP-expressing cells. Estimating that the population expressed EvolvR for six generations, the observed mutation rate is approximately 2.5×10^{-4} mutations per nucleotide per generation. This mutation rate is comparable to the mutation rate observed in *E. coli*.

Example 4

(565) The base-pair editor (BPE) system was adapted for use in eukaryotic cells. BPEs provide for editing both nucleotides of user-defined base-pairs by directing the synthesis of a DNA polymerase across deaminated templates without relying on double-stranded breaks, HDR, genome replication, or redirection of endogenous mismatch repair pathways.

(566) A HEK293T-BFP cell line with a genomically-integrated, constitutively-expressed BFP gene was used. As depicted in FIG. 39A, a BFP gene can become a green fluorescent protein (GFP) gene by undergoing a particular single-nucleotide substitution causing an H67Y missense mutation. The

frequency of GFP-positive cells in a BPE-expressing population was used as a relative proxy for the BPE's editing efficiency.

(567) A BPE expression plasmid was constructed. The BPE expression plasmid was transiently transfected into human cells. The plasmid consists of a CMV promoter-driven fusion cassette, where the cassette encodes the fusion protein: NLS, rAPOBEC1, enCas9, NLS, *E. coli* PolI. Three days after transiently transfecting this plasmid with a gRNA expression plasmid and a mCherry transfection control plasmid into HEK293T-BFP cells, the frequency of GFP-expressing cells was analyzed using flow cytometry.

(568) As depicted in FIGS. 40C-40D, the population of cells transfected with the BPE plasmid showed 22% and 22.31% GFP-positive fractions when co-transfected with an on-target gRNA (FIG. 40C), and 0.37% and 0.41% GFP-positive fractions when co-transfected with an off-target gRNA (FIG. 40D). As shown in FIGS. 40A-40B, a similar plasmid lacking the polymerase showed 6.51% and 5.83% GFP-positive fractions with an on-target gRNA (FIG. 40A), and 0.40% and 0.41% GFP-positive fraction with an off-target gRNA (FIG. 40B). These results show that fusing a DNA polymerase to Cas9/deaminase fusions increases editing efficiencies.

REFERENCES

(569) 1. Camps, M., Naukkarinen, J., Johnson, B. P. & Loeb, L. A. Targeted gene evolution in *Escherichia coli* using a highly error-prone DNA polymerase I. *Proc. Natl. Acad. Sci. U.S.A* 100, 9727-32 (2003). 2. Crook, N. et al. In vivo continuous evolution of genes and pathways in yeast. *Nat. Commun.* 7, 13051 (2016). 3. Esvelt, K. M., Carlson, J. C. & Liu, D. R. A system for the continuous directed evolution of biomolecules. *Nature* 472, 499-503 (2011). 4. Ma, Y. et al. Targeted AID-mediated mutagenesis (TAM) enables efficient genomic diversification in mammalian cells. *Nat. Methods* 13, 1-9 (2016). 5. Hess, G. T. et al. Directed evolution using dCas9-targeted somatic hypermutation in mammalian cells. *Nat. Methods* 13, 1036-1042 (2016). 6. BRENNER, S., STRETTON, A. O. W. & KAPLAN, S. Genetic Code: The 'Nonsense' Triplets for Chain Termination and their Suppression. *Nature* 206, 994-998 (1965). 7. Maheshri, N., Koerber, J. T., Kaspar, B. K. & Schaffer, D. V. Directed evolution of adeno-associated virus yields enhanced gene delivery vectors. *Nat. Biotechnol.* 24, 198-204 (2006). 8. Bryson, D. I. et al. Continuous directed evolution of aminoacyl-tRNA synthetases. *Nat. Chem. Biol.* 13, 1253-1260 (2017). 9. Hubbard, B. P. et al. Continuous directed evolution of DNA-binding proteins to improve TALEN specificity. *Nat. Methods* 12:939 (2015). 10. Costantino, N. & Court, D. L. Enhanced levels of lambda Red-mediated recombinants in mismatch repair mutants. *Proc. Natl. Acad. Sci. U.S.A* 100, 15748-53 (2003). 11. Wang, H. H. et al. Programming cells by multiplex genome engineering and accelerated evolution. *Nature* 460, 894-8 (2009). 12. Barbieri, E. M., Muir, P., Akhuetie-Oni, B. O., Yellman, C. M. & Isaacs, F. J. Precise Editing at DNA Replication Forks Enables Multiplex Genome Engineering in Eukaryotes. *Cell* 171, 1453-1467.e13 (2017). 13. DiCarlo, J. E. et al. Yeast Oligo-Mediated Genome Engineering (YOGE). *ACS Synth. Biol.* 2, 741-749 (2013). 14. Ronda, C., Pedersen, L. E., Sommer, M. O. A. & Nielsen, A. T. CRMAGE: CRISPR Optimized MAGE Recombineering. *Sci. Rep.* 6, 19452 (2016). 15. Findlay, G. M., Boyle, E. A., Hause, R. J., Klein, J. C. & Shendure, J. Saturation editing of genomic regions by multiplex homology-directed repair. *Nature* 513, 120-123 (2014). 16. Mao, Z., Bozzella, M., Seluanov, A. & Gorbunova, V. DNA repair by nonhomologous end joining and homologous recombination during cell cycle in human cells. *Cell Cycle* 7, 2902-2906 (2008). 17. Aguirre, A. J. et al. Genomic Copy Number Dictates a Gene-Independent Cell Response to CRISPR/Cas9 Targeting. *Cancer Discov.* 6, 914-29 (2016). 18. Choi, P. S. & Meyerson, M. Targeted genomic rearrangements using CRISPR/Cas technology. *Nat. Commun.* 5, 4728 (2014). 19. Frock, R. L. et al. Genome-wide detection of DNA double-stranded breaks induced by engineered nucleases. *Nat. Biotechnol.* 33, 179-186 (2014). 20. Mao, Z., Bozzella, M., Seluanov, A. & Gorbunova, V. Comparison of nonhomologous end joining and homologous recombination in human cells. *DNA Repair (Amst)*. 7, 1765-71 (2008). 21. Camps, M., Naukkarinen, J., Johnson, B. P. & Loeb, L. A. Targeted gene evolution in *Escherichia coli* using a

highly error-prone DNA polymerase I. *Proc. Natl. Acad. Sci. U.S.A* 100, 9727-32 (2003). 22. Bambara, R. A. & Choi, T. On the Processive Mechanism of *Escherichia coli* DNA Polymerase I Quantitative Assessment of Processivity. *J. Biol. Chem.* 253, 413-423 (1973). 23. de Boer, J. G. & Ripley, L. S. An in vitro assay for frameshift mutations: hotspots for deletions of 1 bp by Klenow-fragment polymerase share a consensus DNA sequence. *Genetics* 118, (1988). 24. Sarkar, S., Ma, W. T. & Sandri, G. v. H. On fluctuation analysis: a new, simple and efficient method for computing the expected number of mutants. *Genetica* 85, 173-179 (1992). 25. Drake, J. W. A constant rate of spontaneous mutation in DNA-based microbes. *Proc. Natl. Acad. Sci. U.S.A* 88, 7160-4 (1991). 26. Jagessar, K. L. & Jain, C. Functional and molecular analysis of *Escherichia coli* strains lacking multiple DEAD-box helicases. *RNA* 16, 1386-92 (2010). 27. Fijalkowska, I. J. & Schaaper, R. M. Mutants in the Exo I motif of *Escherichia coli* dnaQ: defective proofreading and inviability due to error catastrophe. *Proc. Natl. Acad. Sci. U.S.A* 93, 2856-61 (1996). 28. Wang, H. H. et al. Programming cells by multiplex genome engineering and accelerated evolution. *Nature* 460, 894-8 (2009). 29. Kleinstiver, B. P. et al. High-fidelity CRISPR-Cas9 nucleases with no detectable genome-wide off-target effects. *Nature* 529, 490-495 (2016). 30. Slaymaker, I. M. et al. Rationally engineered Cas9 nucleases with improved specificity. *Science* (80-.). 351, 84-88 (2015). 31. Minnick, D. T. et al. Side chains that influence fidelity at the polymerase active site of *Escherichia coli* DNA polymerase I (Klenow fragment). *J. Biol. Chem.* 274, 3067-75 (1999). 32. Loh, E., Salk, J. J. & Loeb, L. A. Optimization of DNA polymerase mutation rates during bacterial evolution. *Proc. Natl. Acad. Sci. U.S.A* 107, 1154-9 (2010). 33. Truniger, V., Lázaro, J. M., de Vega, M., Blanco, L. & Salas, M. phi 29 DNA polymerase residue Leu384, highly conserved in motif B of eukaryotic type DNA replicases, is involved in nucleotide insertion fidelity. *J. Biol. Chem.* 278, 33482-91 (2003). 34. Wang, Y. et al. A novel strategy to engineer DNA polymerases for enhanced processivity and improved performance in vitro. *Nucleic Acids Res.* 32, 1197-1207 (2004). 35. Whitaker, W. R., Lee, H., Arkin, A. P. & Dueber, J. E. Avoidance of Truncated Proteins from Unintended Ribosome Binding Sites within Heterologous Protein Coding Sequences. *ACS Synth. Biol.* 4, 249-257 (2015). 36. Salis, H. M. The Ribosome Binding Site Calculator. in *Methods in enzymology* 498, 19-42 (2011). 37. Funatsu, G., Schiltz, E. & Wittmann, H. G. Ribosomal proteins. XXVII. Localization of the amino acid exchanges in protein S5 from two *Escherichia coli* mutants resistant to spectinomycin. *Mol. Gen. Genet.* 114, 106-11 (1972). 38. Zheng, X., Xing, X.-H. & Zhang, C. Targeted mutagenesis: A sniper-like diversity generator in microbial engineering. (2017). doi: 10.1016/j.synbio.2017.07.001 39. Timms, A. R., Steingrimsdottir, H., Lehmann, A. R. & Bridges, B. A. Mutant sequences in the rpsL gene of *Escherichia coli* B/r: Mechanistic implications for spontaneous and ultraviolet light mutagenesis. *MGG Mol. Gen. Genet.* 232, 89-96 (1992). 40. Brocklehurst, P. & Peter. Antibiotics for gonorrhoea in pregnancy. in *Cochrane Database of Systematic Reviews* (ed. Kellie, F. J.) (John Wiley & Sons, Ltd, 2002). doi: 10.1002/14651858.CD000098 41. Lee, R. E. et al. Spectinamides: a new class of semisynthetic antituberculosis agents that overcome native drug efflux. *Nat. Med.* 20, 152-158 (2014). 42. Liu, B. & Pop, M. ARDB-Antibiotic resistance genes database. *Nucleic Acids Res.* 37, 443-447 (2009). 43. Lee, M. E., DeLoache, W. C., Cervantes, B. & Dueber, J. E. A Highly-characterized Yeast Toolkit for Modular, Multi-part Assembly. *ACS Synth. Biol.* (2015). doi: 10.1021/sb500366v 44. Koboldt, D. C. et al. VarScan 2: somatic mutation and copy number alteration discovery in cancer by exome sequencing. *Genome Res.* 22, 568-76 (2012). 45. Hall, B. M., Ma, C.-X., Liang, P. & Singh, K. K. Fluctuation analysis CalculatOR: a web tool for the determination of mutation rate using Luria-Delbruck fluctuation analysis. *Bioinformatics* 25, 1564-5 (2009). 46. Rosche, W. A. & Foster, P. L. Determining Mutation Rates in Bacterial Populations. *Methods* 20, 4-17 (2000). 47. Hess, G. T. et al. Directed evolution using dCas9-targeted somatic hypermutation in mammalian cells. *Nat. Methods* 13, 1036-1042 (2016). 48. de Vega, M., Lazaro, J. M., Salas, M. & Blanco, L. Primer-terminus stabilization at the 3'-5' exonuclease active site of phi29 DNA polymerase. Involvement of two amino acid residues highly conserved in proofreading DNA polymerases. *EMBO J.* 15,

1182-92 (1996). 49. Ducani, C., Bernardinelli, G. & Högberg, B. Rolling circle replication requires single-stranded DNA binding protein to avoid termination and production of double-stranded DNA. *Nucleic Acids Res.* 42, 10596-10604 (2014). 50. Povilaitis, T., Alzbutas, G., Sukackaite, R., Siurkus, J. & Skirgaila, R. In vitro evolution of phi29 DNA polymerase using isothermal compartmentalized self replication technique. *Protein Eng. Des. Sel.* 29, 617-628 (2016). 51. Greener, A., Callahan, M. & Jerpseth, B. An efficient random mutagenesis technique using an *E. coli* mutator strain. *Mol. Biotechnol.* 7, 189-195 (1997). 52. Badran, A. H. & Liu, D. R. Development of potent in vivo mutagenesis plasmids with broad mutational spectra. *Nat. Commun.* 6, 8425 (2015).

(570) While the present invention has been described with reference to the specific embodiments thereof, it should be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the true spirit and scope of the invention. In addition, many modifications may be made to adapt a particular situation, material, composition of matter, process, process step or steps, to the objective, spirit and scope of the present invention. All such modifications are intended to be within the scope of the claims appended hereto.

Claims

1. A cell comprising: a) an enzymatically active RNA-guided endonuclease that introduces a break in at least one strand in a target DNA; b) a DNA polymerase that binds to the target DNA 5' to the break; and c) a guide RNA that binds the endonuclease, wherein neither said endonuclease nor said polymerase comprises a deaminase.
2. The cell of claim 1, wherein the cell is a prokaryotic cell.
3. The cell of claim 1, wherein the cell is a eukaryotic cell.
4. The cell of claim 1, wherein the cell is in vitro.
5. The cell of claim 1, wherein the guide RNA comprises a protein-binding segment comprising a nucleotide sequence that binds to the endonuclease, and a target-binding segment comprising a nucleotide sequence that is complementary to a target nucleotide sequence of the target DNA.
6. The cell of claim 1, wherein the guide RNA comprises one or more of: a) a modified base; b) a modified backbone; c) a modified sugar moiety; d) a non-natural internucleoside linkage.
7. The cell of claim 1, wherein the guide RNA guides the endonuclease to introduce the break in the target DNA.
8. The cell of claim 1, wherein the break in at least one strand is a single-stranded break.
9. The cell of claim 1, wherein the DNA polymerase synthesizes a new strand on the target DNA.
10. The cell of claim 9, wherein the DNA polymerase introduces a mutation in the new strand.
11. The cell of claim 10, wherein the DNA polymerase introduces a mutation in the new strand at a distance of from 1 nucleotide to 100 nucleotides from the single-stranded break in the target DNA.
12. The cell of claim 1, wherein the endonuclease comprises a class 2 CRISPR/Cas endonuclease.
13. The cell of claim 12, wherein the class 2 CRISPR/Cas endonuclease comprises a type V or type VI CRISPR/Cas endonuclease.
14. The cell of claim 12, wherein the class 2 CRISPR/Cas endonuclease comprises a Cas9 polypeptide.
15. The cell of claim 1, wherein the endonuclease and the DNA polymerase are fused as parts of a fusion polypeptide.
16. The cell of claim 15, wherein the fusion polypeptide, when complexed with a guide RNA, exhibits a target mutation rate of from 10.sup.-8 to 10.sup.-7, 10.sup.-7 to 10.sup.-6, 10.sup.-6 to 10.sup.-5, 10.sup.-5 to 10.sup.-4, 10.sup.-4 to 10.sup.-3, or 10.sup.-3 to 10.sup.-2 mutations per nucleotide per genome replication event.
17. The cell of claim 15, wherein the fusion polypeptide further comprises a linker connecting the endonuclease and the DNA polymerase.

18. The cell of claim 15, wherein the fusion polypeptide further comprises a nuclear localization signal.

19. The cell of claim 15, wherein the fusion polypeptide comprises, in order from N-terminus to C-terminus: a) the enzymatically active RNA-guided endonuclease; and b) the DNA polymerase.

20. The cell of claim 15, wherein the fusion polypeptide comprises, in order from N-terminus to C-terminus: a) the enzymatically active RNA-guided endonuclease; b) a linker; and c) the DNA polymerase.

21. The cell of claim 15, wherein the fusion polypeptide comprises, in order from N-terminus to C-terminus: a) a nuclear localization signal; b) the enzymatically active RNA-guided endonuclease; and c) the DNA polymerase.

22. The cell of claim 15, wherein the fusion polypeptide comprises, in order from N-terminus to C-terminus: a) a nuclear localization signal; b) the enzymatically active RNA-guided endonuclease; c) a linker; and d) the DNA polymerase.

23. The cell of claim 15, wherein the fusion polypeptide comprises, in order from N-terminus to C-terminus: a) the DNA polymerase; and b) the enzymatically active RNA-guided endonuclease.

24. The cell of claim 15, wherein the fusion polypeptide comprises, in order from N-terminus to C-terminus: a) the DNA polymerase; b) a linker; and c) the enzymatically active RNA-guided endonuclease.

25. The cell of claim 15, wherein the fusion polypeptide comprises, in order from N-terminus to C-terminus: a) a nuclear localization signal; b) the DNA polymerase; and c) the enzymatically active RNA-guided endonuclease.

26. The cell of claim 15, wherein the fusion polypeptide comprises, in order from N-terminus to C-terminus: a) a nuclear localization signal; b) the DNA polymerase; c) a linker; and d) the enzymatically active RNA-guided endonuclease.

27. The cell of claim 1, wherein the cell is in vivo.
